

***Bio-Dimension™ UV/Vis
HPLC Detector
Instruction Manual***

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Section 1 ***Introduction***

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The Bio-Dimension detector is a programmable, high-speed, scanning, multiple wavelength, ultraviolet-visible absorbance detector for high performance liquid chromatography. The unit combines ease of use, flexibility, and power.

1.1 About This Manual

This manual describes the installation, operation, routine maintenance, and service for the detector. In addition, we have included a section on troubleshooting.

The manual is organized in the following manner:

SECTION 1 briefly describes the capabilities and specifications of the unit.

SECTION 2 describes the procedure for unpacking and installing the instrument and for initial operation as a straightforward HPLC detector.

SECTION 3 is a tutorial that provides complete instructions for all facets of instrument operation, arranged by function. New users should familiarize themselves with this section.

SECTION 4 is a reference guide to the operation of the instrument, arranged by menu and command.

SECTION 5 describes routine maintenance

SECTION 6 explains troubleshooting and service

SECTION 7 describes the theory of operation.

1.2 Capabilities

The unit provides a unique combination of ease of use, power, and flexibility. The instrument performs as a:

- Simple, variable wavelength UV/Visible absorbance detector. The monitor may be set to any wavelength in the 190–800 nm range. Real-time chromatographic data is simultaneously transmitted out a recorder channel output (data here can be ranged and filtered) and a fixed-range integrator output (1 volt/AU).
- Real-time multiple wavelength UV/Vis absorbance detector. The detector may be set to monitor up to 32 different wavelengths in the 190 to 800 nm range. Real-time data for four wavelengths may be monitored using any of the four recorder outputs. Additionally, one wavelength may be monitored using the integrator output. Monitored wavelengths are freely changed at will. Any two of the monitored wavelengths absorbencies may be ratioed, for peak purity determination.
- High speed recording spectrophotometer. Spectral data may be acquired and stored in memory for later playback.
- Recording single or multiple wavelength monitor in which chromatographic data is stored and replayed selecting from a variety of ranges and filters. Data may be replayed faster than real-time.
- Time-programmable detector/spectrophotometer in which changes in wavelength, sensitivity, response time, recorder channel output, and spectral scanning are performed.

- Programmable scanning detector in which predefined spectral scans are automatically performed on the upshoulder, apex, and downshoulder of a chromatographic peak. The stored spectra can be replayed and compared to determine peak purity.

Additionally, the diverse selection of superbly designed flow cells allows the monitor to optimally perform single wavelength, multiple wavelength, and scanning applications for the following chromatographic techniques:

Analytical HPLC

Analytical HPLC Requiring Inert Contact Surfaces

Preparative Scale HPLC

Preparative Scale HPLC Requiring Inert Contact Surfaces

For more information regarding these flow cells, please consult the "Options and Accessories" appendix to this manual.

With the addition of optional software, the detector's power is expanded. Used with the software, the Bio-Dimension detector is capable of the following:

- High speed multi-wavelength operation which allows acquisition of continuous spectral data throughout a chromatogram.
- Time Slice operation which permits the display of spectra from multiple wavelength information by slicing the three-dimensional chromatographic data of absorbance vs. time vs. wavelength perpendicularly to the temporal axis. This feature enables the examination of a peak's spectra at any point during its elution, a valuable tool in determining peak purity.
- Batch mode operation compatible with Bio-Rad's HRLC Sample Table for a completely automated system.
- File storage and retrieval of detector programming parameters. In addition to annotating a particular chromatographic run, this feature allows the user to easily load specific detector parameters with minimal effort.
- File storage and retrieval of chromatographic (single and multiple wavelength) and spectral data for future display and data reduction.
- Data reduction including:
 - File subtraction for point-to-point baseline correction and deconvolution of overlapping peaks
 - Wavelength absorbance ratioing
 - First and second derivatives of chromatograms
 - First and second derivatives of spectra
 - Spectra normalization and subtraction for peak purity determination (for discrete or Time Slice spectra)
 - Two and three-dimensional display of reduced chromatographic and spectral data on a computer monitor

- Real-time display of chromatographic data (up to three wavelengths at a time) on a computer monitor
- Chromatogram or spectra hard copy formation using a high speed, high resolution printing function

Consult the *Bio-Dimension Software Owners' Manual* for a complete list and description of software generated features.

Finally, the convenient location of the lamps, photodiodes, and flow cell housing makes changing these components easy.

1.3 Specifications

Additional specifications are in Appendix B

Wavelength Range:	190–800 nm
Band Width:	6.0 nm
Wavelength Accuracy:	+/- 1.0 nm
Wavelength Precision:	0.1 nm
Slew Rate 190–800 nm:	20 milliseconds
Optical Methodology:	Deuterium (190–365 nm) and tungsten (366–800 nm) lamp light sources with servo driven concave holographic grating monochromator and double beam optics provision for purging the monochromator
Range Selections:	0.0005 to 7.9995 AUFS, in 0.0005 increments on any of the chart recorder outputs
Absorbance Ratio:	Displayed as ratio of absorbencies (at the two selected wavelengths) per full scale range; plotted as log of ratio of absorbencies per full scale range
Recorder Outputs:	10 mV, 100 mV, 1.0 V output in any combination of the four recorder outputs, 14 bit resolution
Integrator Output:	1 V/AU, wavelength selectable in multiple wavelength mode
Remote Controls:	Rear panel input for autozero of chart recorder and integrator outputs; rear panel input for event mark.; RS-232C serial interface for computer control and/or data acquisition; DPDT relay closure for optional stop flow device and other controlled devices
Single Wavelength Noise:	$< 2 \times 10^{-5}$ AU/60 second from 220–280 nm with 1.0 second rise time
Drift:	$< 2 \times 10^{-4}$ AU/hour after 30 minute warm-up
Zero Adjust:	Plus or minus any amount by autozero and manual entry
Filter:	Digital second order Bessel filter with 10% to 90% rise times selectable from 0.0 to 9.9 seconds (divide by 2 to convert to an equivalent time constant)

Section 2

Initial Operation

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2.1 Unpacking

Carefully unpack the detector from the shipping container, and inspect both the unit and the packing for any signs of damage. If any damage is noted, contact the shipping company immediately.

In addition to this manual, the shipping container contains a power cord and the detector assembly equipped with a standard flow cell. Carefully check the packing list against the contents of the container. If anything is missing, check the packing materials for overlooked items. If items prove to be missing, contact Bio-Rad's customer service department.

Place the detector on the bench where it will be used and familiarize yourself with the location and function of the controls and connections.

2.2 Location

Place the detector on a laboratory benchtop in close proximity to the LC column outlet. Allow at least 5 inches of clear space between the rear panel of the unit and any wall or obstruction. This provides both access to the rear panel connections and a free flow of cooling air.

2.3 What You Will Need

In addition to the detector itself, you will need the following items for set up and initial operation:

- Strip chart recorder and connecting cables or Bio-Rad's Series 800 HRLC[®] system
- IBM PC AT or compatible computer with 640 K ram, 8 MHz, running Microsoft[®] Windows[®], version 2.03 or later, one hard and one floppy drive (required only if using optional software)
- Liquid chromatograph
- Column
- Standard test mix
- Appropriate solvents, reagents, etc.
- Nuts, ferrules, appropriate to the column end-fitting being used
- Wrenches appropriate to column end fittings
- Connecting tubing and union (if column cannot be connected directly to flow cell)

2.4 Controls and Connections

Front Panel

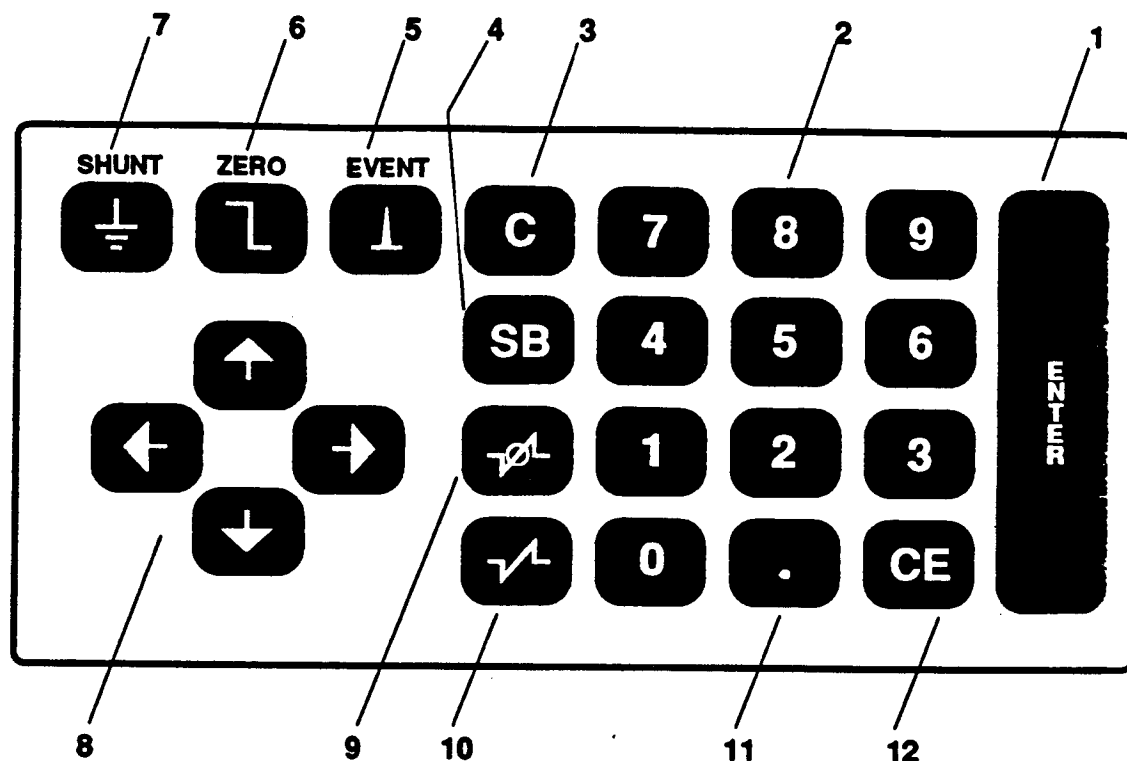


Fig. 2.1. Front panel.

The following controls are located on the front panel of the detector:

1. **ENTER KEY.** Press this key to accept a parameter value you have typed in.
2. **NUMERIC KEYS.** Use this key to enter numeric parameter values.
3. **CLEAR ALL KEY.** Use this key to erase all numeric parameters entered on a given menu. This key function for Polychrome, Schedule Spectra, and Programmed Wavelength Changes menus.
4. **STAND-BY KEY.** Press this key to place the detector in the stand-by status. Under stand-by conditions, the lamps are turned off and the galvanometer drive is locked in a safe position.
5. **EVENT KEY.** Press this key to place a mark (approximately 20% full-scale deflection) on the recorder outputs. This key is also used to advance the detector to the Chromatogram Underway status while in the Programmed Operation Mode.
6. **ZERO KEY.** Press this key to autozero the detector baseline at all wavelengths of operation.
7. **SHUNT KEY.** Press this key to output zero volts to the recorder outputs.
8. **CURSOR KEYS.** The up, down, right and left arrow keys move the cursor around in the various LCD fields. They can also be used to enter a parameter value while simultaneously advancing to the next field.

9. SPECTRA ZERO KEY. Press this key to store absorbance zero values for a defined, discrete spectra.
10. TAKE SPECTRA KEY. Press this key to take a discrete spectra of the flow cell's contents.
11. DECIMAL POINT KEY. Press this key to enter a decimal point.
12. CLEAR ENTRY KEY. Use this key to erase a previously entered numerical parameter.

Rear Panel

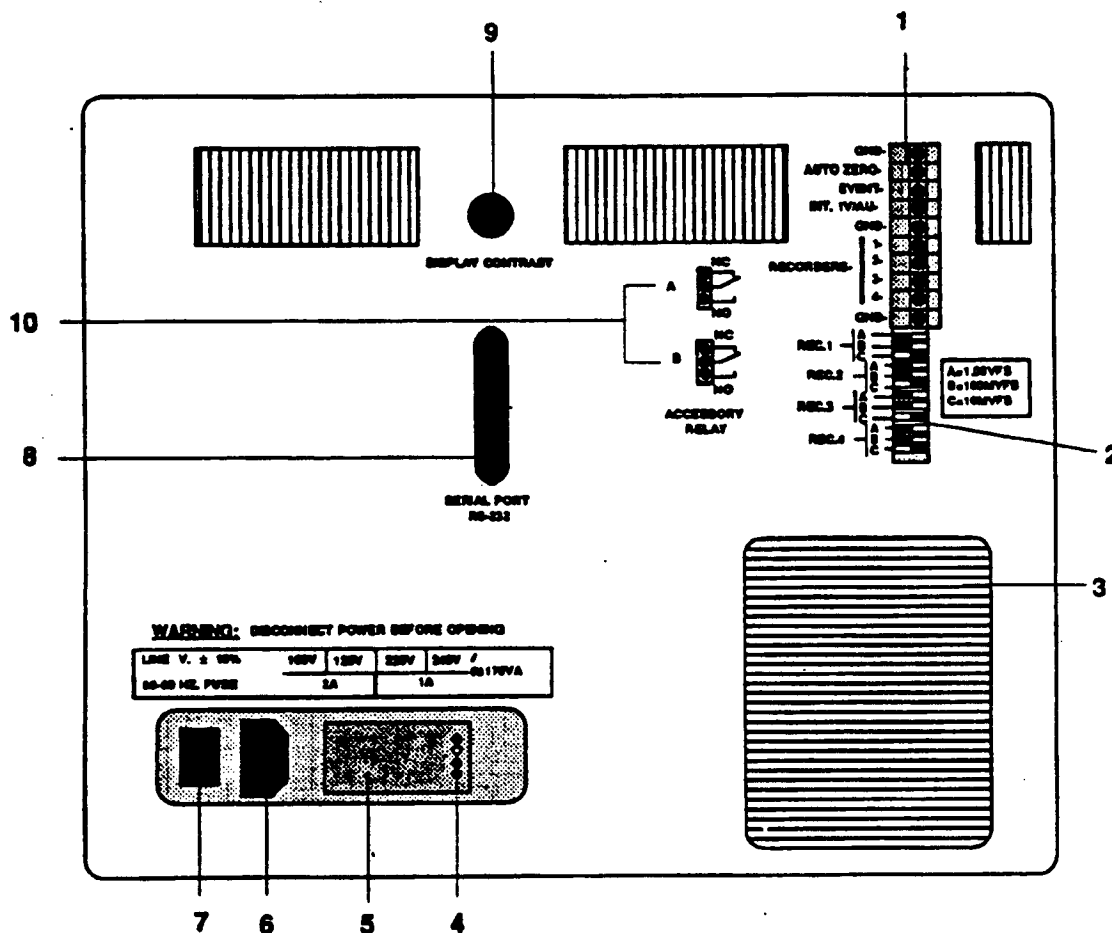


Fig. 2.2. Rear panel.

The following connections are located on the rear panel of the detector:

1. I/O CONNECTIONS. This terminal block allows connection of recorder leads to up to four recorders and one integrator, as well as input connections for remote triggering of the autozero and event marker. Connections are made by pressing in on the orange tabs with a small screwdriver, inserting a stripped wire end into the corresponding hole, and releasing the tab.

2. RECORDER VOLTAGE RANGE SETTING SWITCHES. Allow the output range of the recorder connections to be set to 10, 100, or 1000 mV full scale. The integrator output voltage is fixed at 1 volt/AU.

3. FAN

4. **VOLTAGE SETTING.** Allows the detector operating voltage to be set for 100, 110, 220, or 240 VAC, 50/60 Hz.
5. **FUSE BLOCK.** Pry out this block to allow access to the fuses and to change the operating voltage.
6. **POWER CONNECTOR.** Accepts standard modular power line cord.
7. **ON/OFF SWITCH.** Used to switch the instrument's power on or off.
8. **RS-232C SERIAL INTERFACE.** Twenty five-pin female connector allows RS-232 communication to external computer or terminal. Used with optional software packages.
9. **DISPLAY CONTRAST CONTROL.** Turn to adjust the LCD display contrast for different viewing angles or light conditions.
10. **ACCESSORY RELAY CONNECTOR.** Use these outputs to trigger external events from the detector.

Left Bulkhead

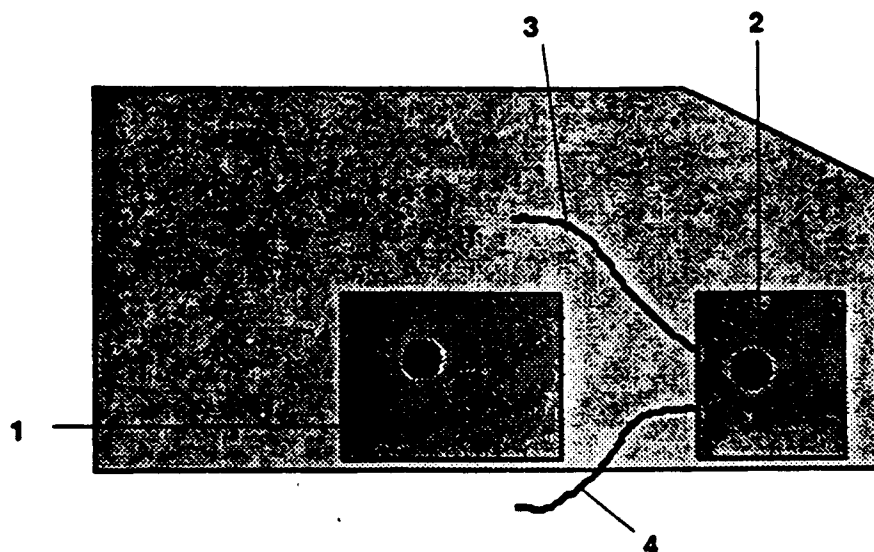


Fig. 2.3. Detector left bulkhead.

The following components are located on the left bulkhead of the detector:

1. **LAMP HOUSING.** Loosen the screw and remove the cover to allow access to the lamps.
2. **FLOW CELL HOUSING.** Loosen the thumbscrew and remove the cover to allow access to the flow cell and sample photodiode.
3. **OUTLET LINE.** A 1/16-inch OD tube for connection to a drain line, backpressure device, or waste reservoir.
4. **INLET LINE.** A 1/16-inch OD tube for connection to the column outlet.

2.5 Setting Up

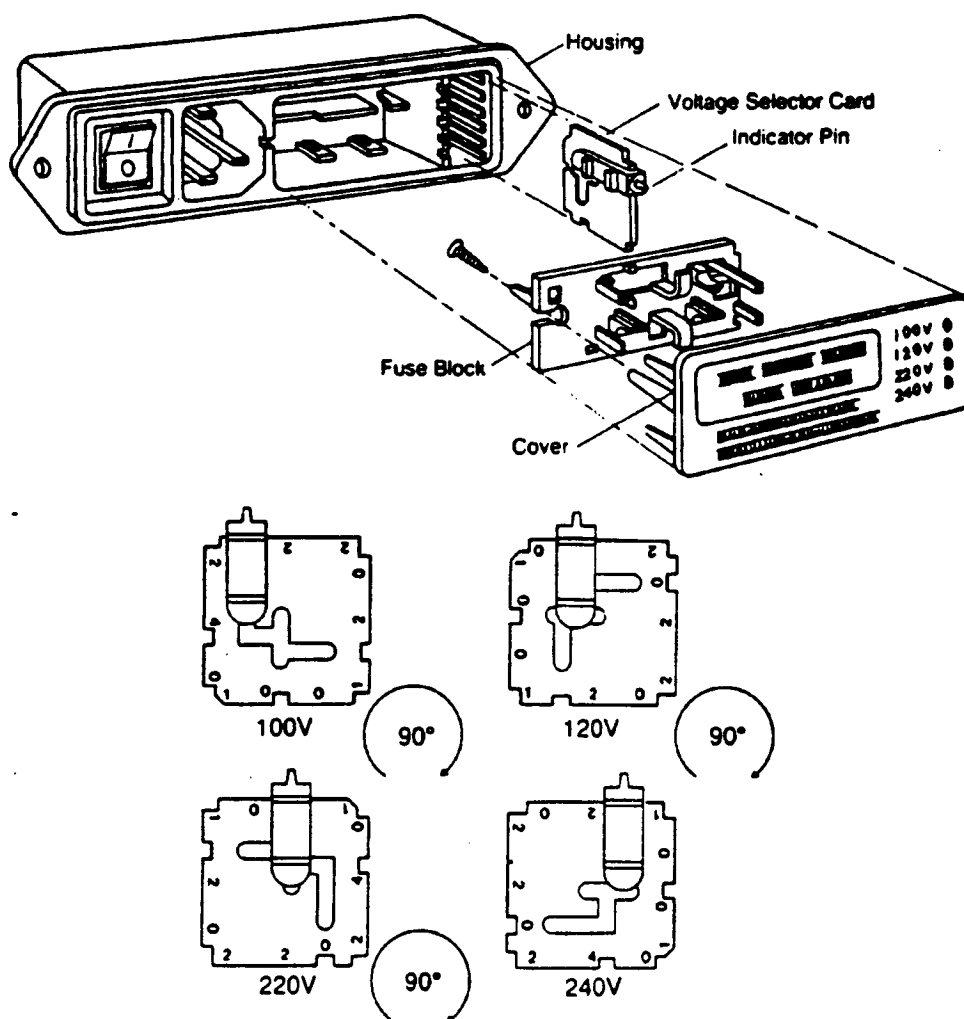
Setting Voltage

Check the voltage selector block located next to the power cord connector on the rear panel. The white plastic pin indicates the voltage for which the instrument has been configured (see figure 2.4). This has been factory set for the voltage appropriate to your country (100, 110, 220, or 240 VAC). If the voltage is not correctly set, you should reset it to the proper value before proceeding further.

Insert the blade of a small screwdriver into the slot next to the power connector and pry open the fuse block. Pull the fuse block straight out. Using tweezers or a pair of long-nose pliers, pull the voltage selector card straight out. Orient the plastic indicator for the proper voltage as indicated in figure 2.4, then press the selector card back into place.

Make sure that the fuse block is oriented so that the single long fuse holder is showing for 100 or 120 V operation, or so that the dual short fuse holders are showing for 220 or 240 V operation. Snap the fuse block back into place, taking care that the white plastic pin protrudes from the hole corresponding to the desired voltage.

Fig. 2.4.



Setting Recorder Range

The detector can accommodate up to four strip chart recorder channels connected simultaneously. The full scale voltage for each channel may be independently set at 10, 100, or 1000 mV using a bank of switches located on the rear panel immediately below the I/O connection block.

This bank contains twelve switches; three (labelled A, B, and C) for each of the recorder channels. The switches correspond to output ranges of 1 V, 100 mV, and 10 mV, respectively. For each channel the switch corresponding to the desired range should be on (facing left). The other two switches for that channel should be off (facing right). Normally, the full scale voltage is set to match the input span of the recorder being used.

To change a range, move the appropriate switches using a small screwdriver or the tip of a pen. Remember, for each recorder output, set one switch to the left and the remaining two, to the right.

Recorder Connections

The recorder cables are connected at the I/O block on the rear panel of the detector (see figure 2.2). The cables should end in 1/4 inch or so of bare wire. To make a connection, press in on the orange tab with the blade of a small screwdriver. Insert the wire end in the hole to the left of the orange tab, and release the tab. To disconnect a wire, simply press the orange tab and pull the wire end out of the hole.

The recorder used should have a "floating" (ungrounded) input. In most cases, the minus (black) input of the recorder should be connected to one of the ground (GND) terminals on the detector. The plus (red) input of the recorder should be connected to the appropriate terminal on the detector.

For the initial, check-out run, you should connect a strip chart recorder to channels one and two. The use of a multiple pen recorder is recommended.

Integrator Connection

The detector provides a fixed-span output (1V/AU) for use with integrators or data acquisition systems. Signals are provided with a resolution ranging from 7 micro AU to 7.9995 AU. Connections are made in the same way as recorder connections are made.

Other Connections

The detector provides for a number of other connections for accessories. You do not need to connect these in order to finish the initial installation and check-out. If you do not wish to connect the Autozero, Event Switch, Relays, or Serial Port at this time, you may proceed directly to INITIAL TEST.

Autozero: The AUTOZERO connection on the I/O block duplicates the function of the ZERO key on the front panel whenever it is connected to ground or a TTL low by a momentary contact closure provided by a remote instrument.

The cable should end in 1/4 inch or so of bare wire. To make the connection, press on the orange tab with the blade of a small screwdriver. Insert the wire end in the hole to the left of the orange tab and release the tab. To disconnect the wire, simply press the orange tab and pull the wire end out of the hole.

If the remote instrument will be using a TTL to trigger the autozero, then the TTL trigger should be connected to the AUTOZERO input of the detector and the grounds of the controlling device should be connected to a ground terminal on the detector. If the remote instrument will be using a contact closure to trigger the autozero, then the common terminal of the relay should be connected to one of the ground terminals of the detector. The normally open terminal on the relay should be connected to the AUTOZERO terminal on the detector.

Event: The EVENT connection on the I/O Block duplicates the function of the EVENT key on the front panel whenever it is connected to a ground or a TTL low by a momentary contact closure provided by a remote triggering device. Connections are made in an identical manner as for the AUTOZERO.

Relays: The detector is provided with two RELAYS for triggering remote devices. Both relays are activated simultaneously. They are rated up to 250 V at 2.5 A. These relays are activated by the Stop-Flow Option of the detector.

RS-232C Serial Port: The instrument is capable of being entirely controlled by a remote computer using commands transmitted via the RS-232C PORT. Data is also transmitted over this port. If you are using the optional software, you should connect one of your computer's serial ports to this interface using a 25-pin straight RS-232 extension. See the *Software Manual* for further details. A complete description of the RS-232 protocol and hardware requirements are provided in the *Bio-Dimension Software Manual*.

2.6 Initial Test

The detector generally requires a 60 minute warm-up period before reaching optimal performance.

Connect the detector to an electrical power circuit and turn the power on using the switch located on the bottom of the rear panel. The liquid crystal display (LCD) should display the MAIN menu.

If the menu does not appear, adjust the DISPLAY CONTRAST control on the rear panel as necessary to provide the best readability for your lighting conditions.

```
Absorbance @ 254 nm = 0.00000 AU : Menu 1
range = 2.0000 AUFS: rise time 1.0 sec
Spectrum: 200 to 360 @ 10 nm ; reps 4
Method: store recall ;
Program ☐multi ☐wavelength ☐spectra
Playback: ☐chromatogram ☐spectra
Menus: ☐start chromatogram ☐auxiliary
Mode: ☐spectra only UV
```

The "UVW" in the lower right-hand corner of the display indicates that the UV lamp is warming up. At the end of the warm-up cycle (about 20 seconds), the display will change to "UV" indicating a lit lamp.

After the UV lamp lights, press the SHUNT key and hold it down while you adjust the pen position on the recorder connected to channel 1. Adjust the pen to the middle of the recorder chart.

Release the SHUNT key and press the ZERO key. The absorbance displayed should be very close to zero, and the recorder pen should be very close to the zero position you just set.

Press EVENT. The recorder pen should momentarily deflect 20% of full scale if the detector's recorder output and input sensitivity are set correctly.

Use the cursor-down key to move the flashing cursor to the RANGE position. If you go too far, use the cursor-up key to move back up.

Type in a more sensitive range setting (0.002 AUFS, for example) by pressing the appropriate numeric keys. If you type in a number incorrectly, use the cursor-left key to back-up, then re-type the correct number. Press the ENTER key to load the new range value.

Turn the detector off and advance to the next section.

2.7 Fluid Connections

The detector inlet is the lower tube which protrudes from the rear wall of the cell compartment. As a general rule, the less tubing between the column outlet and the detector cell, the better. Ideally, the column outlet should be connected directly to the detector inlet line. If this is not feasible (if the column is contained in an oven or heater, for example), you should use a minimum length of narrow-bore (0.005-0.010 inch ID) connecting tubing.

Because different columns use different fittings, the detector is supplied with a bare tube end to allow connection to any column accepting 1/16-inch tubing. You should use nuts and ferrules suitable to your column.

Connect the cell outlet (the upper of the two tubes which protrude from the rear wall of the cell compartment) to a line leading to an appropriate waste reservoir. If bubble formation in the detector cell causes problems, you may wish to connect the cell outlet to a restrictor or backpressure device providing 20 to 60 psi backpressure.

2.8 Operation

Turn on the HPLC system and allow the column to equilibrate with the flowing eluant (the time required will depend on your particular column and eluant). If you have not already done so, turn on the power to the detector using the switch located on the lower right panel. Adjust the contrast of the display using the DISPLAY CONTRAST adjustment as necessary. The MAIN menu should appear on the LCD.

When the UV lamp lights, press the SHUNT key and hold it down while you adjust the pen on the recorder connected to CHANNEL 1. Adjust the pen to your desired position on the recorder chart.

Release the SHUNT key and press the ZERO key. The absorbance display should be very close to the zero position which you just set.

2.9 Setting Parameters

Wavelength

Insure that the cursor is in the ABSORBANCE @ 254 NM position. Enter the wavelength appropriate for your test mix by typing in the numbers and pressing enter. Press the ZERO key to rezero the detector absorbance reading at the new wavelength.

```
Absorbance @ 254 nm = 0.00000 AU : Menu 1
range = 2.0000 AUFS: rise time 1.0 sec
Spectrum: 200 to 360 @ 10 nm ; reps 4
Method: store recall ;
Program ☐multi ☐wavelength ☐spectra
Playback: ☐chromatogram ☐spectra
Menus: ☐start chromatogram ☐auxiliary
Mode: ☐spectra only UV
```

Range

Use the down arrow key to move the cursor to the RANGE = 2.0000 AUFS field. Type in the desired range appropriate for your sample concentration and flow cell pathlength. Press enter.

Rise Time

The detector uses a digital second order Bessel filter. The response of this type of filter is described in terms of a 10% to 90% response to a step input. This response time is called rise time (RT). For a given RT, the response of a Bessel Filter is roughly equivalent to a RC time constant equal to 1/2 the RT. An RT of 1 second will produce approximately the same result as a 0.5 second time constant although the second order Bessel filter produces much less tailing. The default value of RT is set for 1.0 second. This value is adequate for most routine HPLC separations. If you are running a very fast analysis, a shorter RT may be necessary. For very high sensitivity detection, a longer RT may be more appropriate. In general, if the RT is set to be equal to or less than 1/3 the half height peak width, the filter will cause no peak broadening.

Move the cursor to the RISE TIME line. Type in the desired rise time in increments of 0.1 second, and press ENTER. Allowable rise time values range from 0.0 to 9.9 seconds.

2.10 Recording the Chromatogram

Switch the recorder chart on, inject your sample, and press the EVENT key.

You may change the wavelength at any time:

- move the cursor to the ABSORBANCE @ field
- type in the new value
- press the ENTER key
- press the ZERO key

You may change the RANGE at any time:

- move the cursor to the RANGE field
- type in the new value
- press the ENTER key
- press the ZERO key (optional)

2.11 Shutdown

To shut down the system, turn the power switch on the back panel to off. This erases any detector parameters which we have entered. Or you may leave the power on, but shut off the lamps. To do so:

From the MAIN menu, move the cursor to the AUXILIARY field and press ENTER to call up the AUXILIARY menu. Move the cursor to the DEUTERIUM field of the lamp line and press ENTER to turn off the deuterium lamp. If you used the tungsten lamp, move the cursor to the TUNGSTEN field and press ENTER to turn off the tungsten lamp. To turn these lamps back on, move the cursor to the appropriate field and press ENTER.

```

Return to main menu :          MENU 2
Lamps: tungsten OFF  deuterium ON
Auto calibrate
Baseline 0.00 to 3.33 min @ 0.00000 AU
Auto Spectra OFF 1 per peak; sides EARLY
Auto scan relay           [disabled]
Zero upon wavelength change  [ON]
SCROLL to access additional parameters

```

Section 3

Routine Operation Tutorial

Section 2 of this manual described the unpacking, initial set up, and testing of the detector. This section of the manual describes the procedures for using the monitor on an operational basis. The section begins with a discussion of the menu structure and detector key pad and then examines each mode of detector operation, employing a step-by-step tutorial.

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3.1 The Menus

When the instrument is first turned on, the MAIN menu is displayed. The MAIN menu controls the single wavelength mode of the instrument and allows access to all other menus. This menu shares features with all of the other menus.

```
Absorbance @ 254 nm = 0.00000 AU : Menu 1
range = 2.0000 AUFS: rise time 1.0 sec
Spectrum: 200 to 360 @ 10 nm ; reps 4
Method: store recall ;
Program ☐multi ☐wavelength ☐spectra
Playback: ☐chromatogram ☐spectra
Menus: ☐start chromatogram ☐auxiliary
Mode: ☐spectra only UV
```

The display shows a maximum of 8 lines by 32 characters. The first line always holds the title or other fixed information (in this case, the current absorbance at 254 nm). The cursor (the flashing block) can be moved up, down, right, or left using the arrow keys.

From a menu line, you can do one of five things:

1. Move to another field or line using the cursor keys.
2. Change the parameter displayed on a given field by typing in the new value for the parameter and then pressing the ENTER key. For obvious reasons, this is only possible on a line in which a parameter is displayed. If you change your mind, or type a value incorrectly, you may use the CLEAR ENTRY (CE) or cursor left keys to erase your entry. When you press the ENTER key, the new value will be displayed on the appropriate line. You may also enter the value by pressing any of the cursor keys to exit the given field.

If you should enter an invalid value for a particular parameter, you will see a flashing error message on the bottom line. Press any key to clear the error message and enter an appropriate value.

3. Change the setting of a "toggle" between functions by pressing the ENTER key.
4. Move to another menu by pressing ENTER. This is only possible on a field in which no parameter is displayed. On all menus except the MAIN menu, the first choice is "RETURN TO THE MAIN menu." If you should "get lost" in the menu structure for any reason, simply move the cursor to the top line of the current menu and then press ENTER.
5. Perform the operation described on a given field. This is only possible on a field in which no parameter is displayed.

3.2 The Detector Key Pad

There are four different types of keys on the detector key pad:

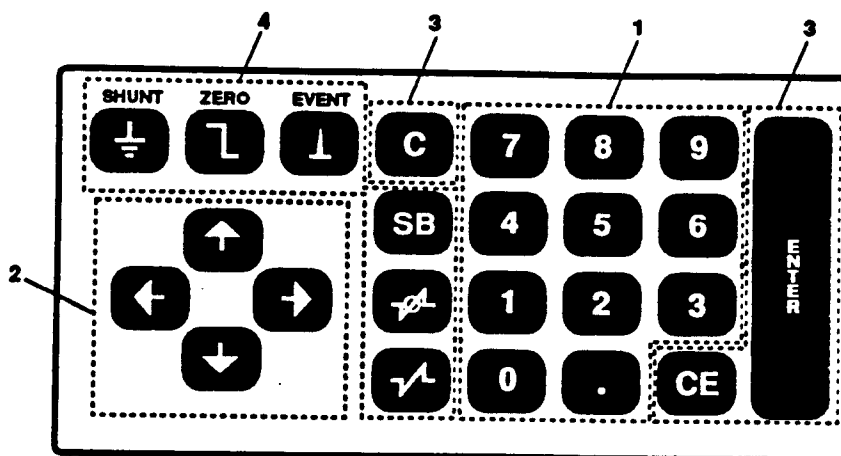


Fig. 3.1.

1. Numeric Keys: These keys are used to enter numerical parameters for a given field.

2. Arrow Keys: These keys are used to move the cursor between lines and fields on a given menu and to erase a numerical entry.

3. Programming Keys: There are three programming keys:

- **Enter Key:** This key is used to enter a given value for a particular field, toggle a function on or off, and advance to another menu.
- **Clear Entry Key:** This key is used to erase an erroneous entered value for a particular field.
- **Clear All Key:** This key is used to erase all numeric parameters listed on a given menu; it functions only on the MULTI, SCHEDULE SPECTRA, and SCHEDULE WAVELENGTH CHANGES menus.

4. Activity Keys: The activity keys are used to perform particular tasks. These keys remain active on all detector menus. There are six ACTIVITY KEYS:

- **Event Key:** The EVENT key is used to signal the beginning of a chromatogram. It sends out a 20% full scale (approximate) spike to the recorder outputs and advances the detector to the UNDERWAY status in the Programmed Operation Mode. If the detector is used with the optional computer software, the EVENT key signals the computer for the start of the run. All event marks are recorded in the the detector's memory for playback purposes. Thus, they may be used as an annotative device.
- **Shunt Key:** The SHUNT key is used to set all recorder outputs to zero volts for the purpose of adjusting recorder pen position. After releasing this key, the zero volt output is maintained for approximately 3.0 seconds.
- **Zero Key:** This key is used to zero the absorbance values at all monitored wavelengths. It does not zero the wavelengths of a programmed spectra.
- **Take Spectrum Key:** This key is used to take a discrete spectra of the flow cell contents.
- **Spectra Zero Key:** This key is used to store background zero values for a programmed, discrete spectra.
- **Stand-By Key:** This key places the monitor in the stand-by status, where all lamps are shut off and the galvanometer drive is locked in a safe position.

3.3 Detector Capabilities

The Bio-Dimension is a high speed, high sensitivity, multiple wavelength, scanning spectrophotometric, UV/Vis monitor. Because it uses both a deuterium and tungsten lamp, it is truly a full spectral scanning instrument.

The monitor gathers data in four different schemes:

- **HIGH SENSITIVITY SINGLE WAVELENGTH:** In high sensitivity single wavelength, the instrument gathers data at 24 points/second for a given single wavelength. This is the wavelength which appears at the top of the detector's MAIN menu. Changes in monitored wavelength are actuated within 10 milliseconds.
- **HIGH SENSITIVITY MULTI-WAVELENGTH:** High sensitivity multi-wavelength allows the instrument to monitor up to 32 wavelengths at a data rate of 24 points/second/wavelength. Therefore, if four wavelengths are monitored, each wavelength would have a data rate of 6 points/second.
- **HIGH SPEED SPECTRA:** High speed spectra permits the scan of the flow cell's contents at a data rate of 96 points/second. Spectra can be gathered from 190 to 365 or 366 to 800 nm with a maximum resolution of 1 nm at a fixed 6 nm band width. Scans are defined using an upper and lower wavelength range along with a step wavelength, which defines the interval between scanned points. A user selectable REPETITION function allows spectra to reflect averaged data of one, two, or four scans. This function must be optimized balancing signal-to-noise (the greater the REPETITION number, the higher the S/N) with peak width (peak speed).

Note: The rapid data rate of high speed spectra exceeds the response speed of many recording devices, hence real-time spectral data is not transmitted out of the recorder channels. Spectra must be stored in the detector's memory and later replayed at a slower speed. Because of this, HIGH SPEED SPECTRA should only be performed while in the PROGRAMMED OPERATION MODE (see next section). If scans are performed while in IMMEDIATE MODE, the command will be carried out but spectral data will not be accessible.

- **HIGH SPEED MULTI-WAVELENGTH:** High speed multi-wavelength is supported only when the detector is used in conjunction with the optional software. High speed multi-wavelength allows continuous full spectral data to be obtained at a data rate of 96 points/second/wavelength. Refer to the *Software Manual* for complete instructions.

3.4 Detector Operation—Immediate Mode and Programmed Operation Mode

The detector has two modes of operations, IMMEDIATE MODE and PROGRAMMED OPERATION MODE. While in IMMEDIATE MODE, a number of operations may be carried out directly from the keyboard in real-time. These operations include:

- Setting detector parameters:
 - Single and multiple wavelength values
 - Full scale range for each monitored wavelength
 - Designation of recorder outputs for up to four of the monitored wavelengths
 - Designation of the integrator output for any one of the monitored wavelengths
 - Wavelength values for ratioing
 - Recorder output for ratio signal
 - Rise time

- Zeroing the detector
- Performing a full spectral autozero
- Setting a zero offset
- Switching lamps on or off
- Setting high speed spectra parameters:
 - Upper and lower wavelength
 - Step size
 - Repetition
- Setting baseline correction parameters
 - Start time
 - End time
 - End absorbance
- Setting memory extender parameters
- Operating the accessory relay
- Playing back stored spectra or chromatograms

Note: Neither chromatograms nor spectra can be stored in the detector's memory during IMMEDIATE MODE OPERATION. Additionally, baseline correction will not function and spectral scans should not be performed while in IMMEDIATE MODE.

If you wish to store a chromatogram in the detector's memory for later replay or perform and store a spectra, you must use the PROGRAMMED OPERATION MODE. This mode also allows for scheduled events, such as Programmed Wavelength Changes and Scheduled Spectra, to be implemented. Additionally, the Baseline Correction and Memory Extender functions of the monitor are activated under this mode of operation.

While in PROGRAMMED OPERATION, you may:

- Schedule detector parameters to change:
 - Wavelength (single wavelength only)
 - Range
 - Recorder output
 - Rise time
 - Schedule end time for a run
- Zero the detector at all wavelengths
- Perform a full spectral autozero
- Perform a spectra and store its data
- Change spectra repetition parameters
- Schedule a predefined spectra to occur and store its data
- Perform a chromatogram and store its data

- Change recorder output for up to four of the monitored wavelengths
- Change range for up to four of the monitored wavelengths
- Change rise time
- Change the integrator wavelength
- Change ratioed wavelengths
- Operate baseline correction
- Operate memory extender
- Operate the stop-flow accessory (if installed)
- Operate auto-spectra

3.5 Setting Detector Parameters (Immediate Mode Operation)

The most commonly used IMMEDIATE MODE detector parameters are set from the MAIN menu.

```
Absorbance @ 254 nm = 0.00000 AU : Menu 1
range = 2.0000 AUFS: rise time 1.0 sec
Spectrum: 200 to 360 @ 10 nm ; reps 4
Method: store recall ;
Program ☐multi ☐wavelength ☐spectra
Playback: ☐chromatogram ☐spectra
Menus: ☐start chromatogram ☐auxiliary
Mode: ☐spectra only UV
```

Wavelength

When you turn on the monitor, the cursor should be located in the wavelength field. Type in the desired value and press enter. Values outside of the 190–800 nm range will not be accepted; the message “INVALID WAVELENGTH ENTERED” will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value.

Range

Move the cursor to the RANGE field and type in the desired value, then press ENTER. Values above 7.9995 AUFS and below 0.0005 AUFS will not be accepted; the message “INVALID RANGE ENTERED” will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value. Values may be entered to four decimal places; the detector will round the fourth decimal place to the nearest 0.0005 AUFS.

Rise Time

Move the cursor to the RISE TIME line and type in the desired value, then press ENTER. The new value will be displayed on the RISE TIME field. Allowable values range from 0 to 9.9 seconds. Values outside of this range will not be accepted; the message "INVALID RISE TIME" will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value.

Spectrum

Move the cursor to the SPECTRUM line. There are four sets of values on this line. The first two define the spectral start and stop wavelengths, respectively. Spectra may be taken using the deuterium lamp (190–365 nm) or the tungsten lamp (366–800 nm). Values outside of the ranges from 190 to 365 nm or 366 to 800 nm will not be accepted; the message "INVALID SCAN PARAMETERS" will appear. To clear the error message, press any key and enter the correct value.

The third number field defines the spectra step wavelength. This is the interval between scanned points. Acceptable values range from 1 to 10 nm. Values outside of this range will not be accepted; the message "INVALID SCAN PARAMETERS" will appear. To clear the error message, press any key and enter the correct value.

The scanning algorithm of the monitor mandates the selection of upper and lower scan wavelength values which are inside the ranges of 190 to 365 and 366 to 800 nm by a factor of at least the step wavelength divided by two. For example, using a step size of 6 nm, the largest scans would be 193 to 362 and 369 to 797 nm. If these parameters are incorrect, the detector will display the message "INVALID SCAN PARAMETERS" on the lower line. Depending upon the entered lower, upper, and step wavelength values, the monitor will correct one of these three when it displays the error message. To clear the error message, press any key and enter the correct value.

The fourth number field lists the repetition number for the defined scan. Valid entries are 1, 2, and 4. Values outside this range trigger an error message which informs the user of the allowed choices. This parameter should be optimized to balance signal-to-noise (the greater the repetition number, the higher the s/n) with peak width/speed.

The Program Line

This line lists fields for entering the MULTI (WAVELENGTH), WAVELENGTH (CHANGES) and (SCHEDULE) SPECTRA menus. To access a menu, for example, the MULTI-WAVELENGTH menu, move the cursor to that field and push enter. To return to the MAIN menu, move the cursor to the RETURN TO PREVIOUS MENU field and push ENTER.

MULTI-WAVELENGTH						MENU 4	
☐ Return to previous menu				☐ sort			
Ratio: 254 over 350						: CH 3	
NO	nm	CH	range	RT	INT		
1	254	1	0.0020	1.0	1	☐ INS	
2	350	2	0.0020	1.0		☐ INS	

The Playback Line

This line allows access to either the CHROMATOGRAM or SPECTRA PLAYBACK menus. To access a menu, move the cursor to one of the fields and push ENTER. To return to the MAIN menu, move the cursor to the RETURN field and push ENTER.

```
CHROMATOGRAM PLAYBACK          MENU 8
ABSORBANCE @ 254 nm =          AU
MODE: spectra only
☐Return      ☐Playback chromatogram
... @          1 times real time ☐stop PB
PB range= 2.0000 AUFS; rise time= 1.0 sec
Baseline  0.83 to 3.33 min @ 0.00000 AU
Strip chart channel no.= 1
```

The Menus Line

The menus line contains fields for entering the AUXILIARY menu and arming the detector for PROGRAMMED OPERATION MODE. To enter these menus, move the cursor to the proper field and push ENTER. To return to the MAIN menu move the cursor to the RETURN TO MAIN menu field and push ENTER.

```
Return to main menu :          MENU 2
Lamps: tungsten ☐OFF    deuterium ☐ON
☐Auto calibrate
Baseline  0.80 to 3.33 min @ 0.00000 AU
Auto Spectra ☐OFF 1 per peak; sides EARLY
Auto scan relay                ☐disabled
Zero upon wavelength change    ☐ON
SCROLL to access additional parameters
```

The Mode Line

This line contains the field used to toggle the detector between SPECTRA ONLY (MONOCHROME), SINGLE WAVELENGTH (PROGRAMMED WAVELENGTH CHANGES), and MULTIPLE WAVELENGTH (POLYCHROME) modes of operation. Operation modes are toggled by moving the cursor to the MODE field and pressing ENTER until the desired mode is displayed.

3.6 Performing a Single Wavelength Chromatogram (Monochrome) (Immediate Mode Operation)

Immediate Mode Monochrome chromatograms are performed using the MAIN menu. Turn on the HPLC system and allow the column to equilibrate with the flowing eluant (the time required will depend on your particular column and eluant). If you have not already done so, turn on the power to the detector using the switch located on the lower rear panel. Adjust the contrast of the display using the DISPLAY CONTRAST control as necessary. The MAIN menu should appear on your screen.

The lamp indicator located in the lower right corner of the screen indicates the warm up status of the deuterium lamp by displaying "UVW". When the UV lamp lights, the display changes to "UV".

Press the SHUNT key and hold it down while you adjust the recorder pen to the desired position.

Release the SHUNT key and press the ZERO key. The absorbance display should be very close to zero, and the recorder pen should be very close to the zero position which you just set.

You may now set the appropriate detector parameters.

Setting Wavelength

Insure that the cursor is on the WAVELENGTH field of the MAIN menu. Type in the desired wavelength, then press the ENTER key. You may also enter the wavelength parameter by using the down arrow key to advance to the RANGE field.

Setting Range

If you have not already done so, use the down arrow key to move to the RANGE field. Enter the RANGE appropriate for your sample concentration and flow cell path length. Use the cursor down key to advance to the RISE TIME field.

Setting Rise Time

The detector uses a digital second order Bessel filter. The response of this type of filter is described in terms of a 10% to 90% response to a step input. This response time is called RISE TIME (RT). For a given RT, the response of a Bessel filter is roughly equivalent to a RC time constant equal to 1/2 the RT. A RT of 1 second will produce approximately the same result as a 0.5 second time constant although the second order Bessel filter produces much less tailing. The default value of RT is set for 1.0 second. This value is adequate for most routine HPLC separations. If you are running a very fast analysis, a shorter RT may be necessary. For very high sensitivity detection, a longer RT may be more appropriate. In general, if the RT is set to be equal to or less than 1/3 the half height peak width, the filter will cause no peak broadening.

Type in the desired RISE TIME in increments of 0.1 second and press the ENTER key.

Recorder and Integrator Channels

While in the single wavelength mode, the monitor transmits ranged and filtered data out of RECORDER CHANNEL 1. Nonranged (unattenuated), filtered data is available from the INTEGRATOR OUTPUT. If you are using a chart recorder, insure that it is connected to RECORDER CHANNEL 1. If an integrator or data acquisition system is to be used, it should be connected to the INTEGRATOR OUTPUT.

Running the Chromatogram

Switch your recording device on, inject your sample, and press the EVENT key. You may change wavelength at any time, to do so:

- move the cursor to the WAVELENGTH line
- type in the new value
- press the ENTER key
- press the ZERO key

You may change RANGE or RISE TIME at any time by moving the cursor to the appropriate field, typing in the new value, and pressing the ENTER key.

3.7 Recording a High Sensitivity Multi-Wavelength Chromatogram (Immediate Mode Operation)

Turn on the HPLC system and allow the column to equilibrate with the flowing eluant (the time required will depend on your particular column and eluant). If you have not already done so, turn on the power to the detector using the switch located on the lower rear panel. Adjust the contrast of the display using the DISPLAY CONTRAST adjustment as necessary. The MAIN menu should appear.

Move the cursor to the MULTI field of the PROGRAM line and press enter. The MULTI-WAVELENGTH menu should be displayed.

You may now set the appropriate detector parameters.

MULTI-WAVELENGTH							MENU 4
☐ Return to previous menu				☐ sort			
Ratio: 254 over 350 : CH 3							
NO	nm	CH	range	RT	INT		
1	214	1	0.0020	1.0	1	☐ INS	
2	350	2	0.0020	1.0		☐ INS	

Multi-Wavelength Parameters

Move the cursor to the NM field of the MULTI-WAVELENGTH menu NO 1 line. Enter the first of the desired wavelengths. If you are unsure of the appropriate wavelengths to use for your sample, use 214 nm. Enter 214 as your first wavelength by typing in the wavelength value and pressing the ENTER key.

Move the cursor across to the CH field and type in the desired RECORDER CHANNEL NUMBER. For organizational sake, you should use CHANNEL 1. Enter the value by advancing the cursor across to the RANGE field.

Type in an appropriate RANGE for your sample mix at this wavelength. Enter the value by advancing the cursor across to the RISE TIME (RT) field.

Type in an appropriate RISE TIME value for your test mix and chromatographic conditions. For most instances, a 1.0 second RISE TIME is sufficient for peaks as short as 10 seconds at their base. A shorter RISE TIME should be used for faster peaks.

Enter the remaining desired WAVELENGTHS, RECORDER CHANNELS, RANGES, and RISE TIMES in the same manner as used above. If you are unsure of the wavelengths to use for your sample, use 254 nm, 280 nm, and 329 nm as the remaining three wavelengths.

Move the cursor to the RATIO line. Select two of the monitored wavelengths to be ratioed. Enter these wavelengths in the NUMERATOR and DENOMINATOR fields. If you are unsure of which two wavelengths to ratio, ratio 214 nm over 280 nm.

MULTI-WAVELENGTH						MENU 4	
☐ Return to previous menu						☐ sort	
Ratio: 214 over 280 : CH 3							
NO	nm	CH	range	RT	INT		
1	214	1	0.0020	1.0	I	☐ INS	
2	254	2	0.0020	1.0		☐ INS	
3	280	3	0.0020	1.0		☐ INS	
4	329	4	0.0020	1.0		☐ INS	

Enter the RECORDER CHANNEL (CH) OUTPUT for the RATIO PLOT. It is recommended that three of the outputs be used for WAVELENGTH ABSORBANCE DATA and the fourth used for the RATIO PLOT.

The RATIO PLOT is the log of the NUMERICAL RATIO.

$$\text{NUMERICAL RATIO} = \frac{\text{ABSORBANCE WAVELENGTH 1} / \text{RANGE WAVELENGTH 1}}{\text{ABSORBANCE WAVELENGTH 2} / \text{RANGE WAVELENGTH 2}}$$

If an integrator or data acquisition system is to be used, you must identify one of the monitored wavelengths for integration. Move the cursor to the INTEGRATOR (INT) field. Using the up or down arrows, move the cursor into the line which contains the desired wavelength for integration and press ENTER. An "I" will appear in that line.

Move the cursor to the RETURN TO PREVIOUS menu field and press ENTER. The MAIN menu should now be displayed.

Move the cursor down to the SPECTRA ONLY field of the MODE line and press ENTER twice to toggle the mode to MULTIPLE WAVELENGTH.

Move the cursor to the MULTI field of the PROGRAM line and push enter. The MULTI-WAVELENGTH menu should appear.

Running the Chromatogram

Switch your recording devices on and press the ZERO key. The absorbance values should be very close to zero. Press the SHUNT key and adjust the pen position on your chart recorder(s). Position the RATIO PLOT pen at 50% full scale. If an integrating device is being used, set the device for the baseline noise of the integrated wavelength.

Inject your sample and press the EVENT key. You may change any of the PARAMETERS during the run by advancing the cursor to the appropriate field, typing in the desired change, and pressing the ENTER key or moving the cursor to another field.

As your peaks elute, notice the RATIO PLOT trace. If the peak is pure (no coelution occurs) the RATIO PLOT will trace a "square wave" pattern. Deviations from this pattern are indicative of coelution and an impure peak. Ratioing is further discussed in the "Theory of Operation" section.

3.8 Recording and Storing a Programmed Single Wavelength Chromatogram with Spectra (Programmed Operation Mode)

Note: The Programmed Operation Mode of the detector provides a convenient means of changing parameters (wavelength, range, rise time, or recorder channel) or of scanning spectra at predefined times after the start of a chromatographic run. In addition, it provides the only mechanism by which chromatographic data may be stored in the detector's memory for later replay. Also, it is the only mode in which spectral data may be accessed after performing a scan. Manual changes of wavelength during Programmed Operation are not allowed.

Turn on the HPLC system and allow the column to equilibrate with the flowing eluant (the time required will depend on your particular column and eluant). If you have not already done so, turn on the power to the detector using the switch located on the lower rear panel. Adjust the contrast of the display using the DISPLAY CONTRAST adjustment as necessary. The MAIN menu should appear on the LCD.

With the cursor at the ABSORBANCE @ field, type in the desired wavelength for your test mix and then advance the cursor to the RANGE field.

Type in a RANGE appropriate for your test mix and flow cell path length. Move the cursor to the RISE TIME field.

Enter a RISE TIME value appropriate for your sample. The default value of 1.0 second is sufficient for peaks as quick as 10.0 seconds at their base. If you expect peaks faster than this, enter a shorter RISE TIME. Move the cursor to the SPECTRUM line.

Enter the upper and lower limits for the spectral scan appropriate for your test mix and eluant. Acceptable values range from 190 to 365 and 366 to 800 nm. Move the cursor to the STEP WAVELENGTH field.

Enter the desired STEP WAVELENGTH value. It is important to insure that the STEP WAVELENGTH is chosen so that the selected upper and lower wavelengths clear their maximum or minimal values by at least a factor of the STEP WAVELENGTH divided by two. In other words, if your programmed upper and lower wavelengths are 364 and 191 nm respectively, you must select a step wavelength of 1 nm. To select a step wavelength of 5 nm, the upper and lower wavelengths must be 362 and 193 nm, respectively. Failure to select an appropriate STEP WAVELENGTH will result in an "INVALID SCAN PARAMETERS" error message and the detector will change either the upper, lower, or STEP WAVELENGTH values to correct the situation.

Move the cursor to the REPS field, and enter the desired REPETITION value. REPETITION controls the number of spectral scans that will be averaged together for a given spectra command. Allowable values are 1, 2, or 4. You should choose a REP value which balances signal-to-noise with scan time (the greater the REP value, the higher the s/n; the greater the REP value, the longer the scan time).

For a single wavelength chromatogram, the detector defaults the ranged, filtered chromatographic data to RECORDER CHANNEL 1 and the nonranged, filtered data to the INTEGRATOR OUTPUT. Insure that your chart recorder is connected to RECORDER CHANNEL 1 and that your integrator/data acquisition device is connected to the INTEGRATOR OUTPUT.

Move the cursor to the START CHROMATOGRAM field of the MENUS line and push the ENTER key. The PUSH EVENT TO START CHROMATOGRAM menu will be displayed.

```
PUSH EVENT TO START CHROMATOGRAM  MENU 6
ABSORBANCE @ 254 nm = 0.00000 AU
MODE: spectra only
☐ return to main menu
non-scheduled spectra reps: 4
range 2.0000 AUFS
rise time = 1.0 sec                               UV
```

Push the ZERO key to zero the absorbance at the monitored wavelength. The absorbance value should be very close to zero.

Push the FULL SPECTRAL ZERO key to zero the background absorbance of your effluent at all of the scanned wavelengths.

Press the SHUNT key and adjust the position of the pen(s) on your recording device. If an integrator/data acquisition system is being used, set this device for the background noise at the monitored wavelength.

Running the Chromatogram

Inject your sample and press the EVENT key. The CHROMATOGRAM UNDERWAY menu will appear.

```
CHROMATOGRAM UNDERWAY  MENU 6
ABSORBANCE @ 254 nm = 0.00000 AU
MODE: spectra only      0.0
☐ STOP chromatogram
spectra reps: 4
range 0.002 AUFS
rise time = 1.0 sec
```

You may change the parameters of RISE TIME, RANGE, and SPECTRA REPS at any time. You may NOT change the monitored wavelength. The elapsed time is indicated by the clock in the upper, right corner of the menu.

When one of the peaks elute, press the TAKE SPECTRA key when the pen is close to the apex of the peak. This will store the spectra for that peak in the detector's memory.

After the last peak elutes, move the cursor to the STOP CHROMATOGRAM field and press ENTER. This halts the data storage process and rearms the detector to start recording a chromatogram. If you wish to replay the chromatogram, advance to the "Chromatogram Playback" section of this manual. If you wish to view the recorder spectra, advance to the "Spectra Playback" section of this manual.

If you wish to record another chromatogram or additional spectra of the peaks in your test mix, it is not necessary to reprogram the monitor. All entered detector parameters are retained, and you need only to inject your sample and press the EVENT key to advance the detector to the UNDERWAY mode. It should be noted that only one chromatogram, with its associated spectra, is stored in the monitor at any time. Advancing the detector to the UNDERWAY status erases all stored chromatographic and spectral data from the previous run.

3.9 Recording and Storing a Programmed High Sensitivity Multi-Wavelength Chromatogram (Programmed Operation Mode)

Turn on the HPLC system and allow the column to equilibrate with the flowing eluant (the time required will depend on your particular column and eluant). If you have not already done so, turn on the power to the detector using the switch located on the lower rear panel. Adjust the contrast of the display using the DISPLAY CONTRAST control as necessary. The MAIN menu should appear on the display.

```
Absorbance @ 254 nm = 0.00000 AU : Menu 1
range = 2.0000 AUFS: rise time 1.0 sec
Spectrum: 200 to 360 @ 10 nm ; reps 4
Method: store recall ;
Program: ☐multi ☐wavelength ☐spectra
Playback: ☐chromatogram ☐spectra
Menus: ☐start chromatogram ☐auxiliary
Mode: ☐spectra only UV
```

Move the cursor to the MULTI field of the PROGRAM line and press ENTER. The MULTI-WAVELENGTH menu should be displayed.

```
MULTI-WAVELENGTH MENU 4
☐Return to previous menu ☐sort
Ratio: 254 over 350 : CH 3
NO nm CH range RT INT
1 214 1 0.0020 1.0 1 ☐INS
2 350 2 0.0020 1.0 ☐INS
```

Move the cursor to the NM field of the MULTI-WAVELENGTH menu NO 1 line. Enter the first of the desired wavelengths. If you are unsure of the appropriate wavelengths to use for your sample, refer to the spectra taken in the previous section, or simply use 214 nm. Enter 214 as your first wavelength. Type in the WAVELENGTH value and then press the ENTER key.

MULTI-WAVELENGTH						MENU 4
☐ Return to previous menu			☐ sort			
Ratio: 214 over 280 : CH 3						
NO	nm	CH	range	RT	INT	
1	214	1	0.0020	1.0	1	☐ INS
2	254	2	0.0020	1.0		☐ INS
3	280	3	0.0020	1.0		☐ INS
4	329	4	0.0020	1.0		☐ INS

Move the cursor across to the CH field and type in the desired RECORDER CHANNEL NUMBER. For organizational purposes, it is recommended to use CHANNEL 1. Enter the value by advancing the cursor across to the RANGE field.

Type in an appropriate RANGE for your sample mix and flow cell path length at this wavelength. Enter the value by advancing the cursor across to the RISE TIME (RT) field.

Type in an appropriate RISE TIME value for your test mix and chromatographic conditions. For most instances, a 1.0 second RISE TIME is sufficient for peaks as short as 10 seconds at their base. A shorter RISE TIME should be used for faster peaks.

Enter the remaining desired WAVELENGTHS, RECORDER CHANNELS, RANGES, and RISE TIMES in the same manner as above. If you are unsure of the wavelengths to use for your sample, refer to the spectra gathered in the previous section or use 254, 280, and 329 nm as the remaining wavelengths.

Move the cursor to the RATIO line. Select two of the monitored wavelengths to be ratioed. Enter these wavelengths in the NUMERATOR and DENOMINATOR fields. If you are unsure of which two wavelengths to ratio, ratio 254 nm over 350 nm.

Enter the RECORDER CHANNEL (CH) OUTPUT for the RATIO PLOT. It is recommended that three of the outputs be used for WAVELENGTH ABSORBANCE DATA and the fourth be used for the RATIO PLOT OUTPUT.

The RATIO PLOT is the log of the NUMERICAL RATIO.

$$\text{NUMERICAL RATIO} = \frac{\text{ABSORBANCE WAVELENGTH 1} / \text{RANGE WAVELENGTH 1}}{\text{ABSORBANCE WAVELENGTH 2} / \text{RANGE WAVELENGTH 2}}$$

If an integrator or data acquisition system is to be used, you must identify one of the monitored wavelengths for integration. Move the cursor to the INTEGRATOR (INT) field. Using the up or down arrows, move the cursor into the line which contains the desired wavelength for integration and press ENTER. An "I" will appear in that line.

MULTI-WAVELENGTH						MENU 4	
☐ Return to previous menu						☐ sort	
Ratio: 214 over 280 : CH 3							
NO	nm	CH	range	RT	INT		
1	214	1	0.0020	1.0	I	☐ INS	
2	254	2	0.0020	1.0		☐ INS	
3	280	3	0.0020	1.0		☐ INS	
4	329	4	0.0020	1.0		☐ INS	

Move the cursor to the RETURN TO PREVIOUS menu field and press ENTER. The MAIN menu should now be displayed.

Move the cursor down to the SPECTRA ONLY field of the MODE line and press ENTER twice to toggle the mode to MULTIPLE WAVELENGTH. Move the cursor to the START CHROMATOGRAM field of the menu line and press ENTER to arm the detector. The PUSH EVENT TO START CHROMATOGRAM menu should be displayed. Allow the monitor a five minute warm up period to reach optimal running conditions.

PUSH EVENT TO START CHROMATOGRAM						MENU 7	
MODE: multi-wavelength							
☐ return to main menu						; spectra reps: 4	
Ratio: 214 over 280 : CH 4							
NO	nm	CH	range	RT	INT		
1	214	1	0.0020	1.0	I	0.00000	
2	254	2	0.0020	1.0		0.00000	
3	280	3	0.0020	1.0		0.00000	
4	329		0.0020	1.0		0.00000	

Push the ZERO key to zero the absorbencies at all monitored wavelengths. All absorbance values should be very close to zero.

Press the SHUNT key and adjust the position of the pen(s) on your recording device. If an integrator/data acquisition system is being used, set this device for the background noise at the monitored wavelength.

Running a Chromatogram

Inject your sample and press the EVENT key. The CHROMATOGRAM UNDERWAY menu will appear.

PUSH EVENT TO START CHROMATOGRAM						MENU 7
MODE: multi-wavelength						
☐ return to main menu ; spectra reps: 4						
Ratio: 214 over 280 : CH 4						
NO	nm	CH	range	RT	INT	
1	214	1	0.0020	1.0	1	0.00000
2	254	2	0.0020	1.0		0.00000
3	280	3	0.0020	1.0		0.00000
4	329		0.0020	1.0		0.00000

You may change the parameters of RISE TIME and RANGE at any time. You may NOT change any of the monitored wavelengths. The elapsed time is indicated by the clock in the upper, right corner of the menu.

After the last peak elutes, move the cursor to the STOP CHROMATOGRAM field and press ENTER. This halts the data storage process and rearms the detector to start recording a chromatogram. If you wish to replay the chromatogram, advance to the "Chromatogram Playback" section of this manual. If you wish to record another chromatogram, it is not necessary to reprogram the detector. All entered parameters are retained, and you need only to inject your sample and press the EVENT key to advance to the UNDERWAY status. It should be noted that only one chromatogram can be stored in the monitor's memory at any time. Advancing the detector to the UNDERWAY status erases all stored chromatographic data from the previous run.

In this example, we did not define or perform any spectra. However, it is possible to take spectra while in the multiple wavelength mode. For details on programming spectral parameters, see the "Recording and Storing a Programmed Single Wavelength Chromatogram With Spectra" section of this manual.

3.10 Playing Back a Chromatogram

A chromatogram stored in memory may be played back at any time except during the recording process. It should be noted that a chromatogram is retained in memory only until another one is recorded or the power is turned off. Just one chromatogram and its associated spectra can be stored within the detector's memory at any time. The details of playback procedure depend to some extent on the conditions under which the chromatogram was originally recorded: Single Wavelength, Multiple Wavelength, and with or without Extended Memory. Stored data may be played back any number of times. Each time the RANGE, RISE TIME or any other parameter may be changed.

Replaying a Single Wavelength Chromatogram

From the MAIN menu, move the cursor to the CHROMATOGRAM field of the PLAYBACK line and press ENTER. The CHROMATOGRAM PLAYBACK menu should now be displayed.

```
CHROMATOGRAM PLAYBACK          MENU 8
ABSORBANCE @ 254 nm = 0.00000 AU  0.0
MODE: spectra only
☐Return    ☐Playback chromatogram
... @      1 times real time ☐stop PB
PB range= 2.0000 AUFS; rise time= 1.0 sec
Baseline  0.83 to 3.33 min @ 0.00000 AU
Strip chart channel no.= 1
```

Move the cursor down to the 1 TIMES REAL TIME field. This field controls the rate at which the monitor will replay the recorded chromatogram. Chromatograms may only be played back as fast or faster than real-time. Acceptable values range from 1 to 20. Enter the desired playback rate.

Move the cursor down to the PB RANGE field and enter a full scale range appropriate for your sample concentration and flow cell path length.

Move the cursor across to the RISE TIME field and enter a rise time appropriate for the recorded chromatographic run.

Move the cursor down to the STRIP CHART CHANNEL NO. field and enter the appropriate value. The monitor will replay unfiltered, nonranged data through the integrator output for use with integrators or data acquisition systems.

Move the cursor up to the PLAYBACK CHROMATOGRAM field. Ready your recording device(s), and press ENTER to replay the chromatogram. The wavelength being replayed, absorbance, and time will be displayed upon the screen.

At the completion of the playback, the parameters of wavelength, absorbance, and time will disappear from the screen.

To stop the playback at any time, move the cursor to the STOP PB field and press ENTER.

Replaying a Multiple Wavelength Chromatogram

From the MAIN menu, move the cursor to the CHROMATOGRAM field of the PLAYBACK line and press ENTER. The CHROMATOGRAM PLAYBACK menu should now be displayed.

MULTI-WAVELENGTH PLAYBACK						MENU 9
☐ Return		☐ Playback chromatogram				
... @		1 times real time		☐ stop PB		
Ratio: 214 over 280 : CH 4						
Baseline 0.83 to 3.33 min @ 0.00000 AU						
NO	nm	CH	range	RT	INT	
1	214	1	0.0020	1.0	I 0.00000	
2	254	2	0.0020	1.0	0.00000	
3	280	3	0.0020	1.0	0.00000	
4	329	4	0.0020	1.0	0.00000	

Move the cursor down to the 1 TIMES REAL TIME field. This field controls the rate at which the monitor will replay the recorded chromatogram. Chromatograms may only be played back as fast or faster than real-time. Acceptable values range from 1 to 20. Enter the desired playback rate.

Move the cursor down to the first wavelength line and enter the desired values for RANGE, RECORDER CHANNEL, and RISE TIME.

Move the cursor down to all subsequent lines and repeat the above procedure. If more than three wavelengths have been recorded, it is necessary to use the down arrow key to scroll the screen and expose all of the other recorded wavelengths.

If an integrator or data acquisition system is connected to the integrator output, you must specify a wavelength for integration. To do so:

Place the cursor on the wavelength line to be integrated, move it across to the INT field, and push ENTER. An "I" will appear in the INT field for the desired wavelength to be integrated.

If you desire to record an absorbance RATIO PLOT of any of the two recorded wavelengths, the following procedure is used:

Move the cursor to the RATIO line and enter the numerator and denominator wavelengths. Move the cursor across to the CH field and enter the recorder channel output for the ratio trace. If you enter a recorder channel output already being used by one of the recorded wavelengths, the monitor will prioritize this channel for the RATIO PLOT.

Ready your recording device(s) and move the cursor to the PLAYBACK CHROMATOGRAM field. Press ENTER to begin the playback process. The PLAYBACK CHROMATOGRAM UNDERWAY menu will appear.

The elapsed time and absorbencies for each recorded wavelength will be displayed. At the completion of the chromatogram replay, time and absorbance values will no longer be displayed.

To stop the playback at any time, move the cursor to the STOP PB field and press ENTER.

Replaying Memory Extended Chromatograms

The procedure used to replay a MEMORY EXTENDED chromatogram, be it single or multiple wavelength, is very similar to replaying an unextended version. Because MEMORY EXTENDER

functions by altering the data storage rate, it affects the playback parameters of playback rate and rise time. The actual playback rate will be the product of the MEMORY EXTENDER MULTIPLIER and the entered playback rate, while the actual rise time will be the product of the multiplier and the entered rise time. For more details regarding MEMORY EXTENDER and CHROMATOGRAM PLAYBACK, consult the "Theory of Operation" section of this manual.

The MEMORY EXTENDER MULTIPLIER is indicated in the PLAYBACK RATE field as an integer and an "X". For example, a MEMORY EXTENDER MULTIPLIER OF 2 would be indicated on the PLAYBACK RATE line as 2 X 1 TIMES REAL-TIME if a playback rate value of 1 was entered.

```

CHROMATOGRAM PLAYBACK                                MENU 8
ABSORBANCE @ 254 nm = 0.00000 AU    0.0
MODE: spectra only
☐ Return      ☐ Playback chromatogram
... @          1 times real time  ☐ stop PB
PB range= 2.0000 AUFS; rise time= 1.0 sec
Baseline  0.83 to 3.33 min @ 0.00000 AU
Strip chart channel no.= 1
  
```

To replay the chromatogram, enter the desired playback rate remembering that the final rate will be the product of the MEMORY EXTENDER MULTIPLIER and the entered PLAYBACK RATE.

Enter the desired rise times remembering that the resultant rise time is the product of the multiplier and entered rise time. For example, if a MEMORY EXTENDER MULTIPLIER of 2 was used, and a 1.0 second rise time is desired, enter the value 0.5 as the rise time parameter.

The remainder of the playback procedure is identical to that of an unextended chromatogram.

3.11 Playing Back a Spectrum

Because the scanning rate of the detector exceeds the frequency response of many recording devices, real-time spectral data is not available. All spectra must be stored in the detector using the PROGRAMMED OPERATION MODE and then replayed, post run, at a slower speed.

To replay a stored spectrum:

From the MAIN menu move the cursor to the SPECTRA field of the PLAYBACK line and press ENTER to bring up the SPECTRA PLAYBACK menu.

```

SPECTRA PLAYBACK                                MENU 10
Absorbance @ 210 nm = 0.00000 AU
Spectrum: 200 to 360 @ 10 nm steps
Spectrum no  1 taken at      1 sec w/4reps
☐ Return      ☐ Playback spectra
☐ Next spectra    playback rate 20 nm/sec
Range  2.0000 AUFS          ch no
  
```

If you have performed the spectra outlined in this tutorial and you have not shut the detector off since the acquisition of those spectra, the spectra playback menu will be displayed after the PLEASE WAIT message disappears. If you have not performed spectra or shut the monitor off since recording the spectra, no spectra will be stored in the instrument's memory, and the error message INVALID SCAN PARAMETERS will be listed.

Note the spectral parameters of lower, upper, and STEP WAVELENGTHS that were programmed for the recorded spectra. These properties are listed on the third line. The fourth line should read "SPECTRUM NO 1 TAKEN AT 1 SEC W/4 REPS." This indicates that the first spectra, number 1, taken at 1 second after the start of the programmed mode run, is ready to be played back. Also note that the REPETITION value of four is displayed, meaning that a total of four repetitive scans were performed and averaged together for the stored spectral data.

Move the cursor down to the PLAYBACK RATE field and enter an appropriate playback rate in nm/second. Choose a rate which will not exceed the response speed of your recording device while simultaneously allowing you to calibrate the wavelength axis. The latter is achieved by coordinating the playback rate with the strip chart paper speed. For example, if a playback rate of 5 nm/second is used with a chart speed of 12 cm/minute, the resultant wavelength (x) axis is scaled to 5 nm for every 2 mm of paper.

Move the cursor to the RANGE field and enter an appropriate RANGE for your test mix and flow cell path length. Allowable values range from 0.0005 to 7.9995 AUFS in multiples of 0.0005 AUFS.

Move the cursor to the CHANNEL NUMBER (CH NO) field and specify the recorder output to which your recording device is interfaced. Allowable outputs are channels 1 through 4. Spectra MAY NOT be replayed through the INTEGRATOR output.

Set the proper pen position for the recording device and start the chart drive. Move the cursor to the PLAYBACK SPECTRA field and press ENTER to begin the playback process. Wavelengths and their absorbance values should be displayed on the second line. Upon completion of the playback, these values will disappear from the screen.

To replay the next spectra, move the cursor to the NEXT SPECTRA field and press ENTER. The fourth line should note "SPECTRA NO. 2... ETC." Proceed with the rest of the playback procedure as previously outlined.

3.12 Programmed Wavelength Changes (Single Wavelength Operation)

This mode of programmed operation allows changes in WAVELENGTH, RANGE, RISE TIME, and RECORDER CHANNEL to be made at predetermined times during the chromatogram. These changes must be scheduled before the beginning of the run. Events may be scheduled in minutes or seconds. The default status is in minutes. For this tutorial, change the status to seconds by advancing to the AUXILIARY menu, scrolling to the bottom line and toggling the DISPLAY TIME BASE field to seconds.

From the MAIN menu, move the cursor to the TIMED MONO field of the PROGRAM line and push ENTER. The SCHEDULED WAVELENGTH CHANGES menu should be displayed.

SCHEDULED WAVELENGTH CHANGES						MENU 3
☐ Return to main menu						
NO	NM	FROM	TO	CH	RANGE	FILTER
1	254	0	1200	1	2.0000	1.0 ☐ INS

Determine the elution time and maximum wavelength of absorbance for each peak in your test mix. Move the cursor to the NM field of the first SCHEDULED WAVELENGTH line and type in the WAVELENGTH of maximum absorbance for the first peak in your test mix. If you are not performing chromatography or are unsure of this value, enter 214 nm. Move the cursor to the TO field of the first line.

SCHEDULED WAVELENGTH CHANGES							MENU 3	
☐ Return to main menu								
NO	NM	FROM	TO	CH	RANGE	FILTER		
1	214	0	210	1	0.0002	1.0	☐ INS	
2	254	211	290	1	0.0100	1.0	☐ INS	
3	280	291	480	1	0.0050	1.0	☐ INS	

Type in a TIME, in seconds, which corresponds to a period in which no peaks are detected directly after the elution time for the first peak. If you are unsure of this value or not performing chromatography, enter 30 seconds.

Move the cursor to the CH field of the first line and enter the desired RECORDER CHANNEL OUTPUT. If you are not performing chromatography, enter 1. Move the cursor to the RANGE field of the first line.

Enter a RANGE appropriate for the first peak of your test mix. If you are unsure of this value or are not performing chromatography, enter 0.002 AUFS. Move the cursor to the FILTER field of the first line.

Enter a RISE TIME appropriate for the first peak and scheduled RANGE. If you are not performing chromatography, enter 1.0 second. Move the cursor down to the next line. You will notice that the second line is now identical to the first.

Enter the parameters of WAVELENGTH, END TIME, CHANNEL, RANGE, and FILTER for the remaining peaks in your test mix as above. If you are not performing chromatography, enter the wavelengths of 254 and 280 nm with their accompanying parameters as depicted in the SCHEDULE WAVELENGTH CHANGES.

Move the cursor to the RETURN TO MAIN menu field and press ENTER. The main menu is now displayed.

Note that the MODE field indicates SINGLE WAVELENGTH. Move the cursor to the START CHROMATOGRAM field of the MENUS line and press PUSH EVENT TO START CHROMATOGRAM menu will be displayed.

```

PUSH EVENT TO START CHROMATOGRAM
ABSORBANCE @ 214 nm = 0.00000 AU
MODE: single wavelength
☐ STOP chromatogram      00:00:00
spectra reps: 4
range 0.0020 AUFS
rise time 1.0 sec
  
```

Press the ZERO key to zero the absorbance at the first monitored wavelength. The absorbance value should be very close to zero.

Press the SHUNT key and adjust the position of the pen(s) on your recording device. If an integrator/data acquisition system is being used, set this device for the background noise at the monitored wavelength.

Inject your sample and press the EVENT key. The CHROMATOGRAM UNDERWAY menu will appear.

CHROMATOGRAM UNDERWAY	MENU 6
ABSORBANCE @ 254 nm = 0.00000 AU	
MODE: spectra only	0.0
☐ STOP chromatogram	
spectra reps: 4	
range 0.002 AUFS	
rise time = 1.0 sec	

Watch the menu and note that changes of wavelength, rise time, range, and channel occur. You may manually change the parameters of rise time and range at any time. You may NOT alter the wavelength value.

At the end time of the last scheduled change, the detector will stop gathering data and reset to the ARMED TO START STATUS. To replay the recorded data, simply move the cursor to the RETURN TO MAIN menu field and press ENTER. Then move the cursor to the CHROMATOGRAM field of the PLAYBACK line and press ENTER. The remaining instructions are found in the "Playback Chromatogram" section of this manual.

Editing the Wavelength Changes Menu

Editing the SCHEDULED WAVELENGTH CHANGES menu is easy. If you decide to delete a scheduled wavelength, move the cursor to any field on that wavelength's line and press the CLEAR key twice.

If you decide to insert a scheduled change between two other scheduled changes, move the cursor to the INSERT field of the scheduled change line which is just before the desired insertion time, and press the ENTER key. This automatically copies the previous line and moves the cursor to the INSERT field of the newly made line. Now, type in the desired wavelength, end time, channel, range, and filter parameters for the new line. You have just inserted a new change.

Zero Upon Wavelength Change

As a default parameter, the monitor performs an AUTOZERO after every change of wavelength, whether it is manually or automatically actuated. This function is controlled on the AUXILIARY menu. While performing SCHEDULED WAVELENGTH CHANGES, it is important to determine if any of the scheduled changes occur during times when a given peak still contributes to the absorbance. For example, if all of your peaks are well resolved, as in the chromatogram depicted in figure 3.2, wavelength changes can be scheduled during times between the peaks when the baseline is flat.

In this case the ZERO UPON WAVELENGTH CHANGE is desirable. If your chromatogram contains poorly resolved peaks, as in figure 3.3, and you schedule a wavelength change to occur between two poorly resolved peaks, a ZERO UPON WAVELENGTH CHANGE would subtract a value equivalent to the sum of the effluent and peak absorbencies. This would result in a negative displacement of the baseline.

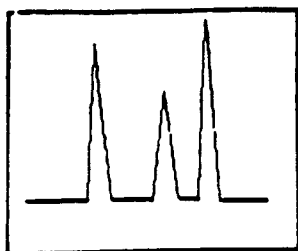


Fig. 3.2.

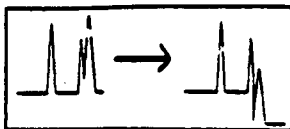


Fig. 3.3.

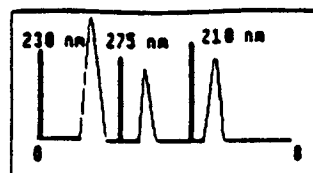


Fig. 3.4.

To avoid this situation, you should deactivate the ZERO UPON WAVELENGTH CHANGE function. To do so:

From the MAIN menu, move the cursor to the AUXILIARY field of the menu line and press ENTER. The AUXILIARY menu should now be displayed. Move the cursor to the ZERO UPON WAVELENGTH CHANGE field and press ENTER to toggle this field to off. Move the cursor to the RETURN TO MAIN menu field and press ENTER. Before beginning a SCHEDULED WAVELENGTH CHANGES run without automatic zeros, you must zero the effluent absorbance at each of the scheduled wavelengths (figure 3.4). To do so: From the MAIN menu, type in the wavelength value for the first scheduled wavelength and press ENTER. Press the ZERO key. Enter the wavelength of the next scheduled wavelength and press the ZERO key. Repeat this procedure for all of the scheduled wavelengths. This stores the proper effluent zero values for all of the scheduled wavelengths, negating the need for a zero upon wavelength change. Advance the detector to the PROGRAMMED OPERATION status in the usual manner.

3.13 Scheduled Spectra Operation

This feature allows the user to program specific times at which a predefined, discrete spectra will be performed. Additionally, it is possible to program different REPETITION values for each scheduled spectra. All of these parameters are entered before the beginning of the chromatographic run. It is also possible to schedule a FULL SPECTRAL AUTOZERO to occur. This feature is desirable for gradient elution applications in which the background absorbance may vary as the run progresses.

From the MAIN menu move the cursor to the SPECTRUM field of the PROGRAM line and press ENTER. The SCHEDULE SPECTRA menu should be displayed.

SCHEDULED SPECTRA			MENU 5	
☐ return to main menu				
NO	TIME	REPS	☐ sort	
1	0.00	1	☐ zero	
				UV

The spectral parameters of lower, upper, and step wavelengths are entered in the usual manner from the MAIN menu. See the SPECTRUM section of the "Immediate Mode Operation" portion of this tutorial for further details. In this case, the spectral default parameters will suffice.

For demonstrative purposes, we are going to schedule three intrinsic spectra to occur at 10, 50, and 200 seconds. The first spectra will have a REPETITION value of 4; the second, 2; and the third, 4. We will schedule a spectra zero to occur at 1.0 seconds.

Move the cursor to the TIME field for the first scheduled spectra and enter 1. Move the cursor across to the Zero field and press ENTER. A "Z" should appear indicating a spectra zero. Press the arrow down key to advance to the second line.

Move the cursor to the TIME field and enter 10 seconds. Move the cursor across to the REPS field and enter 4. Press the arrow down key to advance to the third line.

Move the cursor to the TIME field and enter 50 seconds. Move the cursor across to the REPS field and enter 2. Press the arrow down key to advance to the fourth line.

SCHEDULED SPECTRA			MENU 5	
☐ return to main menu				
NO	TIME	REPS	☐ sort	
1	1.00	2	☐ zero	
2	10.00	4	☐ zero	
3	50.00	2	☐ zero	
4	200.00	4	☐ zero	UV

Move the cursor to the TIME field and enter 200 seconds. Move the cursor across to the REPS field and enter 2. The SCHEDULED SPECTRA menu should look like the one above.

To begin the SCHEDULED SPECTRA program, move the cursor to the RETURN TO MAIN menu field and press ENTER.

From the MAIN menu, move the cursor to the START CHROMATOGRAM field of the menu line and press ENTER. The ARMED TO START menu should appear.

To begin the program, press the ENTER key. All scheduled spectra will be performed as programmed. To stop the program, move the cursor to the STOP CHROMATOGRAM field and push ENTER.

Editing the Schedule

The schedule of spectra may be edited at any time except when a chromatogram is UNDERWAY. To delete any scheduled spectra, move the cursor to the spectra of interest and press the CLEAR ENTRY key (CE) twice. This can be accomplished either from the TIME or REPS fields. To erase the entire screen of SCHEDULED SPECTRA, press the CLEAR ALL (C) key.

To insert a scheduled spectra between two other spectra, simply move the cursor down to next available line and enter the desired values of TIME and REPS for the inserted spectra. Move the cursor up to the SORT field and press ENTER. The monitor will automatically sort the schedule in the proper order of increasing time.

3.14 Baseline Correction

The detector incorporates a baseline correction program in its software. The baseline correction capability is enabled whenever parameters other than zero are entered in the baseline correction fields located on the AUXILIARY menu and the monitor is activated under the PROGRAMMED OPERATION MODE.

The BASELINE CORRECTION algorithm subtracts a linear function from the measured absorbance prior to sending the chromatogram to the output channel. The BASELINE CORRECTION is defined by START and END times, and an END ABSORBANCE that you specify.

With these three parameters, the detector performs a linear interpolation and subtracts the corresponding absorbance from the measured absorbance as a function of time. This correction is performed only on REAL-TIME data acquired under the PROGRAMMED OPERATION MODE. Raw data stored by the monitor for later replay is NOT baseline corrected. However, replayed data may be baseline corrected by entering the appropriate BASELINE CORRECTION parameters on the AUXILIARY menu just prior to playback.

Correction parameters are specified on the AUXILIARY menu.

From the MAIN menu, move the cursor to the AUXILIARY field of the menu line and press ENTER. The AUXILIARY menu should now be displayed.

```
Baseline 0.80 to 3.33 min @ 0.00000 AU
Auto Spectra ☐OFF 1 per peak; sides EARLY
Auto scan relay ☐disabled
Zero upon wavelength change ☐ON
SCROLL to access additional parameters
Zero value @ 254 nm 0.00000 AU
Memory extend 1 times
Display time base ☐min
```

Move the cursor to the appropriate fields and enter the desired values for START TIME (START @), END TIME (END @), and END ABSORBANCE (FINAL VALUE).

- **START @:** This field is used to enter the time coordinate for the first point defining the beginning of the BASELINE CORRECTION line SEGMENT. The time selected here should correspond to a point just prior to the start of the baseline change to be corrected. The absorbance coordinate for this point is predefined as 0.00000 AU and cannot be changed by the user.
- **END @:** This field is used to enter the time coordinate defining the last point of the BASELINE CORRECTION line SEGMENT. The time selected here should correspond to the end point of the baseline change to be corrected.
- **FINAL VALUE (END ABSORBANCE):** This field is used to enter the absorbance coordinate for the last point of the BASELINE CORRECTION line SEGMENT. Values entered should correspond to the absorbance value for the selected END TIME. Negative values are entered by typing in a 9 before the absorbance value. For example a -0.0123 AU is entered as 9.0123. Upon pressing the ENTER key, the value will be accepted and displayed as -0.123 AU.

An Example

Consider the example of a gradient separation in which the "B" solvent has a slightly lower absorbance than the "A" solvent. The result is a consistent downward drift in the baseline during a portion of the chromatogram (figure 3.5).

For this example:

1. START @ is entered as 60 seconds
2. END @ is entered as 300 seconds
3. FINAL VALUE is entered as 9.002 AU

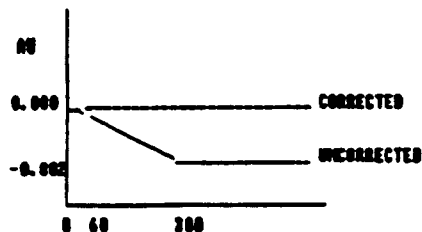


Fig. 3.5.

3.15 Memory Extender

The detector is equipped with a MEMORY EXTENDER feature which increases the monitor's capacity for absorbance vs. time data in order to allow long chromatograms to be recorded. Under the default conditions, this feature is not active and the monitor will store approximately 60 minutes of chromatographic data. After 60 minutes of operation, the detector will display "CMF" (chromatogram memory full) in place of elapsed time on the CHROMATOGRAM UNDERWAY menu. Although subsequent data are still output to the recorder in real-time, they are NOT stored. ONLY THE FIRST 60 MINUTES IS STORED FOR SUBSEQUENT PLAYBACK.

The MEMORY EXTENDER feature works by varying the data storage rate while leaving the data acquisition rate unchanged. MEMORY EXTENDER is activated by entering a MEMORY EXTENDER MULTIPLIER VALUE on the AUXILIARY menu. The default value for the multiplier is one: the detector stores every data point. When the multiplier is changed to an integer, n, the detector stores every nth point. A multiplier value of 5, for example, stores every fifth data point and effectively extends the storage time by a factor of five. For more information regarding this feature, consult the THEORY OF OPERATION section of this manual.

To enter a MEMORY EXTENDER MULTIPLIER:

From the MAIN menu, move the cursor to the AUXILIARY field of the menu line and push ENTER. The AUXILIARY menu should be displayed.

```
Baseline  0.80 to 3.33 min @ 0.00000 AU
Auto Spectra ☐OFF 1 per peak; sides EARLY
Auto scan relay                               ☐disabled
Zero upon wavelength change                   ☐ON
SCROLL to access additional parameters
Zero value      @ 254 nm 0.00000 AU
Memory extend   1 times
Display time base ☐min
```

Use the down arrow key to scroll to the MEMORY EXTENDER line and move the cursor to the 1 TIMES field. Type in the desired multiplier value (for demonstrative purposes, enter 2). Using the cursor up arrow key, move the cursor to the RETURN TO MAIN menu line and push ENTER. Initiate a PROGRAMMED MODE CHROMATOGRAM in the usual fashion.

3.16 Performing Auto-Spectra

The monitor has a peak detection program that is used to trigger the acquisition of discrete spectra. One, two, or three spectra may be taken on a given peak. Spectra may be taken early or late on a peak's up-shoulder, at its apex, and early or late on the down-shoulder.

To activate the AUTO-SPECTRA function, advance to the AUXILIARY menu.

```

Return to main menu :          MENU 2
Lamps: tungsten ☐OFF  deuterium ☐ON
Auto calibrate
Baseline  0.00 to 3.33 min @ 0.00000 AU
Auto Spectra ☐OFF 1 per peak; sides EARLY
Auto scan relay                ☐disabled
Zero upon wavelength change    ☐ON
SCROLL to access additional parameters

```

Move the cursor down to the cursor field of the AUTO-SPECTRA line. Press the ENTER key to toggle this function to on. Move the cursor one position to the right and enter the number of spectra you wish to gather for each peak. Allowable choices are 1, 2, 3.

Move the cursor one position to the right and press the ENTER key to toggle between the EARLY and LATE detection schemes for spectra taken on the peak's shoulders. For more details regarding this program, the user is directed to the "Auto-Spectra" section of the "Theory of Operation" section.

Note: Now initiate a Programmed Mode Chromatogram in the usual manner. IT SHOULD BE STRESSED THAT AUTO-SPECTRA FUNCTIONS ONLY IN THE PROGRAMMED MODE OF DETECTOR OPERATION.

3.17 Storing and Recalling a Detector Program

The detector has provisions for storing up to four different Programmed Operation Mode schemes. To do so, enter all parameters for the program and return to the MAIN menu.

Move the cursor to the STORE field and enter a number (from 1 to 4) to identify the program. To recall the program, move the cursor to the RECALL field and enter the number for the desired program. All detector parameters will automatically download. Simply advance to the ARM TO START status and begin the run in the usual manner.

3.18 Performing an Auto-Calibration

The monitor is equipped with an AUTO-CALIBRATION program designed to insure proper wavelength accuracy. When triggered, this program searches for distinct atomic lines of the deuterium lamp in the visible region. These lines are monitored using the sample photodiode to avoid any deleterious effects stemming from the flow cell's contents.

It is recommended that the user allow the detector a 30 to 60 minute warm-up before activating the auto-calibration function.

From the MAIN menu, access the AUXILIARY menu.

Move the cursor to the AUTO-CALIBRATE field and press ENTER.

```

Return to main menu :      MENU 2
Lamps: tungsten ☐OFF  deuterium ☐ON
☐Auto calibrate
Baseline  0.00 to 3.33 min @ 0.00000 AU
Auto Spectra ☐OFF 1 per peak; sides EARLY
Auto scan relay                ☐disabled
Zero upon wavelength change    ☐ON
SCROLL to access additional parameters

```

A "PLEASE WAIT" message will appear. When it disappears, the auto-calibration procedure is complete.

Section 4

Command Reference Guide

Section 3 of this manual provided a tutorial guide to the operation of the instrument. That section was arranged according to function (running a chromatogram, scanning a spectrum, etc.) and described the series of steps required to accomplish each goal.

Section 4 covers the same material, but gathers the information according to menu screen and command. Also further details of each option are explained.

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4.1 The Menus

When the instrument is first turned on, the MAIN menu is displayed. The MAIN Menu controls the single wavelength mode of the instrument and allows access to all other menus. This menu shares the following features with all of the other menus.

```
Absorbance @ 254 nm = 0.00000 AU : Menu 1
range = 2.0000 AUFS: rise time 1.0 sec
Spectrum: 200 to 360 @ 10 nm ; reps 4
Method: store recall ;
Program ☐multi ☐wavelength ☐spectra
Playback: ☐chromatogram ☐spectra
Menus: ☐start chromatogram ☐auxiliary
Mode: ☐spectra only UVW
```

The display shows a maximum of 8 lines by 32 characters. The first line always holds the title or other fixed information (in this case, the current absorbance at 254 nm). The cursor (the flashing block) can be moved up, down, right, or left using the arrow keys.

From a menu line, you can do one of five things:

1. Move to another field or line using the arrow keys.
2. Change the parameter displayed on a given field by typing in the new value for the parameter and then pressing the ENTER key. For obvious reasons, this is only possible on a line in which a parameter is displayed. If you change your mind, or type a value incorrectly, you may use the CLEAR (C) or arrow left keys to erase your entry. When you press the ENTER key, the new value will be displayed on the appropriate line. You may also enter the value by pressing any of the arrow keys to exit the given field.

If you should enter an invalid value for a particular parameter, you will see a flashing error message on the bottom line. Press any key to clear the error message, and enter an appropriate value.

3. Change the setting of a "toggle" between functions by pressing the ENTER key.
4. Move to another menu by pressing ENTER. This is only possible on a field in which no parameter is displayed. On all menus except the MAIN menu, the first choice is "RETURN TO THE MAIN MENU". If you should get lost in the menu structure for any reason, simply move the cursor to the top line of the current menu and press ENTER.
5. Perform the operation described on a given field. This is only possible on a field in which no parameter is displayed.

4.2 The Detector Key Pad

There are four different types of keys on the detector key pad:

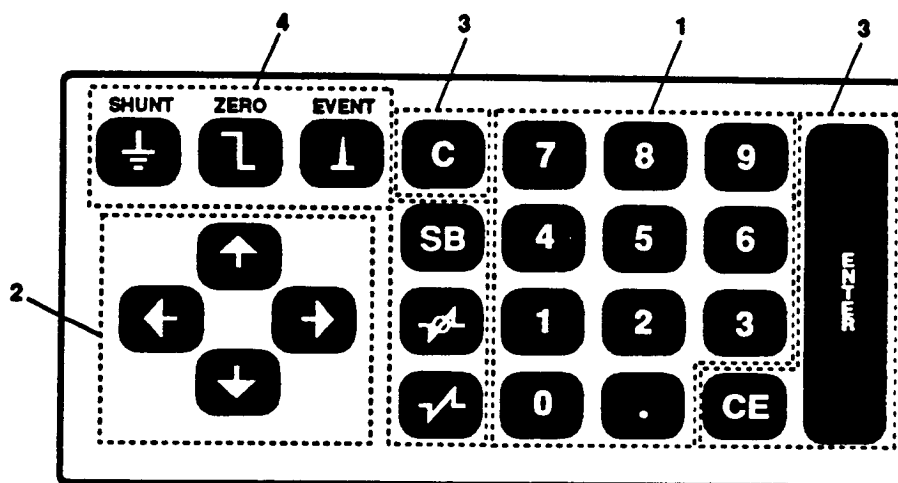


Fig. 4.1. The detector key pad.

1. Numeric Keys: These keys are used to enter numerical parameters for a given field.

2. Arrow Keys: These keys are used to move the cursor between lines and fields on a given menu and to erase a numerical entry.

3. Programming Keys: There are three programming keys:

Enter Key: This key is used to enter a given value for a particular field, to toggle a function on or off, and to advance to another menu.

Clear Entry Key: This key is used to erase an erroneous entered value for a particular field.

Clear All Key: This key is used to erase all numeric parameters listed on a given menu. This key functions only on the POLYCHROME, SCHEDULED SPECTRA, and SCHEDULED WAVELENGTH CHANGES menus.

4. Activity Keys: The activity keys are used to perform particular tasks. These keys remain active on all detector menus. There are six ACTIVITY keys:

Event Key: The EVENT key is used to signal the beginning of a chromatogram. It sends out a 20% full scale (approximate) spike to the recorder outputs and advances the detector to the UNDERWAY status in the Programmed Operation Mode. If the detector is used with the optional computer software, the EVENT key signals the computer for the start of the run. All event marks are recorded in the the detector's memory for playback purposes. Thus, they may be used as an annotative device.

Shunt Key: The SHUNT key is used to set all recorder outputs to 0 V for the purpose of adjusting recorder pen position. After releasing this key, the 0 V output is maintained for approximately 3.0 seconds.

Zero Key: This key is used to zero the absorbance values at all monitored wavelengths. It does not zero the wavelength of a programmed spectra.

Take Spectrum Key: This key is used to take a discrete spectra of the flow cell contents.

Spectra Zero Key: This key is used to store background zero values for a programmed, discrete spectra.

Stand-By Key: This key places the monitor in the stand-by status, in which all lamps are shut off and the galvanometer drive is locked in a safe position.

4.3 Detector Capabilities

The Bio-Dimension detector is a high speed, high sensitivity, multiple wavelength, scanning spectrophotometric, UV/Vis monitor. Because it uses both a deuterium and tungsten lamp, it is truly a full spectral scanning instrument.

The monitor gathers data in four different schemes:

- **HIGH SENSITIVITY SINGLE WAVELENGTH (Monochrome):** In high sensitivity monochrome, the instrument gathers data at 24 points/second for a given single wavelength. This is the wavelength which appears at the top of the detector's MAIN menu. Changes in monitored wavelength are actuated within 10 milliseconds.

- **HIGH SENSITIVITY MULTI-WAVELENGTH:** This mode allows the instrument to monitor up to 32 wavelengths at a data rate of 24 points/second/ wavelength. Therefore, if four wavelengths are monitored, each wavelength would have a data rate of 6 points/second.

- **HIGH SPEED SPECTRA:** High speed spectra permits the scan of the flow cell's contents at a data rate of 96 points/second. Spectra can be gathered from 190 to 365 or 366 to 800 nm with a maximum resolution of 1 nm at a fixed 6 nm band width. Scans are defined using an upper and lower wavelength range along with a step wavelength, which defines the interval between scanned points. A user selectable REPETITION function allows spectra to reflect averaged data of one, two, or four scans. This parameter must be optimized balancing signal-to-noise (the greater the REPETITION number, the higher the S/N) with peak width (peak speed).

The rapid data rate of HIGH SPEED SPECTRA exceeds the response speed of many recording devices, hence real-time spectral data is not transmitted out of the recorder channels. Spectra must be stored in the detector's memory and later replayed at a slower speed. Because of this, HIGH SPEED SPECTRA should only be performed while in the PROGRAMMED OPERATION MODE (see next section). If scans are performed while in IMMEDIATE MODE, the command will be carried out but spectral data will not be accessible.

- **HIGH SPEED MULTI-WAVELENGTH:** This mode is supported only when the detector is used in conjunction with the optional software. High speed multi-wavelength allows numerous wavelengths to be scanned at a data rate of 96 points/second/wavelength. Refer to the *Software Manual* for complete instructions.

4.4 Detector Operation — Immediate and Programmed Operation Modes

The detector has two modes of operation, IMMEDIATE MODE and PROGRAMMED OPERATION MODE. While in IMMEDIATE MODE, a number of operations may be carried out directly from the keyboard in real-time. These operations include:

- Setting detector parameters:
 - Single and multi-wavelength values
 - Full scale range for each monitored wavelength
 - Designation of recorder outputs for up to four of the monitored wavelengths
 - Designation of the integrator output for any one of the monitored wavelengths
 - Wavelength values for ratioing
 - Recorder output for ratio signal
 - Rise time
- Zeroing the detector
- Performing a full spectral autozero
- Setting a zero offset
- Switching lamps on or off
- Setting high speed spectra parameters:
 - Upper and lower wavelength
 - Step size
 - Repetition
- Setting baseline correction parameters:
 - Start time
 - End time
 - End absorbance
- Setting memory extender parameters
- Operating the accessory relay
- Playing back stored spectra or chromatograms

Neither chromatograms nor spectra can be stored in the detector's memory during IMMEDIATE MODE OPERATION. Additionally, baseline correction will not function and spectral scans should not be performed while in IMMEDIATE MODE.

If you wish to store a chromatogram in the detector's memory for later replay or perform and store a spectra, you must use the Programmed Operation Mode. This mode also allows for scheduled events, such as Programmed Wavelength Changes and Scheduled Spectra, to be implemented. Additionally, the Baseline Correction and Memory Extender functions of the monitor are activated under this mode of operation.

While in PROGRAMMED OPERATION, you may:

- Schedule detector parameters to change:
 - Wavelength (single wavelength only)
 - Range
 - Recorder output
 - Schedule end time for a run
- Zero the detector at all wavelengths
- Perform a full spectral autozero
- Perform a spectra and store its data
- Change spectra repetition parameters
- Schedule a predefined spectra to occur and store its data
- Perform a chromatogram and store its data
- Change recorder output for up to four of the monitored wavelengths
- Change range for up to four of the monitored wavelengths
- Change rise time
- Change the integrator wavelength
- Change ratioed wavelengths
- Operate baseline correction
- Operate a memory extender
- Operate the stop-flow accessory (if installed)
- Perform auto spectra

4.5 The Main Menu

```
Absorbance @ 254 nm = 0.0000 AU : Menu 1
range = 2.0000 AUFS: rise time 1.0 sec
Spectrum: 200 to 360 @ 10 nm ; reps 4
Method: store recall ;
Program ☒multi ☒wavelength ☐spectra
Playback: ☒chromatogram ☐spectra
Menus: ☐start chromatogram ☐auxiliary
Mode: ☐spectra only UVW
```

The MAIN menu is the most frequently used menu on the detector. From it you can:

- Enter detector parameters (wavelength, range, etc.) for the single wavelength mode
- Store and recall programs
- Access other menus
- Switch between modes of operation (single wavelength, multiple wavelength, and programmed wavelength changes).

.The MAIN menu fields are:

Wavelength

When you turn on the monitor, the cursor should be located in the wavelength field (ABSORBANCE @ 254 NM = 0.00000). The default wavelength which is displayed upon power-up is 254 nm. To change wavelength, type in the desired value and press ENTER. Acceptable wavelengths extend from 190 to 800 nm. Values outside of this range are not accepted; the message "INVALID WAVELENGTH ENTERED" will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value.

Range

This field is used to set the full scale range in absorbance units (AUFS) for IMMEDIATE MODE or PROGRAMMED MODE monochrome operation. It does not function for multiple wavelength (POLYCHROME) operation. In single wavelength operation, all absorbance data is transmitted out of recorder CHANNEL 1. Thus, this field controls the range for that channel. The default value for this parameter is 2.0000 AUFS. To change range, move the cursor to the RANGE field, type in the desired value, and then press ENTER. Acceptable values range from 0.0005 AUFS to 7.9995 AUFS in multiples of 0.0005 AUFS. Values may be entered to four decimal places; the detector will round the 4th decimal place to the nearest 0.0005 AUFS. Values outside of the allotted range will not be accepted; the message "INVALID RANGE ENTERED" will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value. The RANGE setting does not affect the INTEGRATOR OUTPUT.

Rise Time

This field is used to set the RISE TIME FILTER for IMMEDIATE or PROGRAMMED MONOCHROME operation. It does not function for multiple wavelength (POLYCHROME) operation. The RISE TIME parameter controls a progressive averaging, digital filter which conserves peak area at all settings. The filter may distort the shape of the peak, however area is always conserved.

As a general rule, a rise time should be selected which is 1/3 to 1/5 the half-height width of the fastest peak in a given separation. The default rise time setting of 1.0 second will suffice for peaks as fast as 3 seconds at half height. The RISE TIME filter may be totally bypassed by entering a RISE TIME value of 0.0 seconds.

Allowable RISE TIME values range from 0.0 to 9.9 seconds in intervals of 0.1 second. Values outside of this range will not be accepted; the message "INVALID RISE TIME" will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value.

The RISE TIME filter functions for both the recorder and integrator outputs.

Spectrum

The SPECTRUM line contains four fields used to program the parameters for a discrete spectra. Because the scanning speed of the monitor exceeds the response speed of analog recording devices, no real-time spectral data is transmitted out of the unit's four recorder channels. Spectra must be stored in the monitor's memory under PROGRAMMED MODE OPERATION conditions and then later replayed using the SPECTRA PLAYBACK function. Subsequently, spectra should only be taken while in the PROGRAMMED OPERATION MODE. If spectra are taken while under IMMEDIATE MODE OPERATION, the scan command will be carried out, but no spectral data will be available.

The first two fields of the SPECTRUM line define the spectral start and end wavelengths, respectively. Values outside of the allowed 190 to 365 and 366 to 800 nm range will not be accepted; the message "INVALID SCAN PARAMETERS" will appear. To clear the error message, press any key and enter the correct value. The detector scans from low to high wavelengths. Thus you must specify an end wavelength value which is greater than the start wavelength value.

The third number field defines the spectra STEP WAVELENGTH. This is the interval between scanned points. Acceptable values range from 1 to 10 nm. Values outside of this range will not be accepted; the message "INVALID SCAN PARAMETERS" will appear. To clear the error message, press any key and enter the correct value.

The scanning algorithm of the monitor mandates the selection of lower and upper wavelength values which are inside the range of 190 to 365 and 366 to 800 nm respectively, by a factor of at least the step wavelength divided by two. If these parameters are incorrect, the detector will display the message "INVALID SCAN PARAMETERS" on the lower line. Depending upon the entered lower, upper, and step wavelength values, the monitor will correct one of these three when it displays the error message. To clear the error message, press any key. For more information regarding the monitor's high speed spectral scanning algorithm, consult the "Theory of Operation" section of this manual.

The fourth field lists the REPETITION number (REPS) for the defined scan. This parameter determines the number of consecutive scans that will be performed after initiation of the TAKE SPECTRA command. Data derived from these scans are then averaged and stored in the monitor's memory for later playback.

Averaging spectral scans together improves the resultant signal-to-noise ratio by a factor equivalent to the square root of the number of averaged scans. A good rule of thumb is to select a REP value which allows all scans to be performed at the upper portion of a given peak. Spectral scans should be initiated just prior to the peak's apex. In this case, scans may be performed near the apex on the up shoulder, at the apex, and near the apex on the down shoulder.

Valid entries for the REP field are 1, 2, and 4. Values outside this range trigger an error message that informs the user of the allowed choices.

The Method Line

The METHOD line is used to store or recall a given program for PROGRAMMED OPERATION of the monitor.

The Program Line

The PROGRAM line is used to access the MULTI (wavelength), WAVELENGTH (CHANGES), and (SCHEDULE) SPECTRA menus. To access any of these menus, move the cursor to the field of the desired menu and press ENTER. These menus are discussed individually, in greater detail later in this section.

```
Absorbance @ 254 nm = 0.00000 AU : Menu 1
range = 2.0000 AUFS: rise time 1.0 sec
Spectrum: 200 to 360 @ 10 nm ; reps 4
Method: store recall ;
Program ☐multi ☐wavelength ☐spectra
Playback: ☐chromatogram ☐spectra
Menues: ☐start chromatogram ☐auxiliary
Mode: ☐spectra only UVW
```

The Playback Line

This line allows access to either the CHROMATOGRAM or SPECTRA PLAYBACK menus. To access the CHROMATOGRAM PLAYBACK menu, you must first store a chromatogram in the instrument's memory. Failure to do so results in a "NO CHROMATOGRAM IN MEMORY" error message. To remove the message, press any key.

To access the SPECTRA PLAYBACK menu, you must first store spectra in the instrument's memory. Failure to do so results in a "NO SPECTRA IN MEMORY" error message. To remove the message, press any key.

The menus are discussed in detail below.

The Menus Line

The MENUS line contains fields for entering the AUXILIARY menu and arming the detector for PROGRAMMED OPERATION. To call up the AUXILIARY menu, move the cursor to the AUXILIARY field and press ENTER. This menu is further discussed in later sections of this manual.

In order for the detector to enter the PROGRAMMED OPERATION MODE, in which you may run programs for SCHEDULED WAVELENGTH CHANGES and SCHEDULED SPECTRA and store chromatographic and/or spectral data for later replay, you must ARM THE INSTRUMENT TO START. This is accomplished by moving the cursor to the START CHROMATOGRAM field and pressing ENTER. You should complete all programming steps before arming the monitor.

The Mode Line

The MODE line contains the field used to toggle the detector between SPECTRA ONLY (MONOCHROME), SINGLE WAVELENGTH (PROGRAMMED WAVELENGTH CHANGES), and MULTIPLE WAVELENGTH (POLYCHROME) modes of operation. Of these three, only TIME CHANGES must be operated while in the PROGRAMMED OPERATION MODE. Toggling between these three modes is accomplished by moving the cursor to the MODE field and pressing ENTER until the desired mode is displayed.

It is important to check the status of the mode line when performing IMMEDIATE or PROGRAMMED MODE multi-wavelength. Entering the wavelength, range, recorder channel, rise time, ratio, and integrator channel outputs on the MULTI-WAVELENGTH menu does NOT automatically toggle the mode line to the MULTIPLE WAVELENGTH status. You must manually change to this mode. It is recommended that you program all desired polychrome wavelengths on the MULTI-WAVELENGTH menu before toggling the MODE line to the MULTIPLE WAVELENGTH status.

In most case, programming the SCHEDULED WAVELENGTH CHANGES menu automatically advances the mode status to TIMED CHANGES. However, it is recommended that you check the status of this field before arming the detector.

4.6 The Multi-Wavelength Menu

The MULTI-WAVELENGTH menu is used to program detector parameters for HIGH SENSITIVITY MULTI-WAVELENGTH detection in both IMMEDIATE and PROGRAMMED modes of operation. The menu may be present in two forms.

- IMMEDIATE MODE MULTI-WAVELENGTH MENU
- PROGRAMMED MODE MULTI-WAVELENGTH CHROMATOGRAM UNDERWAY MENU

The MULTI-WAVELENGTH menu is accessed by moving the cursor to the POLY field of the MAIN menu's PROGRAM line and pressing ENTER. The following fields provide the following functions for the MULTI-WAVELENGTH menu:

MULTI-WAVELENGTH						MENU 4	
☐ Return to previous menu				☐ sort			
Ratio: 254 over 350 : CH 3							
NO	nm	CH	range	RT	INT		
1	254	1	0.0020	1.0	1	☐ INS	
2	350	2	0.0020	1.0		☐ INS	

Return to Previous Menu

When the cursor is positioned in this field and the ENTER key is pushed, the LCD screen reverts back to the MAIN menu. Multiple wavelength operation is maintained, however the monitored wavelengths and their present absorbencies can not be seen. While in IMMEDIATE MODE, this field is operational at all times. Upon advancing to PROGRAMMED OPERATION, this field is eliminated.

Ratio

This line is used to enter the values for two of the monitored wavelengths to be ratioed. Ratioing the absorbencies at two particular wavelengths may be used as a means of determining the purity of a given peak. The ratio value at any given time is the quotient of the molar absorptivities of the peak at the two ratioed wavelengths. If the peak is pure, that is, no compound coelution (solute and impurity present at the same time) occurs, the ratio should remain constant during the peak's elution period. A changing ratio during a peak's elution period is indicative of solute coelution, and the peak is impure. A word of caution, however, ratioing will only detect compound coelution if the impurity absorbs at both of the ratioed wavelengths. Also, if both the impurity and solute of interest display identical elution profiles (equal ascending and descending slopes resulting in an identical increase and decrease in concentration as a function of time), the measured ratio becomes the sum of the molar absorptivities of the solute and impurity, and no ratio change will be seen during the peak's elution period.

Calculated ratios are not the ratio of the absolute absorbencies for the two ratioed wavelengths, but rather the ratio of their absorbencies as a percentage of full scale. Thus, the calculated numerical ratio is:

$$\text{NUMERICAL RATIO} = \frac{\text{ABSORBANCE WAVELENGTH 1} / \text{RANGE WAVELENGTH 1}}{\text{ABSORBANCE WAVELENGTH 2} / \text{RANGE WAVELENGTH 2}}$$

Manipulating the data in this fashion restricts the possible range for the calculated ratio.

Since it is of no value to ratio baseline absorbance data, the ratio calculation is only performed in the presence of a peak or when the user activates the EVENT key. This is accomplished by using the range parameter to control ratio activation. The ratio calculation is activated when BOTH wavelength absorbencies exceed 3% of full scale. Ratios are no longer calculated when these values fall below a full scale level of 2%.

The NUMERICAL RATIO which is displayed on the detector's LCD is not the ratio transmitted out of the RECORDER OUTPUT. The recorder output transmits a RATIO PLOT which is the log of the

NUMERICAL RATIO. By calculating the NUMERICAL RATIO as a percent of full scale and tracing it as a log function, the overall RATIO PLOT is easily displayed on a strip chart recorder, eliminating a ratio full scale setting. If the recorder's pen is positioned at 50% full scale for a ratio of 1.0, then 100% full scale indicates a ratio of 10.0 and 0% full scale corresponds to a 0.1 ratio. The NUMERICAL RATIO may be calculated from the RATIO PLOT by using the following equation.

$$\text{Numerical Ratio} = 10^{\left(\frac{X-50}{50}\right)}$$

where X is the full scale deflection of the recorder pen.

Some Examples:

During the elution of a given peak, the RATIO PLOT moved from 50% full scale to 30% full scale. The corresponding NUMERICAL RATIO (NR) is:

$$\text{Numerical Ratio} = 10^{\left(\frac{30-50}{50}\right)} = 10^{-2/5} = 0.398$$

During the elution of another peak, the RATIO PLOT moved from 50% full scale to 75% full scale. The corresponding NUMERICAL RATIO (NR) is:

$$\text{Numerical Ratio} = 10^{\left(\frac{75-50}{50}\right)} = 10^{1/2} = 3.162$$

MULTI-WAVELENGTH						MENU 4	
☐ Return to previous menu				☐ sort			
Ratio: 254 over 350 : CH 3							
NO	nm	CH	range	RT	INT		
1	254	1	0.0020	1.0	1	☐ INS	
2	350	2	0.0020	1.0		☐ INS	

The first and second fields of the RATIO line are used to enter the numerator and denominator wavelengths for the ratio, respectively. Values are entered by moving the cursor to the appropriate field, typing in the wavelength values, and pressing the ENTER key. Acceptable wavelengths are any of the monitored wavelengths in the 190 to 800 nm range.

The third field of the RATIO line controls the recorder output for the RATIO PLOT. Acceptable values are 1, 2, 3, and 4. The user should avoid entering a RATIO PLOT OUTPUT which is already specified as a RECORDER CHANNEL for one of the monitored wavelengths. If duplication occurs, the RATIO PLOT will be transmitted out of the specified RECORDER CHANNEL.

Wavelength Lines

The remaining horizontal lines contain a number of parameters for the monitored wavelengths entered in a given multiple wavelength application. These are the WAVELENGTH lines. As a stand-alone instrument, the detector can monitor up to 32 different wavelengths in its HIGH SENSITIVITY MULTI-WAVELENGTH MODE. Under this mode of operation, the detector gathers data at a rate of 24 points per second per wavelength. Thus, if four wavelengths are monitored, each would have a data acquisition rate equivalent to 6 points per second.

The following fields are located in each WAVELENGTH line:

Wavelength (NM)

This field is used to enter the wavelength value, in nm, for one of the monitored wavelengths. Acceptable values range from 190 to 365 and 366 to 800 nm. Values outside of this range are not accepted; the message "INVALID WAVELENGTH ENTERED" will flash on the bottom line. When this occurs, press any key to clear the error message and enter the correct value.

MULTI-WAVELENGTH						MENU 4	
☐ Return to previous menu				☐ sort			
Ratio: 254 over 350						: CH 3	
NO	nm	CH	range	RT	INT		
1	214	1	0.0020	1.0	I	☐ INS	
2	254	2	0.0020	1.0		☐ INS	
3	280	3	0.0020	1.0		☐ INS	
4	329	4	0.0020	1.0		☐ INS	

Recorder Channel (CH)

This field is used to enter the RECORDER CHANNEL OUTPUT for a defined wavelength. Acceptable values range from 1 to 4. Duplication of recorder outputs result in assigning the channel to the wavelength whose output was last specified.

Range

This field is used to set the full scale range in absorbance units (AUFS) for a defined wavelength. The default value for this parameter is 0.002 AUFS. To change range, move the cursor to the RANGE field, type in the desired value, and then press ENTER. Acceptable values range from 0.0005 AUFS to 7.9995 AUFS in multiples of 0.0005 AUFS. Values may be entered to four decimal places; the detector will round the fourth decimal place to the nearest 0.0005 AUFS. Values outside of the allotted range will not be accepted; the message "INVALID RANGE ENTERED" will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value.

Rise Time (RT)

This field is used to set the RISE TIME FILTER for a defined wavelength. The RISE TIME parameter controls a progressive averaging, digital filter which conserves peak area at all settings. The filter may distort the shape of the peak, however area is always conserved.

As a general rule, a rise time should be selected which is 1/10th the half-height width of the fastest peak in a given separation. The default RISE TIME setting of 1.0 second will suffice for peaks as fast as 10 seconds at half height. The RISE TIME filter may be totally bypassed by entering a RISE TIME value of 0.0 seconds.

Allowable RISE TIME values range from 0.0 to 9.9 seconds in intervals of 0.1 second. Values outside of this range will not be accepted; the message "INVALID RISE TIME" will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value.

The RISE TIME filter functions for both the recorder and integrator outputs.

MULTI-WAVELENGTH						MENU 4	
☐ Return to previous menu						☐ sort	
Ratio: 254 over 350						: CH 3	
NO	nm	CH	range	RT	INT		
1	214	1	0.0020	1.0	I	☐ INS	
2	254	2	0.0020	1.0		☐ INS	
3	280	3	0.0020	1.0		☐ INS	
4	329	4	0.0020	1.0		☐ INS	

Integrator (INT)

This field selects the INTEGRATOR OUTPUT WAVELENGTH. The default status places an "I" in the line of the first wavelength. To select a different wavelength to be sent to the INTEGRATOR OUTPUT, move the cursor to the INT field of the desired wavelength and press the ENTER key. The "I" should now be displayed in the desired wavelength line. Data channeled to the INTEGRATOR OUTPUT is transmitted at a fixed 1 volt/AU scale and is filtered by the RISE TIME circuit. To obtain unfiltered data, enter a 0.0 second RISE TIME for the integrated wavelength.

The Insert Function (INS)

The INS field is used to insert a wavelength between two other monitored wavelengths. To do so:

Move the cursor to the INS field of the wavelength line that is to precede the inserted wavelength and press ENTER. An exact copy of this line will appear immediately below.

Move the cursor to the newly inserted line and enter the desired parameters for the inserted wavelength.

It is not necessary to enter wavelengths in increasing or decreasing order. The monitor automatically sorts the entered wavelengths and scans them in order of increasing wavelength (this sorting function is not indicated by the LCD). Because only five wavelengths and their absorbance values may be viewed on the MULTI-WAVELENGTH menu at any time, the INSERT function was included to allow the user to enter a wavelength within a group of other wavelengths, so that all may be simultaneously displayed.

Immediate Mode and Programmed Mode Multi-Wavelength Menus

All previous examples of the MULTI-WAVELENGTH menu displayed in this section depicted the IMMEDIATE MODE MULTI-WAVELENGTH menu. This menu is used during IMMEDIATE

OPERATION and to program the parameters for a PROGRAMMED MULTI-WAVELENGTH CHROMATOGRAM.

The PROGRAMMED MODE MULTI-WAVELENGTH menu is displayed when the detector performs HIGH SENSITIVITY MULTI-WAVELENGTH in the ARMED TO START or CHROMATOGRAM UNDERWAY status. To arm the detector:

Enter all desired values on the IMMEDIATE MODE MULTI-WAVELENGTH menu.

Move the cursor to the RETURN TO PREVIOUS MENU field and press ENTER.

From the MAIN menu, move the cursor to the MODE field and press the ENTER key until MULTIPLE WAVELENGTH is displayed.

Move the cursor to the START CHROMATOGRAM field of the MENUS line and press ENTER. The MULTI-WAVELENGTH ARMED TO START menu should be displayed.

PUSH EVENT TO START CHROMATOGRAM						MENU 7
MODE: multi-wavelength						
☐ return to main menu ; spectra reps: 4						
Ratio: 214 over 280 : CH 4						
NO	nm	CH	range	RT	INT	
1	214	1	0.0020	1.0	I	0.00000
2	254	2	0.0020	1.0		0.00000
3	280	3	0.0020	1.0		0.00000
4	329		0.0020	1.0		0.00000

Push the EVENT key to advance the detector to the UNDERWAY status. The POLYCHROME CHROMATOGRAM UNDERWAY menu should be displayed.

4.7 The Scheduled Wavelength Changes Menu

The SCHEDULED (PROGRAMMED) WAVELENGTH CHANGES menu is accessed from the MAIN menu by moving the cursor to the TIMED MONO field of the PROGRAM line and pressing the ENTER key.

SCHEDULED WAVELENGTH CHANGES						MENU 3
☐ Return to main menu						
NO	NM	FROM	TO	CH	RANGE	FILTER
1	254	0	1200	1	2.0000	1.0 ☐ INS

The first field of this menu is RETURN TO MAIN MENU. When the cursor is in this position and the ENTER key is pressed, the MAIN menu will be displayed on the detector LCD.

The remaining fields comprise the various components for the SCHEDULED WAVELENGTH lines. A maximum of 20 wavelength changes may be entered. Attempts at entering more than 20 changes result in the error message "PROGRAM MEMORY FULL." This is displayed on the bottom line and will obscure the last (20th) scheduled wavelength change. To clear the error message, press any key.

The Wavelength Line Fields are:

Wavelength (NM)

This field is used to enter the wavelength value for a particular wavelength change. Acceptable wavelengths extend from 190 to 365 and 366 to 800 nm. Values outside of this range will not be accepted; the message "INVALID WAVELENGTH ENTERED" will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value.

From...To...

These two fields allow entry of the elapsed time range during which a given wavelength will be monitored. To change the time, type in the new value and press the ENTER key. The new value will be displayed. Allowed values are integers between 1 and 65534 second. Values outside this range are ignored.

Recorder Channel (CH)

This field is used to enter the RECORDER CHANNEL OUTPUT for real-time absorbance data monitored for the defined wavelength change. Acceptable values range from 1 to 4.

Range

This field is used to set the full scale range in absorbance units (AUFS) for a given wavelength change. The default value is 2.0000 AUFS. Acceptable values range from 0.0005 AUFS to 7.9995 AUFS in multiples of 0.0005 AUFS. Values may be entered to four decimal places; the detector will round the fourth decimal place to the nearest 0.0005 AUFS. Values outside of the allotted range will not be accepted; the message "INVALID RANGE ENTERED" will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value.

Filter (Rise Time)

This field is used to set the RISE TIME FILTER for a given scheduled wavelength. This setting controls a progressive averaging, digital filter that conserves peak area at all settings.

As a general rule, a rise time should be selected which is 1/3 to 1/5 the half-height width of the fastest peak in a given separation. The default setting of 1.0 second will suffice for peaks as fast as 3 seconds at their half height. The filter may be totally bypassed by entering a RISE TIME value of 0.0 second.

Allowable RISE TIME values range from 0.0 to 9.9 second intervals of 0.1 second. Values outside of this range will not be accepted; the message "INVALID RISE TIME" will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value. THE RISE TIME FILTER FUNCTIONS FOR BOTH THE RECORDER AND INTEGRATOR OUTPUT.

SCHEDULED WAVELENGTH CHANGES						MENU 3
☐ Return to main menu						
NO	NM	FROM	TO	CH	RANGE	FILTER
1	254	0	1200	1	2.0000	1.0 ☐ INS

Insert (INS)

This field is used to insert a change between two previously entered wavelength changes. To do so:

Move the cursor to the INS field for the line which is to precede the inserted wavelength change and press the ENTER key. This generates a new line which, with the exception of run time, is a copy of the previous line.

Enter the parameters for the inserted wavelength.

Editing Scheduled Wavelength Changes Menu

If you wish to delete a given SCHEDULED WAVELENGTH, move the cursor to any field on the appropriate wavelength line and press the CLEAR ENTRY (CE) key twice.

If you wish to erase the entire schedule of wavelength changes, press the CLEAR ALL key (C).

If you wish to insert a wavelength change between two other scheduled wavelengths, follow the procedure outlined in the "Insert" section which precedes this part of the manual.

4.8 The Scheduled Spectra Menu

This menu is used to schedule a number of spectra to be taken automatically at predetermined times. The menu is accessed from the MAIN menu by moving the cursor to the SPECTRUM field of the PROGRAM line and pressing ENTER.

SCHEDULED SPECTRA				MENU 5
☐ return to main menu				
NO	TIME	REPS	☐ sort	
1	0.00	1	☐ zero	UU

The first field of this menu is RETURN TO MAIN menu. When the cursor is in this position and the ENTER key is pressed, the MAIN menu will be displayed on the detector LCD.

The remaining fields comprise the various components of the SCHEDULE SPECTRA lines. A maximum of 32 SCHEDULED SPECTRA may be entered. Attempts at entering more than 32 spectra result in the

error message "PROGRAM MEMORY FULL." This is displayed on the bottom line and will obscure the last (32nd) scheduled spectra. To clear the error message, press any key. IT IS NOT NECESSARY TO ENTER SCHEDULED SPECTRA IN CHRONOLOGICAL ORDER; THE MONITOR HAS A SORTING FUNCTION.

The Scheduled Spectra Fields are:

Time

This field is used to enter the time, in seconds, during which a given scheduled spectra will be performed. Allowable times range from 1 to 16383 seconds. Values outside of this range will not be accepted; the error message "INVALID SPECTRA TIME" will appear on the bottom line. To remove the message, press any key.

Reps

This field controls the REPETITION number for a given scheduled spectra. This parameter determines the number of consecutive scans that will be performed during spectral acquisition. The resultant scans are averaged together and stored in the monitor's memory for later playback (see the SPECTRUM section of the MAIN menu portion of the COMMAND REFERENCE GUIDE for further details).

Allowable values for the REPS field are 1, 2, and 4. Values other than these will not be accepted; an error message listing acceptable REPS values will appear. To clear the message, press any key.

Zero

Press ENTER to schedule a Spectra Zero.

Sort

This field is used to sort the scheduled spectra in chronological order after entering them in the menu. To do so, move the cursor to the SORT field and press ENTER. The scheduled spectra should now be listed in chronological order.

4.9 The Playback Chromatogram Menu

This menu is used to play back chromatographic data stored under the PROGRAMMED OPERATION mode. There are two types of PLAYBACK CHROMATOGRAM menus: Single Wavelength and Multiple Wavelength. Both are accessed from the MAIN menu by moving the cursor to the CHROMATOGRAM field of the PLAYBACK line and pressing ENTER.

```
CHROMATOGRAM PLAYBACK          MENU 8
ABSORBANCE @ 254 nm = 0.00000 AU  0.0
MODE: spectra only
[Return]    [Playback chromatogram]
... @      1 times real time  [stop PB]
PB range= 2.0000 AUFS; rise time= 1.0 sec
Baseline  0.83 to 3.33 min @ 0.00000 AU
Strip chart channel no.= 1
```

The second line of this menu displays the monitored wavelength along with the playback absorbance value. The playback elapsed time is indicated in the far right corner of the second line. All times are listed in seconds.

The third line contains the MODE indicator, which indicates the nature of the recorded chromatogram (i.e. spectra only, timed changes, or multiple wavelength). This is not a user interactive field.

The fourth line contains the RETURN and PLAYBACK CHROMATOGRAM fields. Moving the cursor to the RETURN field and pressing the ENTER key returns the display to the MAIN menu. This is the only menu in which the return to main menu function is not listed in the first line.

Moving the cursor to the PLAYBACK CHROMATOGRAM field and pressing the ENTER key begins the playback process. When this occurs, the monitored wavelength, its absorbance, and the elapsed time will be displayed. Upon completion of the playback, these parameters will disappear from the screen.

The fifth line contains the MEMORY EXTENDER MULTIPLIER, PLAYBACK RATE INDICATOR, and STOP PLAYBACK field. The MEMORY EXTENDER MULTIPLIER is only present if the chromatogram was recorded using the MEMORY EXTENDER feature. The example below indicates a chromatogram that was stored using a MEMORY EXTENDER feature of 2 (doubling the effective detector memory).

```
CHROMATOGRAM PLAYBACK          MENU 8
ABSORBANCE @ 254 nm = 0.00000 AU  0.0
MODE: spectra only
Return      Playback chromatogram
... @      1 times real time  stop PB
PB range= 2.0000 AUFS; rise time= 1.0 sec
Baseline  0.03 to 3.33 min @ 0.00000 AU
Strip chart channel no.= 1
```

It should be noted that MEMORY EXTENDER effects both PLAYBACK RATE and RISE TIME. The effective playback rate is equal to the product of the memory extender multiplier and the entered playback rate. The resultant rise time for the replayed chromatogram is equal to the product of memory extender multiplier and the entered rise time. See the "Theory of Operation" section for further details.

The PLAYBACK RATE field (x TIMES REAL-TIME) is used to replay the chromatogram faster than real-time. Allowable values range from 1 to 20. It is possible to program the playback menu so that the monitor's microprocessor does not have sufficient time to perform the range, rise time, and playback rate calculations. In such a situation, the error message "PROCESSOR OVERLOAD" will appear. If this occurs, press any key to clear the error message and try more moderate terms of playback rate, rise time, and range in this order.

As the processor approaches overload status, it will avoid unnecessary functions such as updating the screen of absorbance, wavelength, and time. If you do not see these values on the screen, it simply means that the microprocessor is using all the time it has to get the chromatogram out.

The STOP PLAYBACK (STOP PB) field is used to interrupt the replay of a stored chromatogram. This is accomplished by moving the cursor to this field and pressing the ENTER key.

The sixth line contains the PLAYBACK RANGE and PLAYBACK RISE TIME parameters. Allowable values for range extend from 0.0005 to 7.9995 AUFS in multiples of 0.0005 AUFS. Allowable RISE TIME values extend from 0.0 to 9.9 seconds in multiples of 0.1 second. Values outside of these allotted ranges will trigger an error message.

The final line contains the STRIP CHART CHANNEL OUTPUT NUMBER field. This field controls the recorder outputs for the playback process. Allowable values range from 1 to 4.

If you wish to send filtered, but nonranged data to a stand-alone integrator or data acquisition system during the playback process, you must use the integrator output. The data is automatically transferred to the integrator output upon initiation of the playback process.

4.10 Multi-Wavelength Playback

MULTI-WAVELENGTH PLAYBACK						MENU 9
☐ Return		☐ Playback chromatogram				
... @		1 times real time		☐ stop PB		
Ratio: 214 over 280 : CH 4						
Baseline 0.83 to 3.33 min @ 0.00000 AU						
NO	nm	CH	range	RT	INT	
1	214	1	0.0020	1.0	1	0.00000
2	254	2	0.0020	1.0		0.00000
3	280	3	0.0020	1.0		0.00000
4	329	4	0.0020	1.0		0.00000

The second line contains the RETURN and PLAYBACK CHROMATOGRAM fields. Moving the cursor to the RETURN field and pressing ENTER returns the display to the MAIN menu. Moving the cursor to the PLAYBACK CHROMATOGRAM field and pressing the ENTER key begins the replay process. Chromatogram replay is indicated by the appearance of absorbance values (in the wavelength lines) and elapsed time (in the left side of the first line) upon the menu. All times are listed in seconds.

The third line contains the MEMORY EXTENDER MULTIPLIER VALUE, PLAYBACK RATE VALUE, and STOP PLAYBACK (PB) fields. The user is instructed to consult the previous section for an explanation of these features.

The fourth line contains the RATIO WAVELENGTHS and RATIO PLOT OUTPUT fields. The first of the two RATIO WAVELENGTH fields is the numerator wavelength and the second is the denominator wavelength. For more information regarding the monitor's ratio function, the user is referred to Ratio section of the Polychrome Portion of the Command Reference Guide. The CHANNEL (CH) field of this line selects the recorder output for the RATIO PLOT. Allowable values range from 1 to 4. The user must insure against specifying a RATIO PLOT output which is concomitantly used as one of the wavelength recorder outputs. If this occurs, the recorder output will be assigned to the last field in which the repeated output value was entered and the other field will not specify an output.

The next several lines list parameters for all of the recorded wavelengths.

NM (Wavelength)

These fields list the wavelengths that were stored during the polychrome run. These are not user interactive fields.

MULTI-WAVELENGTH PLAYBACK						MENU 9
☐ Return		☐ Playback chromatogram				
... @		1 times real time				☐ stop PB
Ratio: 214 over 280 : CH 4						
Baseline 0.83 to 3.33 min @ 0.00000 AU						
NO	nm	CH	range	RT	INT	
1	214	1	0.0020	1.0	I	0.00000
2	254	2	0.0020	1.0		0.00000
3	280	3	0.0020	1.0		0.00000
4	329	4	0.0020	1.0		0.00000

CH (Channel)

These fields assign a recorder output channel to the corresponding wavelength. Allowable values extend from 1 to 4. If the integrator output is used, a maximum of five wavelengths can be played back at any time. Thus if more than five wavelengths were monitored, it is necessary to repeat the replay process until all wavelengths have been played back.

RANGE

These fields control the playback range (AUFS) for each recorded wavelength. Acceptable values extend from 0.0005 AUFS to 7.9995 AUFS in multiples of 0.0005 AUFS.

RT (Rise Time)

These fields control the playback rise time filter for each recorded wavelength. Acceptable values extend from 0.0 to 9.9 seconds in intervals of 0.1 second.

INT (Integrator Output Wavelength)

This field selects the wavelength to be channeled to the INTEGRATOR OUTPUT during the playback process. Wavelengths are selected by moving the cursor to the INT field of the desired wavelength, and pressing ENTER. The integrator output will transmit filtered, nonranged data. Filtering may be bypassed by entering a 0.0 second rise time for the integrated wavelength.

4.11 The Spectra Playback Menu

The SPECTRA PLAYBACK menu is used to play out any spectra stored during a given PROGRAMMED OPERATION chromatographic run. This menu is accessed from the MAIN menu by moving the cursor to SPECTRA field of the PLAYBACK line and pressing the ENTER key.

SPECTRA PLAYBACK		MENU 10
Absorbance @ 210 nm = 0.00000 AU		
Spectrum: 200 to 360 @ 10 nm steps		
Spectrum no 1 taken at 1 sec w/4reps		
☐ Return	☐ Playback spectra	
☐ Next spectra	playback rate 20 nm/sec	
Range 2.0000 AUFS	ch no	

The second line of this menu displays the wavelengths and their absorbance values measured in the recorded spectra. Before playback begins, this line will remain blank. Upon initiating the playback process, these fields are displayed.

The third line displays the programmed spectral parameters of lower, upper, and step wavelengths for the stored spectra. There are no user interactive fields on this line.

The fourth line contains information about the stored spectra. Stored spectra are all numbered in sequence of their acquisition and are recalled, by using the NEXT SPECTRA field, in numerical order. This line displays the spectra number, the time (in seconds) during the run in which it was taken, and the number of repetitive scans averaged in the stored data. There are no user interactive fields on this line.

The next line contains the RETURN and PLAYBACK SPECTRA fields. Moving the cursor to the RETURN field and pressing the ENTER key, returns the display to the MAIN menu. Moving the cursor to the PLAYBACK field and pressing the ENTER key begins the spectra playback process. Playback is indicated by the appearance of wavelength and absorbance data on the second line. Upon completion of the playback, these values will disappear from the menu.

The fifth line contains the NEXT SPECTRA and PLAYBACK RATE fields. The NEXT SPECTRA field is used to advance through the stored spectra in numerical order. Moving the cursor to this field and pressing the ENTER key, advances the menu to display the parameters for the next stored spectra. Spectra are stored in order of their acquisition. This newly displayed spectra is replayed by moving the cursor to the PLAYBACK SPECTRA field and pressing ENTER.

The PLAYBACK RATE field specifies the rate of spectra playback in a nm/second basis. Allowable rates extend from 1 to 255 nm/second in 1 nm/second intervals. Spectra playback rates should be chosen which do not exceed the frequency response of the recording device. Matching the playback rate to a given strip chart paper speed, allows calibration of the wavelength axis.

The final line lists the fields for SPECTRA PLAYBACK RANGE and CHANNEL NUMBER (CH NO). Spectra may be replayed at any range within the 0.0005 AUFS to 7.9995 AUFS range in multiples of 0.0005 AUFS. Spectra may be played out of any of the four recorder outputs.

4.12 The Auxillary Menu

The AUXILIARY menu is accessed from the MAIN menu by moving the cursor to the AUXILIARY field of the MENUS line and pressing the ENTER key.

```

Return to main menu :          MENU 2
Lamps: tungsten OFF  deuterium ON
Auto calibrate
Baseline  0.80 to 3.33 min @ 0.00000 AU
Auto Spectra OFF 1 per peak; sides EARLY
Auto scan relay           disabled
Zero upon wavelength change      ON
SCROLL to access additional parameters

```

The first line of this menu is RETURN TO MAIN MENU. Moving the cursor to this field and pressing the ENTER key returns the display to the MAIN menu.

The second line is used to turn the lamps on or off. This is accomplished by moving the cursor to the appropriate lamp field and using the ENTER key to toggle its status.

The third line controls the detector's auto-calibration function.

The fourth line lists the parameters for Baseline Correction Programs. The user is directed to the "Baseline Correction" section of the "Routine Operation Tutorial" for further details.

The fifth line lists parameters for the Auto-Spectra Program. The user is directed to the "Auto-Spectra" section of the tutorial for further details.

The sixth line controls the status of the Auto Scan Relay. The default status is to disable this function. To activate the Auto Scan Relay, move the cursor to this field and press ENTER to toggle it to the ENABLED status.

The seventh line controls the Zero Upon Wavelength Change function.

The ninth line displays and controls the high sensitivity wavelength zero values stored in the monitor. To view the zero value for a given wavelength, move the cursor to the wavelength field of this line, type in the appropriate wavelength, and press ENTER. The new zero value should be displayed. To program an offset, either positive or negative, for a given wavelength, you should:

Enter the desired wavelength value in the wavelength field.

Type in the desired offset value in AU. Acceptable values range from 0.000002 to 8.000002 AU in multiples of 0.000002 AU. A negative value is entered by typing a 9 before the desired offset. For example, -0.02 AU is entered as 9.02 AU.

The tenth line controls the MEMORY EXTENDER multiplier. The default status is 1 (no memory extension). Allowable values range from 1.0 to 255 in multiples of 1. For more information regarding MEMORY EXTENDER refer to the "Theory of Operation" section of this manual.

The final line is used to select between seconds or minutes for timing purposes.

Baseline 0.80 to 3.33 min @ 0.00000 AU
Auto Spectra ☐OFF 1 per peak; sides EARLY
Auto scan relay ☐disabled
Zero upon wavelength change ☐ON
SCROLL to access additional parameters
Zero value @ 254 nm 0.00000 AU
Memory extend 1 times
Display time base ☐min

Section 5

Routine Maintenance

Contents

This section of the manual describes the few routine service and maintenance procedures required for the detector.

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	The Deuterium Lamp	70
	The Tungsten Lamp	70
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5.1 Changing Cells

The detector accommodates a variety of flow cells with different path lengths, illuminated volumes, and wetted materials for different applications. A complete listing of the available cells is provided in Appendix A. All flow cells are provided pre-mounted in a holder assembly to minimize alignment problems.

Detailed instructions specific to the various flow cells are included with the flow cells themselves. This section of the manual provides general guidelines for flow cell service and maintenance.

The flow cell is located in the forward housing on the left-hand side of the detector (fig. 5.1).

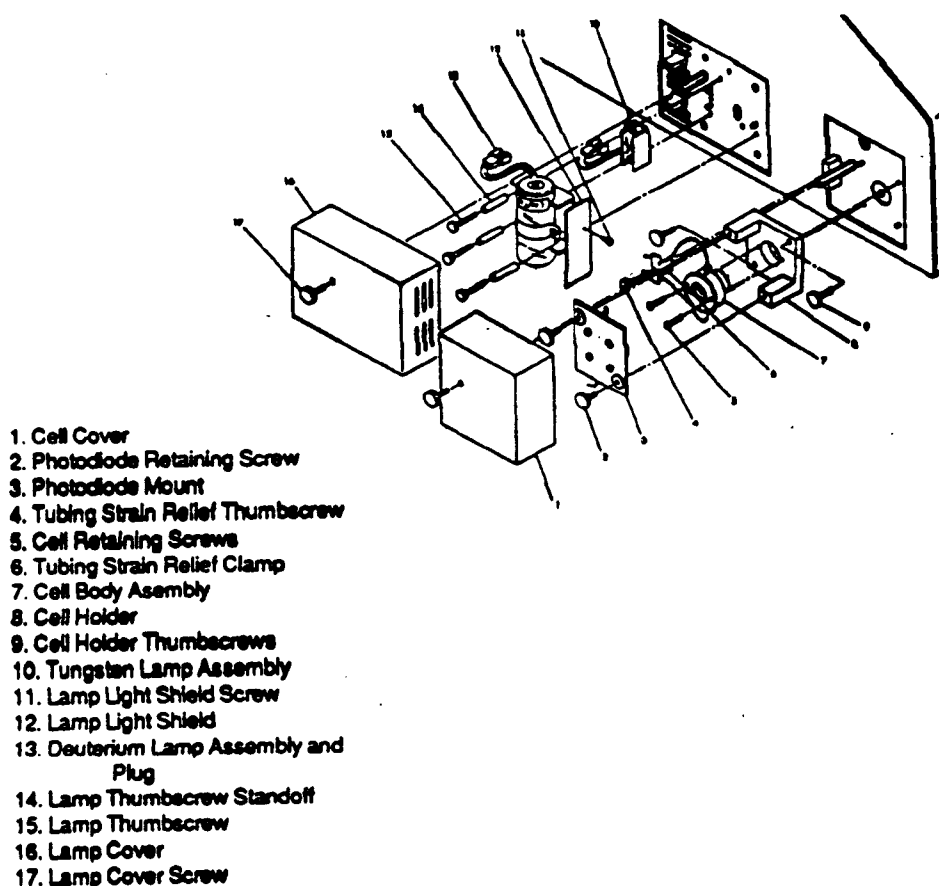


Fig. 5.1. Flow cell and lamp assemblies.

To change a flow cell:

- Make sure the power cord is disconnected from the rear panel of the detector.
- Disconnect the cell inlet tube from the column, stop flow assembly, or connecting tube and free the outlet tubing.
- Remove the cell cover by unscrewing the thumbscrew and pulling the cover straight back to expose the photodiode mount (See Figure 5.2).

- Unscrew the two thumbscrews on the photodiode mount and pull the photodiode mount straight back. The connecting cable is long enough to allow the photodiode mount to rest on the benchtop.
- Loosen the thumbscrew holding the tubing strain relief clamp in place and gently pull the clamp toward you far enough to disengage the tubing from the clamp.
- Unscrew the two thumbscrews holding the cell holder assembly in place and pull the cell holder assembly straight back toward you to remove it.
- Replacement cells are mounted pre-aligned in a cell holder assembly. Installation reverses the removal process.
- Slide the cell holder assembly onto the alignment dowels. The inlet line should come into the bottom of the cell in order to provide efficient bubble flushing. Securely fasten the cell holder assembly with two thumbscrews.
- Slip the inlet and outlet tubes into the slots in the strain relief clamp and tighten the thumbscrew holding the clamp in place.
- Replace the photodiode mount (the cable should come from the right side) and fasten it securely with the two thumbscrews.
- Replace the cell cover (be careful not to pinch the cable or the tubing) and fasten it securely with the thumbscrew.
- Reconnect the inlet line to the column, stop-flow accessory, or the connecting tubing and reconnect the outlet tubing to function collection, back pressure device, or waste container.
- Reconnect the power cord to the rear panel of the detector.

Note: Be careful to avoid putting fingerprints on the photodiode. The photodiode surface may be cleaned with spectroscopic grade methanol and lint-free lens paper.

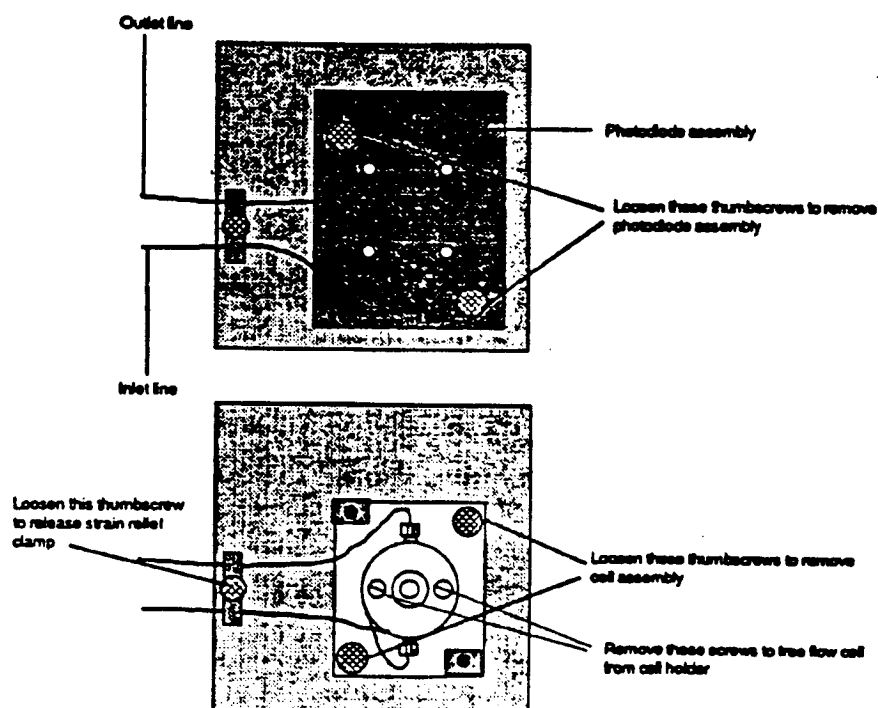


Fig. 5.2.

5.2 Changing Lamps

The detector is provided with two light sources in order to provide an optimum signal-to-noise ratio across the entire UV/visible spectrum.

Caution: UV light can damage eyes and skin. Always disconnect the power cord before working in the vicinity of the lamp.

The Deuterium Lamp

The deuterium lamp is rated for 1000 hours of life to 1/2 the original intensity (this does not mean that the lamp will burn out after 1000 hours, merely that its intensity will be reduced to 50% of the original output). Replacement lamps are provided pre-aligned on a lamp mount for easy installation.

To remove the deuterium lamp:

Make sure that the power cord is disconnected from the rear panel of the detector.

Unscrew the screw and remove the lamp housing (the rear housing on the side detector) to expose the lamps.

Disconnect the UV lamp lead from the detector (the lead has 3 wires and connects to the upper of the lamp terminals) by gently pulling straight back toward you. Do not twist the connector while pulling (See figure 5.3).

Unscrew the two thumbscrews holding the lamp mount in place, and pull the lamp mount straight back toward you. Be careful not to lose the two aluminum standoffs on the thumbscrews. Be careful not to get fingerprints on the lamp.

Note: Never loosen the screw holding the lamp to the mount, and never attempt to rotate or move the lamp up or down in the mount. Doing so will degrade the system performance. The lamp is provided pre-aligned in the mount.

To install the deuterium lamp:

Slide the lamp mount onto the alignment dowel located to the right of the aperture (the mount has a pre-drilled hole to accommodate the dowel). The lamp leads should emerge from the top of the lamp. Use the thumbscrews and aluminum standoffs to attach the lamp mount to the case. Connect the lamp lead to the upper of the two terminals in the lamp compartment.

Replace the lamp compartment cover and fasten with the screw.

Note: No harm will be done if the lamps are plugged into the wrong connectors, but the lamps will fail to light.

The Tungsten Lamp

The lifetime of the tungsten lamp, is approximately 2500 hours. The tungsten lamp, like the deuterium lamp, is provided pre-aligned on a lamp mount for easy installation.

To remove the tungsten lamp:

Make sure that the power cord is disconnected from the rear panel of the detector.

Unscrew the screw and remove the lamp housing (the rear housing on the right side of the detector) to expose the lamps.

Disconnect the tungsten lamp lead from the detector (the lead has 2 wires and connects to the lower of the lamp terminals) by gently pulling straight back toward you. Do not twist the connector while pulling (See figure 5.3).

Unscrew the thumbscrew holding the lamp mount in place, and pull the lamp mount straight back toward you. Be careful not to lose the aluminum standoff on the thumbscrew. Be careful not to get fingerprints on the lamp.

Note: Never loosen the screws holding the lamp to the mount, and never attempt to move the lamp up or down in the mount. Doing so will degrade the system performance. The lamp is provided pre-aligned in the mount.

To install the tungsten lamp:

Slide the lamp mount onto the alignment dowels located above and below the aperture (the mount has pre-drilled holes to accommodate the dowels). The lamp leads should emerge from the right and slightly below the lamp. Use the thumbscrew and aluminum standoff to attach the lamp mount to the case. Connect the lamp lead to the lower of the two terminals in the lamp compartment.

Replace the lamp compartment cover and fasten with the screw.

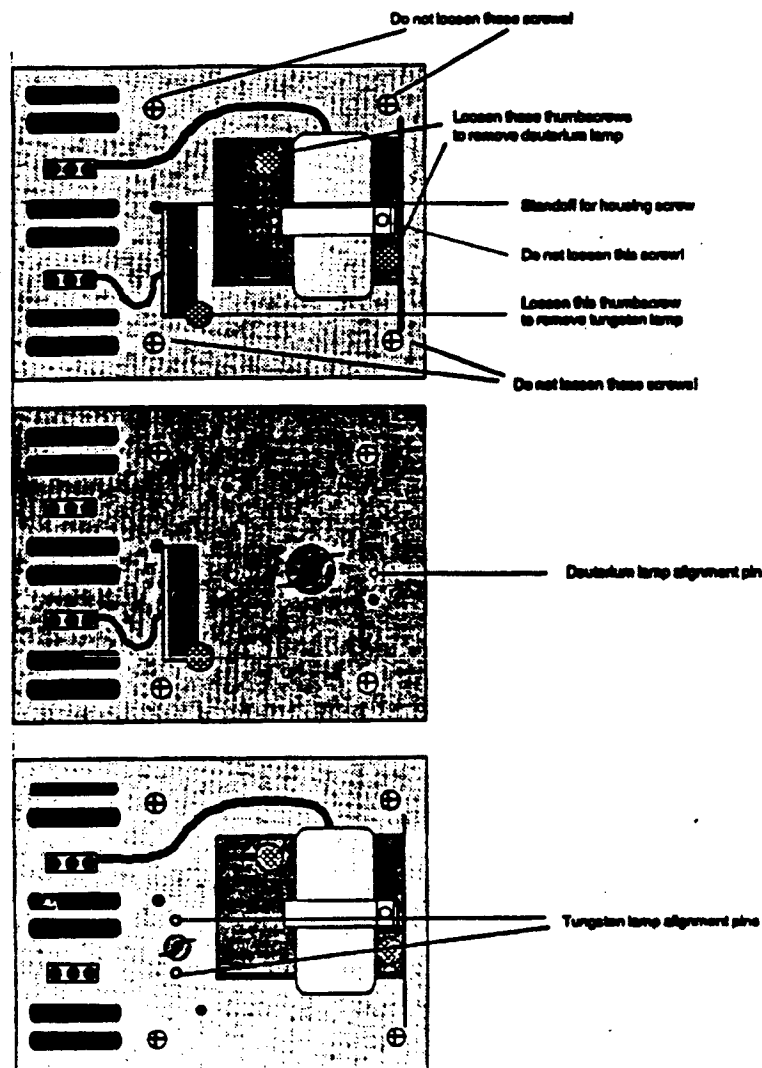


Fig. 5.3.

5.3 Cleaning and Routine Service

All of the stainless steel cells (except for the preparative cell) are constructed similarly (figure 5.4).

The following procedure should be used to disassemble and clean the stainless steel cells:

- Remove the cell and holder from the instrument as previously described.
- Remove the cell from the holder by unscrewing the two screws holding the cell in place and pulling the cell straight back out of the holder.
- If the tubing inlet to the cell is clogged, the tubing may be removed by unscrewing the heat exchanger retaining screw and its washer. The heat exchanger tubing may be loosened from the groove by gently pulling. The zero dead volume inlet fitting may then be unscrewed. If the tubing is not clogged, it should not be removed.
- To remove the windows, gently unscrew the cell nut with a large flat blade screwdriver.
- Carefully note the orientation of the spring washer. The "dished" side should face away from the window. It should be replaced in the same orientation.
- Remove the window gently pulling it upwards. Be careful not to scratch the gasket. Note the orientation of the gasket in the cell. The gasket should be placed such that the oval hole in the gasket exposes both the illuminated path hole (thruhole) and the small feed hole.
- The cell parts may be cleaned with a detergent solution in an ultrasonic bath for 20 minutes, followed by a thorough rinsing with deionized water.
- To reassemble the cell, reverse the disassembly procedure. Be very careful about the orientation of the cell window gaskets (the oval hole in the gasket should expose both the illuminated path hole and the small feed hole) and about the orientation of the spring washer above the window (the "dished" side should face away from the window). The cell nut should be tightened to 15 inch-pounds of torque.
- For cleaning and routine service procedures for Kel-F and adjustable pathlength preparative flow cells, refer to the owner's manuals for the respective cells.

Note: If you removed the heat exchanger inlet tube during the disassembly process, you should coat it with grease during reassembly to provide good thermal contact with the cell body.

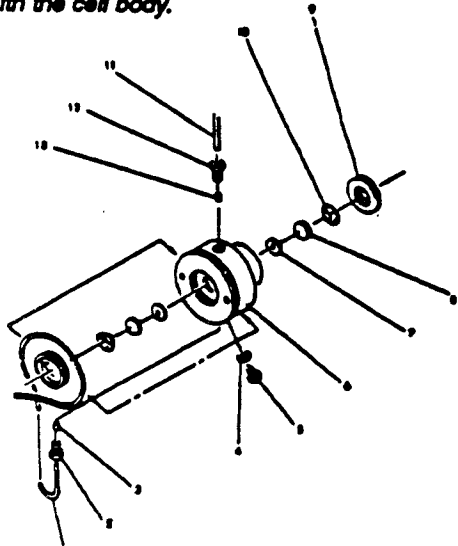
- 
1. Stainless Steel Heat Exchanger Inlet Tube
 2. 1/16" Male Tubing Nut
 3. 1/16" Stainless Steel Ferrule
 4. Heat Exchanger Retaining Washer
 5. Heat Exchange Retaining Screw
 6. Cell Body
 7. Cell Gasket
 8. Cell Window
 9. Cell Nut
 10. Spring Washer

Fig. 5.4

Section 6

Troubleshooting and Service

Contents

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6.4	General Problems Flow Chart for Troubleshooting	78

Many problems associated with HPLC detectors are caused by other parts of the chromatographic system. Noisy and drifting baselines, poor reproducibility in quantitative analysis, and similar problems are most often caused by dissolved air bubbles, contaminated eluants, dirty samples, or damaged columns rather than by the detector itself.

6.1 Baseline Problems

A major mistake can be made when using Full Spectral Autozero. This can manifest itself by baseline noise and offsets of spectra. The full spectral autozero reads and zeros only those wavelengths which are part of the currently defined spectrum. Intermediate wavelengths are neither read nor zeroed.

For example, a spectrum defined as 254 to 350 nm in 5 nm steps would consist of the wavelengths 254, 259, 264, . . . nm. When this spectrum is zeroed, wavelengths such as 255, 256, and 257 are not part of the spectrum. If the spectrum is changed and you do not rezero it, it is possible for the new wavelength values to have no corresponding zero values.

The result can be a noisy baseline. The way to avoid this problem is to **ALWAYS RUN A FULL SPECTRAL AUTOZERO AFTER DEFINING A NEW SPECTRUM.**

Another possible baseline problem is a completely flat or nonresponsive baseline. Possible causes include:

- **RECORDER NOT CONNECTED**
(Ensure wires are connected and not pinching the wire insulation)
- **RECORDER NOT TURNED ON**
- **OUTPUT DIRECTED TO WRONG RECORDER CHANNEL**
- **RECORDER RANGE SWITCHES IMPROPERLY SET**
(Only one switch per channel should be on)

6.2 Error Messages

Error messages are given as flashing characters on the bottom line of the display. Most such error messages result from attempts to enter data outside the acceptable range of the instrument. To clear the error message, press any key, and enter the correct value. Other Important error messages are described in Section 7.9

Error messages include:

"CANNOT ENTER STAND-BY WHEN ACTIVE": It is not possible to enter the stand-by mode while running a method. To engage stand-by mode, first press STOP CHROMATOGRAM then press the STAND-BY key twice.

"CANNOT ENTER PB MENU DURING SPECTRA": The detector will not allow the playback mode to be engaged while collecting spectra.

"CANNOT INS/DEL DURING CHROMATOGRAM": The detector will not allow the insertion or deletion of method parameters (e.g. a new wavelength etc.) while a method is running

"CANNOT FIND D2 LINE": This indicates a problem with the grating being out of calibration. For a corrective procedure see LOW LIGHT FROM DEUTERIUM LAMP.

"CANNOT MODIFY COMPLETED CHROMATOGRAM": The detector will not allow modification of wavelength values when in the playback mode.

"CANNOT STORE/RECALL WHILE ACTIVE": The detector is unable to recall data while a method is being run. Wait until the method is over before attempting to store or recall data.

"CANNOT START DURING SPECTRA": The detector will not allow the starting of a chromatographic run while a spectral run is in progress.

"CANNOT CALIBRATE IN DYNAMIC MODES": It is not possible to calibrate the detector when running a method. You must first press stop chromatogram.

"DUAL LAMP MULTI NOT POSSIBLE": It is not possible to scan between the Deuterium lamp (190 to 365 nm) and the Tungsten lamp (366 to 800 nm) while running a method.

"D2 LINE TO FAR FROM EXPECTED VALUE": This indicates a problem with the grating being out of calibration. For a corrective procedure see **LOW LIGHT FROM DEUTERIUM LAMP**.

"GRATING ACTUATOR ERROR": If the grating mechanism becomes out of sync, this error appears followed by **"GRATING ERROR RECOVERY"**. This is a self correcting problem. If this message occurs frequently, contact your factory representative.

"GRATING CONTROLLER ERROR": If this message occurs, turn the detector's power off and on again. The message should not recur. If it does, contact your factory representative.

"INVALID WAVELENGTH": An attempt to enter a wavelength below 180 nm or above 800 nm.

"INVALID RANGE ENTERED": An attempt to enter a range of zero (0) or above 7.9995.

"INVALID CHANNEL NUMBER": An attempt to enter a recorder number below 1 or above 4.

"INVALID RISE TIME": An attempt to enter a rise time greater than 9.9 second.

"INVALID PLAYBACK RATE": An attempt to enter a playback rate slower than real-time or faster than an allowed time factor of twenty.

"INVALID METHOD NUMBER": An attempt to enter a method number greater than four.

"INVALID SPECTRA TIME": You have entered a time value greater than allowed.

"INVALID TIME": You have entered a time value greater than allowed.

"LOW LIGHT FROM DEUTERIUM LAMP": Either the deuterium lamp has failed to light on the second try or else the lamp has lit, but the output is below acceptable limits. The reference photodiode looks for key atomic wavelines and intensities. If these are not adequate the first time the lamp is lit, the detector will relight the lamp. If on the second attempt the lamp fails, this message is displayed.

If this occurs you should switch the power off for five seconds then power-up the detector again. If the error message recurs, disconnect the power cord, open the lamp housing (refer to "Section 5" on service) and check that the deuterium lamp lead is connected to the upper of the two receptacles. If the lead is properly connected, replace the deuterium lamp assembly.

"LOW LIGHT FROM TUNGSTEN LAMP": Follow procedure for Deuterium lamp.

"MULTI MEMORY IS FULL": No further data can be stored in the detector's memory. One solution would be to rerun the chromatogram with the memory extender feature. To clear the memory, either start a new run or turn off the power.

"NO SPECTRA IN MEMORY": An attempt to playback a spectrum when one is not in the detector's memory.

"NO CHROMATOGRAM IN MEMORY": An attempt to playback a chromatogram when one is not in the detector's memory.

"NO DATA STORED IN THIS METHOD": When powering-up the detector with a key pressed (e.g. when entering auto-calibration) all methods in storage are deleted. If an attempt is made to recall a method after resetting the detector an error message will occur.

"PRESS KEY AGAIN TO ENTER STAND-BY": It is necessary to press the stand-by key twice to engage this function. If you do not wish to enter stand-by mode, then press any key but the stand-by key. This should return you to the main menu.

"PROGRAM MEMORY FULL": The number of wavelengths entered in a method exceeds the number allowed.

"PROCESSOR OVERLOAD": This results from attempting a faster playback rate than the detector can handle.

"PROCESSOR OVERLOAD: BUFFER": See "processor overload" above.

"PLAYBACK DRAM INCONSISTENCY": This occurs with a memory failure (i.e. the data stored in memory is unrecoverable). Contact your service representative.

"RUN AUTO-CALIBRATION": This message occurs after resetting the detector (turning power on while holding down any key). This message will appear every thirty seconds to remind you to engage the auto-calibration function after resetting the detector. To do this toggle to the auxiliary window and press ENTER. Toggle to the auto-calibrate function and press ENTER. This message also occurs if for some reason certain test information stored in memory is not correct.

"SCAN ERROR: AMBIGUOUS PEAK": See "cannot find D2 line" above.

"TURN TUNGSTEN LAMP ON": Toggle the highlight to the auxiliary screen, press ENTER, then scroll to the tungsten lamp on/off box and press ENTER.

"TURN DEUTERIUM LAMP ON": See above.

6.3 Checking Sample and Reference Light Intensities

A noisy baseline may be the result of a weak lamp or dirty flow cell. The detector provides a sample and reference light menu for on-board diagnostics. To reach this menu:

From the MAIN menu, ensure that the cursor is in the top line (wavelength) position. Push the arrow right key once and then the ENTER key. The LIGHT LEVEL menu should appear.

□ RETURN	06-X X26X13X 5	262C99A8
0041A9E6	0042C45B	004219A3
004170AB		
1690	1818	1840
1670		
267786	269778	267777
269222		
1691	1820	1840
1671		
267788	269888	267898
269323		
1692	1820	1841
1672		
267789	269878	267888
269343		

REFERENCE SAMPLE REFERENCE

This menu contains four vertical rows of numbers. The number immediately following the return command is your current firmware version. For diagnostic purposes, we are interested in the two leftmost

rows. The left-hand row represents the relative light intensity from the reference photodiode. The next-to-leftmost row represents the relative light intensity from the sample photodiode. You should use only the four-digit numbers.

In order to diagnose the condition of the flow cell and both lamps, this menu should be invoked at both 254 nm (deuterium lamp) and 546 nm (tungsten lamp). For the standard stainless steel, 6 mm, 4 ul HPLC flow cell, the following rules apply:

- At 254 nm:
 - Reference numbers should be greater than 1100
 - Ratio of sample to reference numbers should be equal to or greater than 0.4
- At 546 nm:
 - Reference numbers should exceed 1200
 - Ratio of sample to reference numbers should exceed or match 0.6

The reference numbers should be the same regardless of flow cell. The sample/reference ratios for the microbore, capillary, Kel-F and preparative flow cells will be lower.

In general, a dirty flow cell will yield acceptable reference levels and poor sample to reference ratios (be careful, however, excessive solvent absorbance can generate the same problems). A weak lamp will yield a poor reference level with acceptable or near-acceptable sample to reference ratio.

To exit from the diagnostic menu, move the cursor to the RETURN position and press ENTER.

6.4 General Problems

Remaining noise and drift problems can be due to various sources within and external to the system. The following table provides a checklist of symptoms, possible causes, and some remedies.

<i>Symptom</i>	<i>Cause</i>	<i>Remedy</i>
1. Spikes on recorder baseline	a. Bubbles passing through cell	a. Degas solvent and/or supply back pressure to the sample cell, also check all high pressure fittings for leaks (both liquid and gases)
	b. External event marker is picking electrical noise	b. Check electrical lines for good connection and/or interference from broadcast radiation
	c. Extremely large supply voltage transients on the AC line	c. Remove systems that consume high power from the AC line
2. Noisy baseline on recorder	a. Sample cell windows are contaminated	a. Flush cell with solvents (methanol, acetone, water, nitric acid (6N), water, methanol) and check for leaks
	b. Sample input line has a leak	b. Check all lines from the output of the sample to the input of the sample cell for leaks
	c. Bubble trapped in the sample cell	c. Increase flow rate and/or back pressure on cell
	d. Recorder or integrator is grounded and is causing a "ground loop" problem	d. Check recorder with voltmeter to see if either of the signal inputs is grounded to case or check in the owners manual of the recording instrument
	e. Photodiode is dirty or not held down properly to the cell holder	e. Remove and clean photodiode
	f. Sample cell is not screwed down to the main unit	f. Check sample cell mounts and cell holder assembly
	g. Output span of the detector does not match input range of recorder or integrator	g. Press event mark to see if the spike is approximately 20% of full scale
	h. Triggering device for remote event mark or auto zero is causing a ground loop problem	h. Use only triggering device with ground isolated from earth ground

Symptom	Cause	Remedy
3. Baseline drifts excessively	a. Unit is not warmed up	a. Allow 1 hour warm up for optimum performance
	b. Contamination of sample cell window has occurred	b. Clean cell by flushing with solvents (methanol, acetone, water, nitric acid (5N), water, methanol). Inspect cell and photodiode for fingerprints and smudges and clean if necessary
	c. Solvent from column is changing absorption	c. Column is filled with UV absorbers that are bleeding - replace column, impure solvent is equilibrating with the column - replace solvent with more pure grade, switch to a longer wavelength so that background absorption does not fluctuate as much
	d. Leakage in the lines from sample to flow cell	d. Check lines for leakage
	e. Tiny bubble trapped in the sample cell	e. Increase flow rate and/or back-pressure
	f. Output span of detector does not match input span of recorder or integrator	f. Press event mark to see if a spike 20% of full scale occurs
	g. Large temperature fluctuations are occurring	g. Remove detector from the source of drafts of hot and cold air
	h. Flow cell, photodiode assembly, or flow cell cover is loose	h. Tighten thumbscrews fastening flow cell holder, photodiode assembly, and flow cell cover
4. Detector's display panel is not readable	a. Display contrast potentiometer is not set correctly	a. Reach around to the rear panel and turn pot till display is readable (Figure 2.2 #9)
	b. Fuse is blown	b. Check voltage selector card on power inlet module, then check fuse if 7979 power voltage input is correct.

Section 7

Theory of Operation

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7.1 Theory

Figure 7.1 illustrates the optical system for the detector. The light sources are continuum lamps that jointly provide consistent intensity across the entire spectrum from 190 to 800 nm. Two baffles are used to minimize stray light. Wavelength selection is provided by a concave holographic grating actuated by a 6803 microprocessor-controlled servo system.

True double beam operation is provided by a fiber optic beam splitter. A reference photodiode continuously monitors the light from one leg of the beam splitter. The other leg is imaged by a lens through the sample cell onto the sample photodiode.

The photodiodes are connected to individual preamplifiers (see figure 7.2). The preamp output voltages are directly proportional to the light intensity at the photodiode active surfaces. The output of the preamplifiers are sent to an analog-to-digital (A/D) converter.

The A/D converter provides a 19 bit resolution digital word. Depending on the mode of operation, 24 or 96 readings are taken per second. The digitized data is processed by the 6809 microprocessor where all filtering and ranging is carried out. Filtered but nonranged data is transferred out the integrator output using a sliding scale, 14 bit, digital to analog (D/A) converter yielding an effective resolution of 20 bits. Filtered and ranged data are transmitted out of the recorder outputs using a 14 bit D/A.

Sense lines from the 6809 microprocessor to the autozero and event marker connections provide flexibility for remote and/or automatic operation. The key pad on the front panel allows direct operator control over all instrument functions.

An RS-232C interface allows remote control of all parameters used or controlled by the computer. In addition, status information and either raw or processed data can be transmitted through the interface. Bio-Dimension software controls the instrument by connecting to a personal computer via the RS-232 port. For more information, see the *Software Manual*.

7.2 Spectra and Scanning

For discrete spectra and HIGH SPEED POLYCHROME OPERATION, the instrument sums the absorbancies at all data points between defined step wavelengths, averages them, and outputs the resultant data at the defined point. For example, consider a scan defined to run from 195 nm to 360 nm at 5 nm steps. Defined data points would be established for 195, 200, 205, 210 nm, etc. However, the instrument scans from 193, 194, 195, 196, and 197 nm, averages the total absorbance across that range, and assigns the resultant figure as the absorbance at 195 nm. From this it can be seen the user must specify upper and lower wavelength values that are inside the respective 190 to 365 and 366 to 800 nm boundaries by a factor at least as large as the step wavelength value divided by 2. If he or she fails to do so, the monitor will alter the upper, lower, or step wavelength value so that it complies with this requirement.

7.3 The Repetition Parameter for Spectral Scans

The define spectra sections of the MAIN and the SCHEDULE SPECTRA menus contain provisions for entering a scanning repetition value. This parameter determines the number of consecutive scans that will be performed after initiation of the TAKE SPECTRA command. Data derived from these scans are then averaged and stored in the monitor's memory for later playback.

Averaging spectral scans together improves the resultant signal-to-noise ratio by a factor equivalent to the square root of the number of averaged scans. A good rule of thumb is to select a REP value which allows all scans to be performed at the upper portion of a given peak. Spectral scans should be initiated just prior to the peak's apex. In this case, scans may be performed near the apex on the up shoulder, at the apex, and near the apex on the down shoulder. Valid entries for the REP field are 1, 2, or 4 scans.

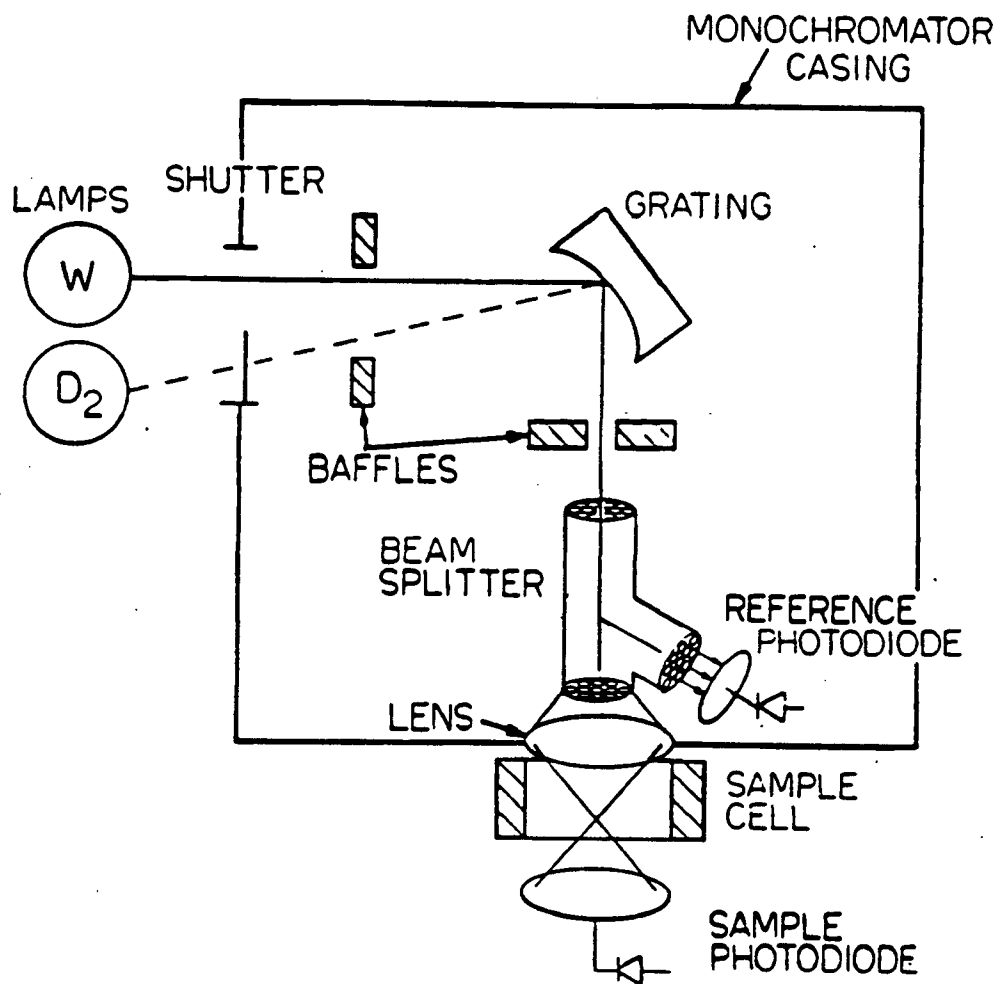


Fig. 7.1. The detector optical sytem.

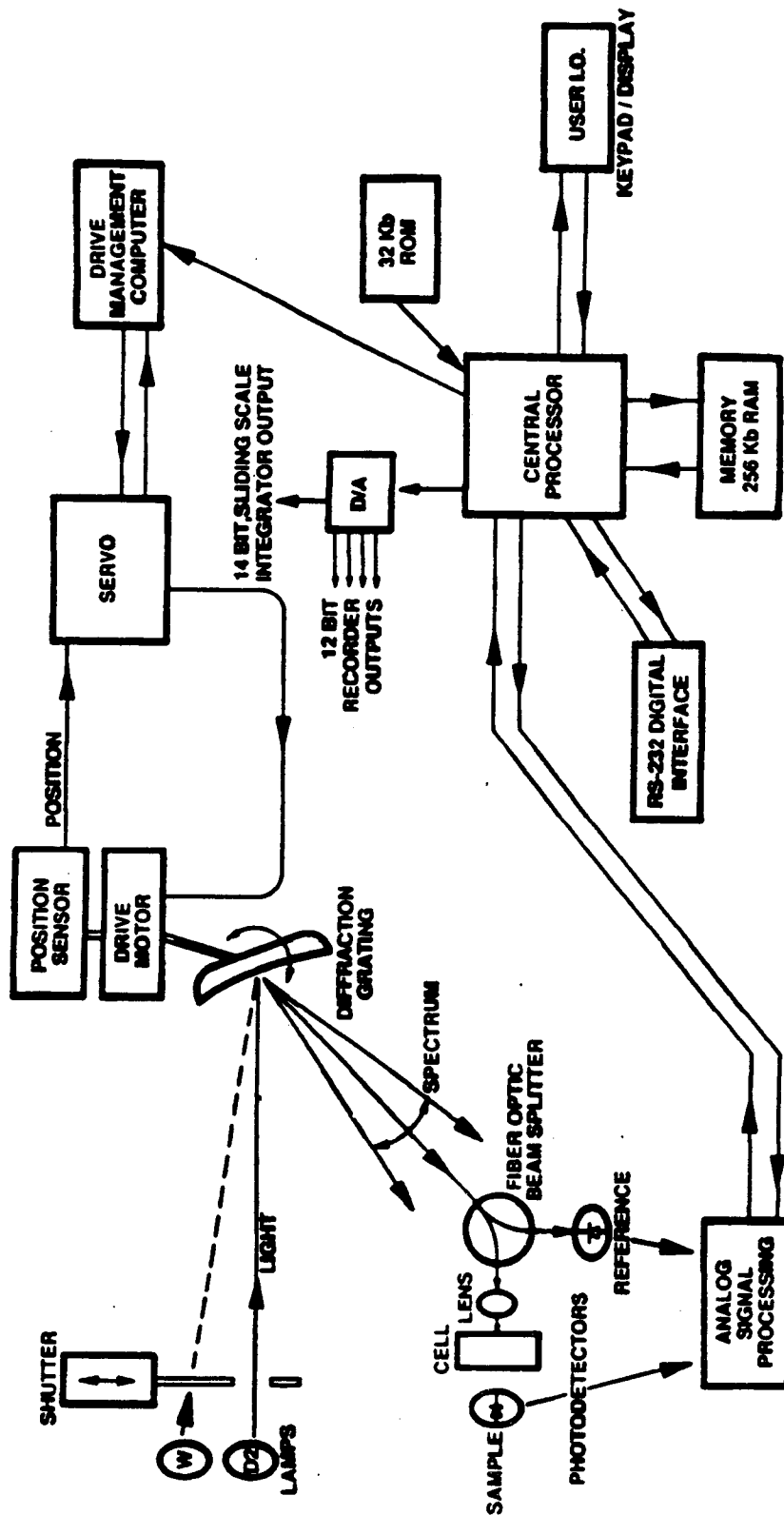


Fig. 7.2.

7.4 Ratio Function

Ratioing the absorbancies at two particular wavelengths may be used as a means of determining the purity of a given peak. The ratio value at any given time is the quotient of the molar absorptivities of the peak at the two ratioed wavelengths. If the peak is pure, that is, if no compound coelution occurs, the ratio should remain constant during the peak's elution period and the resultant ratio trace would be a square wave. A changing ratio during a peak's elution period is indicative of solute coelution, and the peak is impure. Ratioing will only detect compound coelution if the impurity absorbs at both of the ratioed wavelengths. Additionally, if both the impurity and solute of interest display identical elution profiles (equal ascending and descending slopes resulting in an identical increase and decrease in concentration as a function of time), the measured ratio becomes the sum of the molar absorptivities of the solute and impurity, and no ratio change will be seen during the peak's elution period.

7.5 The Detector's Ratio Algorithm

Calculated ratios are NOT the ratios of the absolute absorbancies for the two ratioed wavelengths, but rather the ratios of their absorbancies as a percentage of full scale. Thus, the calculated numerical ratio is:

$$\text{NUMERICAL RATIO} = \frac{\text{ABSORBANCE WAVELENGTH 1} / \text{RANGE WAVELENGTH 1}}{\text{ABSORBANCE WAVELENGTH 2} / \text{RANGE WAVELENGTH 2}}$$

Since it is of no value to ratio baseline absorbance data, the ratio calculation is only performed when BOTH wavelength absorbancies exceed 3% of full scale. Ratios are no longer calculated when these values fall below a full scale level of 2%. The chart recorder ranges for the wavelengths that are being ratioed should be set so that the peaks of interest fall between 20% and 100% of full scale at both wavelengths.

The NUMERICAL RATIO which is displayed on the detector's LCD is not the ratio transmitted out of the RECORDER OUTPUT. The recorder output transmits a RATIO PLOT which is the log of the NUMERICAL RATIO. By calculating the NUMERICAL RATIO as a percent of full scale and tracing it as a log function, the overall RATIO PLOT is easily displayed on a strip chart recorder, eliminating a ratio full scale setting. If the recorder's pen is positioned at 50% full scale for a ratio of 1.0, then 100% full scale indicates a ratio of 10.0 and 0% full scale corresponds to a 0.1 ratio. The NUMERICAL RATIO may be calculated from the RATIO PLOT by using the following equation:

$$\text{Numerical Ratio} = 10^{\left(\frac{X-50}{50} \right)}$$

where X is the full scale deflection of the recorder pen.

Some Examples:

During the elution of a given peak, the RATIO PLOT moved from 50% full scale to 30% full scale. The corresponding NUMERICAL RATIO (NR) is:

$$\text{Numerical Ratio} = 10^{\left(\frac{30-50}{50} \right)} = 10^{-2/5} = 0.398$$

During the elution of another peak, the RATIO PLOT moved from 50% full scale to 75% full scale. The corresponding NUMERICAL RATIO (NR) is:

$$\text{Numerical Ratio} = 10^{\left(\frac{75-50}{50} \right)} = 10^{1/2} = 3.162$$

7.6 A Detailed Explanation of Memory Extender

The Memory Extender function located on the Auxiliary menu allows the user to increase the detector's storage capacity for absorbance vs. time data in order to allow long chromatograms to be recorded.

The total amount of storage time for a chromatogram is approximately 60 minutes at a data rate of 24 points/second/wavelength. After 60 minutes or so of operating in the Programmed Operation Mode, the detector will display "CMF" (chromatogram memory full) next to the elapsed time. Although subsequent data are still output to the recorder in real-time, they are not stored. **ONLY THE FIRST 60 MINUTES ARE STORED FOR SUBSEQUENT PLAYBACK.**

The Memory Extender functions by altering the data storage rate. The default value for the Memory Extender is 1: the detector stores every data point. When the value of the extender is changed to an integer, n , the detector stores every n th data point. A Memory Extender multiplier value of 5 will store every fifth data point and effectively extend the storage time by a factor of 5.

The Memory Extender function should be used carefully. Too large a value can result in distortion or even non-detection of fast peaks. As a rule of thumb, accurate measurement of a peak profile requires 20 data points. Thus, the maximum tolerable Memory Extender value is related to the width of the narrowest peak.

The Memory Extender function also affects the Rise Time and speed of the chromatogram playback. The detector stores all data in "raw" form: unfiltered and nonranged. The filtering and playback algorithms are based on a fixed number of data points. Thus, a Rise Time of 1 second reflects data gathered for 24 points. If the Memory extender value is set at 2, the same rise time would reflect 24 points gathered over a 2 second real-time interval. Thus, the actual rise time is the product of the playback rise time and memory extender value.

Similarly, the playback rate is based on playback of a certain number of data points per second. If the data points were originally stored over a longer time, the result will be a proportional increase in the playback rate. **THE ACTUAL PLAYBACK RATE IS THE PRODUCT OF THE MEMORY EXTENDER MULTIPLIER AND THE NOMINAL PLAYBACK RATE.**

7.7 A Detailed Explanation of Auto-Spectra

The Auto-Spectra function provides the user with the ability to automatically acquire up to three discrete spectra for a given peak. These spectra may be compared to determine peak purity. The Auto-Spectra program functions by monitoring the output of strip chart channel (1) at intervals equivalent to the designated rise time. Thus, it should be noted that the wavelength, rise time, and range values selected for channel (1) affect auto-spectra.

There are two, user-selectable detection schemes for Auto-Spectra: Early and Late Detection. In the Early Scheme peaks are detected for:

The Up-Shoulder: When the second derivative value for the D to A changes decreases to less than $1/2$ of the maximum achieved value.

The Peak Apex: When the first derivative decreases by less than $1/4$ of the maximum value.

The Down-Shoulder: When the absorbance value is less than the first value following the triggering of the Up-Shoulder spectra.

For the Late Detection Scheme, peaks are detected for:

The Up-Shoulder: When the first derivative falls to $7/8$ ths of the maximum attained value.

The Peak Apex: Performed immediately after the Up-Shoulder spectra.

The Down-Shoulder: When the absorbance value is less than the first value following the triggering of the Up-Shoulder spectra.

The user may choose to gather one, two, three spectra for each peak.

7.8 The Auto-Calibration Function

The monitor is equipped with an AUTO-CALIBRATION program designed to insure proper wavelength accuracy. When triggered, this program searches for discrete lines of the deuterium lamp in the visible region. These lines are monitored using the reference photodiode to avoid any deleterious effects stemming from the flow cell's contents. The calibration parameters are stored in the nonvolatile memory of the detector and will remain after the detector has been disconnected from power. See Section 3.18.

It is recommended that the user allow the detector a 60 minute warm-up before activating the auto-calibration function.

7.9 Important Error Messages

If the user should enter invalid parameters on any of the detector's menus, an ERROR MESSAGE will appear on the bottom line. Each message informs the user of his error, and they may be erased by pressing any keyboard key. Most of these messages are self explanatory and do not warrant further discussion. However, two possible error messages need be addressed:

"Non-Volatile Ram Error": See Manual - This message indicates that the battery-backed up random access memory (RAM) has failed. If this occurs, turn off the monitor and reset the RAM by powering up the detector while holding down any of the front panel keys. A RAM reset is indicated by a 256 nm displayed on the MAIN MENU. After a 30 to 60 minute warm-up, activating the AUTO-CALIBRATION function is recommended.

"Autocal Value Invalid": See Manual - This message indicates that the autocalibration function did not find the proper Deuterium spectral lines. Repeat the Auto-Calibration procedure.

Continued autocalibration failure may be due to a bad deuterium lamp. See the CHECKING SAMPLE AND REFERENCE LIGHT INTENSITIES portion of the troubleshooting section for further details.

Appendix A

Options & Accessories

Bio-Dimension Ordering Information

167-0940	Bio-Dimension UV/Vis Monitor, with S.S Flow Cell
167-0941	Bio-Dimension UV/Vis Monitor, with Kel-F® Flow Cell
167-0944	Bio-Dimension Software*
167-0905	Deuterium Lamp Assembly, for the Bio-Dimension monitor
167-0906	Tungsten Lamp Assembly, for the Bio-Dimension monitor
167-0910	Stainless Steel Analytical Flow Cell, for the Bio-Dimension monitor
167-0915	Kel-F® Analytical Flow Cell, for the Bio-Dimension monitor
167-0912	Variable Pathlength Flow Cell, for the Bio-Dimension monitor
167-0916	Sapphire Windows, for the Bio-Dimension monitor
167-0925	RS-232 Cable, 25 pin male-female

* Requires Microsoft Windows to operate.

Replacements Parts

167-0905	Deuterium Lamp Assembly, prealigned
167-0906	Tungsten Lamp Assembly
167-0911	Sapphire Windows, for Analytical Stainless Steel Flow Cells
167-0916	Sapphire Windows, for Analytical Kel-F Flow Cell
100-4500	Analytical Flow Cell Rebuild kit, contains 2 windows and 2 gaskets
100-4510	Preparative Flow Cell Rebuild kit, contains 2 windows and 2 gaskets

Appendix B Specifications

Electrical Specifications

Wavelength Range	190 - 800 nm
Bandwidth	6.0 nm
Wavelength Accuracy	+/- 1.0 nm
Wavelength Precision	0.1 nm
Slew Time 190 - 800 nm	20 milliseconds
Optical methodology	Deuterium(190-365 nm) and tungsten (366-800 nm) lamp light sources with galvanometer-driven concave holographic grating monochromator and double beam optics, provision for purging the monochromator
Range Selections	0.0005 to 7.9995 AUFS, in 0.0005 increments on any of the chart recorder outputs
Absorbance Ratio	Displayed as ratio of absorbances (at the two selected wavelengths) per full scale range; plotted as log of ratio of absorbances per full scale range
Recorder Outputs	10 mV, 100 mV, 1.0 V output in any combination of the four recorder outputs, 12 bit resolution
Integrator Output	1 volt/AU; wavelength selectable in multiple wavelength mode, 19 bit sliding scale resolution
Remote Controls	Rear panel input for autozero of chart recorder and integrator outputs; rear panel input for event mark; RS-232C serial interface for computer control and/or data acquisition; DPDT relay closure for optional stop flow device and other controlled devices
Single Wavelength Noise	$< 2 \times 10^{-5}$ AU/60 sec from 220-280 nm with 1.0 sec. rise time
High Sensitivity Multiple Wavelength Operation	
Two wavelengths from 220- 330 nm, 400-800 nm	$< 2 \times 10^{-5}$ AU/60 sec with 1.0 sec. rise time

Four wavelengths 220-330 nm, 400-800 nm	$< 3 \times 10^{-5}$ AU/60 sec with 1.0 sec rise time
Data acquisition Rate	24 points/seconds/wavelength
High Speed Multiple Wavelength Operation	
Typical Noise with 2.0 sec. rise time for a scan from 195 to 800 nm	195-210 nm $< 8 \times 10^{-5}$ AU 210-300 nm $< 3 \times 10^{-5}$ AU 300-355 nm $< 4 \times 10^{-5}$ AU 355-430 nm $< 8 \times 10^{-5}$ AU 430-600 nm $< 4 \times 10^{-5}$ AU 600-800 nm $< 3 \times 10^{-5}$ AU
Data Acquisition Rate	96 points/second/wavelength
Typical Noise of Discrete Spectra	$< 5 \times 10^{-5}$ AU/nm
Typical Noise of Time Slice Spectra	Dependent upon noise of Multiple Wavelength data -4
Drift	$< 2 \times 10^{-5}$ AU/hour after 30 minute warm-up
Zero Adjust	Plus or minus any amount by auto-zero and manual entry
Chart Recorder Filter	Digital active filter with rise times selectable from 0.0 to 9.9 seconds (divide by 2 to convert to an equivalent time constant)

Appendix C Standard Flow Cell Information

Introduction

This appendix describes the proper use of, and maintenance procedures for, the flow cell used in the Bio-Dimension detector: 167-0910 6 mm pathlength, 9 μ l volume, stainless steel analytical flow cell. Section 5.3 has more information on the flow cell.

Unpacking

Carefully unpack the flow cell from the shipping container and inspect both the cell and packing for any signs of damage. If any signs of damage are noted, contact the shipping company immediately.

If you have ordered this flow cell with a detector, it is already mounted and you need not read the next section.

Installation

1. Disconnect the inlet and outlet tubing of the flow cell presently mounted on your detector from the chromatographic column, waste reservoir, or other devices.
2. Remove the flow cell housing cover by unscrewing its retaining thumbscrew and pulling the cover straight back (see Figure C.1).
3. Free the inlet and outlet tubes from the tubing restraint by loosening the thumbscrew and freeing the tubes from under the clamp.
4. Remove the photodiode assembly by removing its two thumbscrews. Avoid scratching or putting fingerprints on the optical surface of the photodiode assembly. The photodiode assembly should be cleaned with spectroscopic grade methanol and lint-free lens paper.

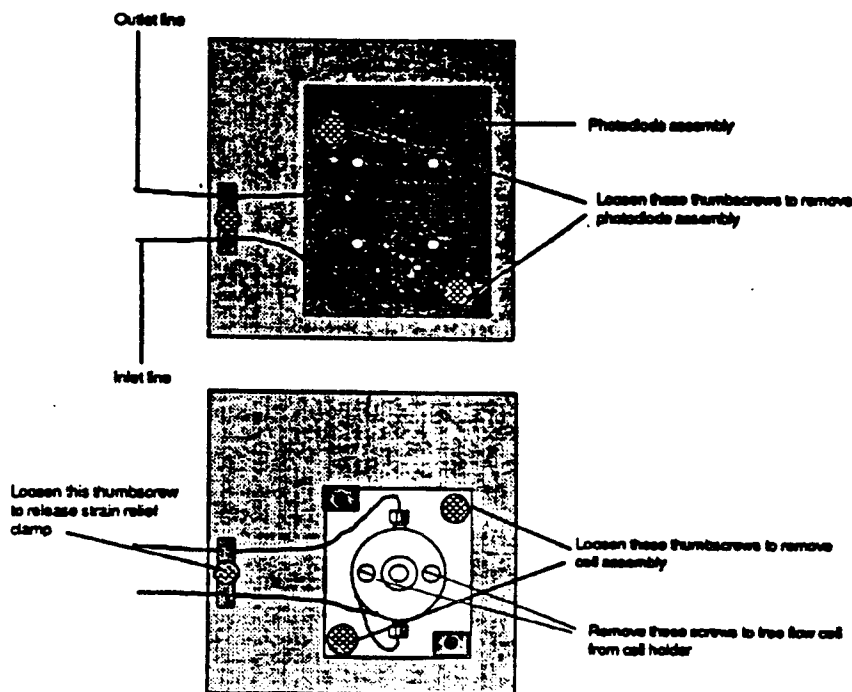


Fig. C.1.

5. Remove the two thumbscrews securing the flow cell assembly to the detector and pull the flow cell assembly straight back along the alignment dowels until it is free from the optical bench.
6. Orient the new flow cell assembly so that the dowel rods, mounting screw holes, and tubing are in the proper position. Mount the cell so the inlet tube (wrapped around the flow cell body) is the lower of the two tubes. Avoid scratching or placing fingerprints on the flow cell windows. These windows should be cleaned with spectroscopic grade methanol and line-free lens paper. They may be dusted by blowing off debris with filtered, oil-free air.
7. Secure the flow cell assembly to the optical bench using the two thumbscrews.
8. Lock the inlet and outlet tubing into place by tightening them down with the tubing restraint.
9. Replace the photodiode assembly.
10. Replace the flow cell housing cover by replacing its retaining screw.
11. Connect the flow cell inlet and outlet tubing to the appropriate components.

Flow Cell Operation

A. Applications

The performance characteristics of the flow cell are dependent on its application. Flow cell Model 167-0910 is intended for use in analytical HPLC applications. Its 6 mm pathlength yields superior sensitivity, while its 9 ml volume maintains excellent peak shape. This cell is designed to be used with 1/16" O.D. x 0.010" ID stainless steel tubing.

B. Static Pressure Ratings and Flow-Pressure Characteristics

The flow cell is rated up to 1,000 PSI nominal static pressure. Pressures exceeding 1,000 PSI may result in cell leakage. If leakage occurs, the cell gaskets should be replaced (See the "Flow Cell Maintenance" section for cell disassembly and repair).

Pressure vs. flow rate data for flow cell is presented in figure C.2. This data reflects typical results obtained for the described flow cell with the standard length of appropriate ID stainless steel tubing used for the cell inlet and outlet lines with water as the solvent. To calculate the expected backpressure for a different solvent (P_{solvent}) in your system:

1. Determine the flow rate to be used.
2. Interpolate the pressure drop for water using the graph (PH₂O).
3. Multiply that pressure drop figure by the viscosity of the solvent in question (Cps).

$$P_{\text{solvent}} = (PH_2O) (Cps)$$

Under typical conditions, laminar flow exists in flow cell. In this case, the following relationship holds:

$$\text{Pressure} = (\text{Flow Rate}) (\text{Resistance})$$

The following table is included for your convenience for determining total backpressure for a system consisting of a given length of tubing and given flow cell.

Tubing ID (inches)	Max Flow Rate for Laminar Flow (ml/min.) With Water as Solvent	Resistance/Length (PSI (min) (ml)
0.005 11.54	3.780	
0.010 23.10	0.112	
0.030 69.20	4×10^{-3}	
Flow Cell	Resistance of Flow Cell (less tubing)	(PSI) (min) ml

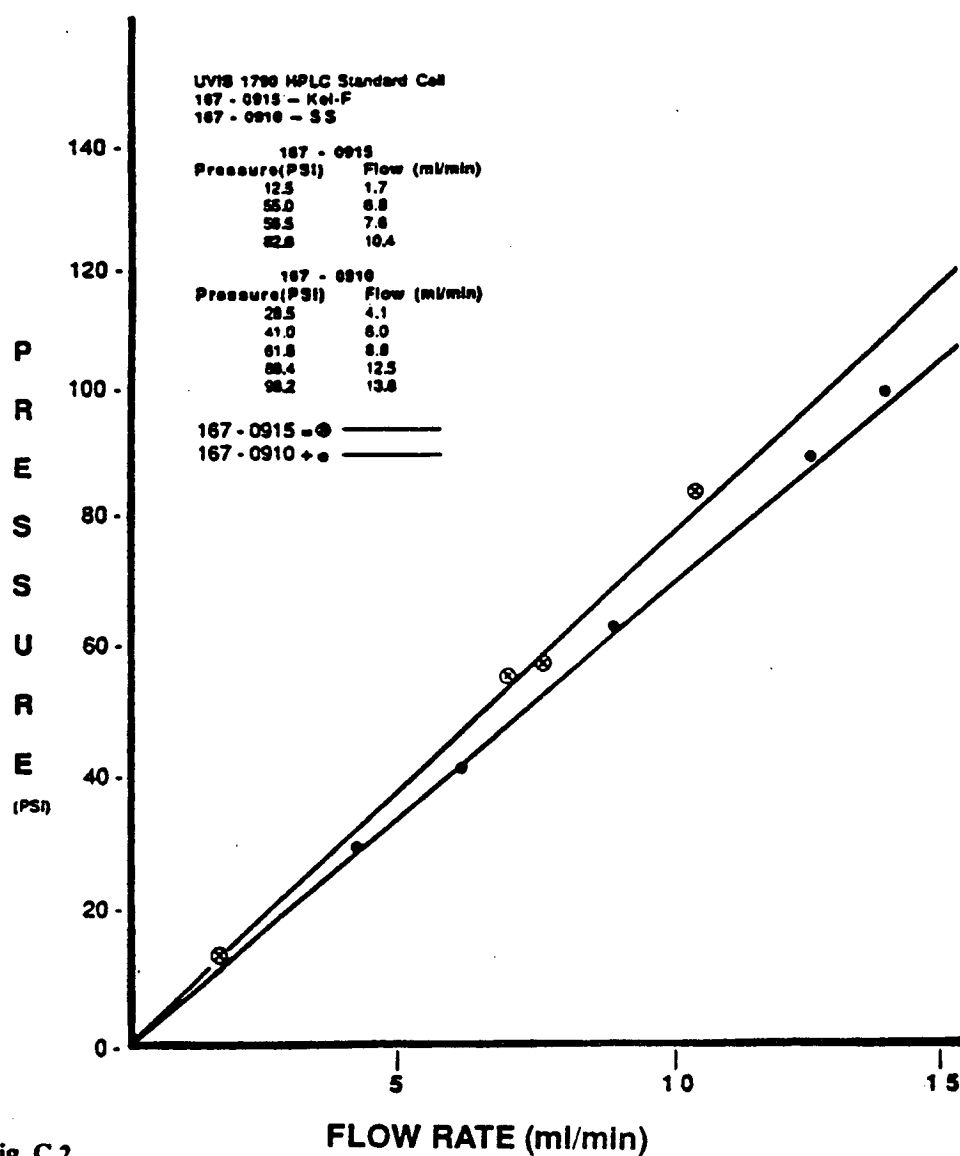


Fig. C.2.

Flow Cell Maintenance

Cell Cleaning

Do not disassemble flow cells for routine cleaning purposes. Most cells can be adequately cleaned by flushing with several milliliters of appropriate solvent. We recommend the following solvents for this purpose.

- Methanol
- Tetrahydrofuran
- Methylene chloride
- HPLC grade water
- 6N nitric acid followed by flushing with HPLC grade water

Cell Disassembly

If flushing proves inadequate for cleaning, or if the flow cell becomes leaky requiring gasket replacement, the following procedure should be followed for flow cell disassembly: (see Figure C.3)

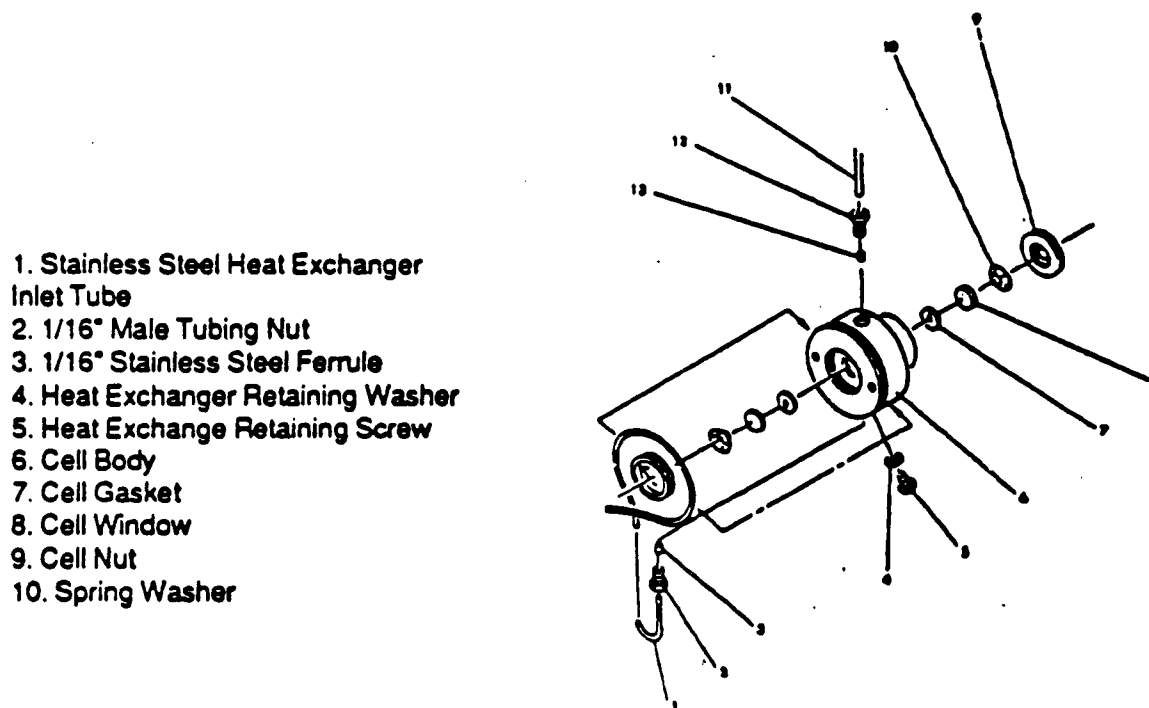


Fig. C.3.

1. Remove the flow cell assembly from the detector as described in "Installation."
2. Remove the two screws which secure the flow cell body to the flow cell mount and free the flow cell from its mount.
3. Using a wide, flat-blade screwdriver, remove the window retaining nut.

4. Remove the retaining washer. Note that it is installed concave-side up.
5. With a pair of fine forceps, gently lift out the flow cell window . Be careful not to scratch these windows. Clean the window using spectroscopic grade methanol and lint-free lens paper.
6. Note the orientation of the tear drop on the flow cell gasket . It should be installed so that the optical bore (the large hole) and fluid bore (the small hole) are not covered. Remove the cell gasket with a pair of fine forceps.
7. Repeat steps 3-6 for the other side of the cell.
8. If the inlet or outlet tubes are clogged, remove them as follows:
 - a. Remove the exit tubing by unscrewing its fitting and pulling the tube straight forward.
 - b. Unscrew the heat exchanger restraint screw and remove the screw and washer.
 - c. Unwind the inlet tubing from around the cell body.
 - d. Remove the inlet tubing by unscrewing its fitting and pulling the tube straight out.
 - e. Remove the remaining heat conductive epoxy out of the heat exchanger groove which circumscribes the cell body.

The flow cell body may be cleaned by soaking it in spectroscopic grade methanol. For best results, an ultrasonic bath should be used.

Flow Cell Reassembly (See Figure C.3)

Note: For best results, replace the flow cell gaskets each time the cell is disassembled.

1. With a pair of fine forceps, replace the cell gasket in its proper orientation so that both the optical bore (large hole) and fluid bore (small hole) are exposed by the tear drop.
 2. Carefully replace the cell window.
 3. Install the retaining washer so that its concave surface faces away from the cell body.
 4. Replace the window retaining nut. Tighten to 14 inch pounds. To avoid possible damage to the windows, do not exceed 14 inch pounds of torque.
 5. Repeat steps 1-4 for the other side of the flow cell.
 6. If the cell tubing was removed, reinstall as follows:
 - a. Replace the cell inlet tubing by replacing its fitting in the inlet hold (the one next to the hold for the heat exchanger restraint).
 - b. Bend the tubing around the cell so that it circumscribes the cell body in the heat exchanger groove before it approaches the hole for the heat exchanger restraint.
 - c. To improve heat exchange between the inlet tubing and flow cell body, a heat conductive epoxy should be applied into the groove before the tubing is installed.
- Note:** We recommend the use of Loctite Thermal Conductive Adhesive (17099) and Activator (17100), Loctite Corp., Newington, CT 06111.
- d. After the epoxy has been applied, secure the tubing in the heat exchanger restraining screw and washer.
 - e. Replace the cell outlet tubing hole.
 9. Reinstall the flow cell body on the flow cell holder using the two mounting screws. Insure that the protruding nose piece of the flow cell is inserted into the cell mount.
 10. Reinstall the flow cell onto the detector as outlined in the installation section of this manual.

Flow Cell Troubleshooting Guide

This section is organized in a symptom, problem, and remedy format.

Symptom	Cause	Solution
1. Noisy baseline	a. Bubble trapped in flow cell	a. Increase flow rate and/or back-pressure.
	b. Exterior surface of flow	b. Clean flow cell window with

	cell window is dirty	spectroscopic grade (S.G.) methanol and lens paper
	c. Photodiode assembly is dirty	c. Clean photodiode assembly with S.G. methanol and lens paper
	d. Flow cell is dirty	d. Flush flow cell with several milliliters of appropriate solvent
		d. Disassemble and clean flow cell
2. Excessive baseline drift	a. Detector is warming up	a. Allow 60 minute warm-up period
	b. Column is equilibrating	b. Allow several column volumes to pass through the system
	c. Tubing is leaking	c. Check for leaks in fittings and tighten
	d. Flow cell is leaking	d. Check flow cell for leaks around windows; check torque on window retaining nut; replace flow cell gaskets
	e. Flow cell windows are dirty	e. Clean exterior of windows with S.G. methanol and lens paper
		e. Flush flow cell with appropriate solvent
		e. Disassemble and clean flow cell
	f. Photodiode assembly dirty	f. Clean photodiode assembly with S.G. methanol and lens paper
3. Excessive backpressure	a. Exit tubing is clogged	a. Replace exit tubing
	b. Entrance tubing is clogged	b. Replace entrance tubing
	c. Flow cell fluid and/or optical bores are clogged	c. Disassemble flow cell and clean in 6N Nitric acid followed by wash in HPLC water

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- Programmable scanning detector in which predefined spectral scans are automatically performed on the upshoulder, apex, and downshoulder of a chromatographic peak. The stored spectra can be replayed and compared to determine peak purity.

Additionally, the diverse selection of superbly designed flow cells allows the monitor to optimally perform single wavelength, multiple wavelength, and scanning applications for the following chromatographic techniques:

Analytical HPLC

Analytical HPLC Requiring Inert Contact Surfaces

Preparative Scale HPLC

Preparative Scale HPLC Requiring Inert Contact Surfaces

For more information regarding these flow cells, please consult the "Options and Accessories" appendix to this manual.

With the addition of optional software, the detector's power is expanded. Used with the software, the Bio-Dimension detector is capable of the following:

- High speed multi-wavelength operation which allows acquisition of continuous spectral data throughout a chromatogram.
- Time Slice operation which permits the display of spectra from multiple wavelength information by slicing the three-dimensional chromatographic data of absorbance vs. time vs. wavelength perpendicularly to the temporal axis. This feature enables the examination of a peak's spectra at any point during its elution, a valuable tool in determining peak purity.
- Batch mode operation compatible with Bio-Rad's HRLC Sample Table for a completely automated system.
- File storage and retrieval of detector programming parameters. In addition to annotating a particular chromatographic run, this feature allows the user to easily load specific detector parameters with minimal effort.
- File storage and retrieval of chromatographic (single and multiple wavelength) and spectral data for future display and data reduction.
- Data reduction including:
 - File subtraction for point-to-point baseline correction and deconvolution of overlapping peaks
 - Wavelength absorbance ratioing
 - First and second derivatives of chromatograms
 - First and second derivatives of spectra
 - Spectra normalization and subtraction for peak purity determination (for discrete or Time Slice spectra)
 - Two and three-dimensional display of reduced chromatographic and spectral data on a computer monitor

2.4 Controls and Connections

Front Panel

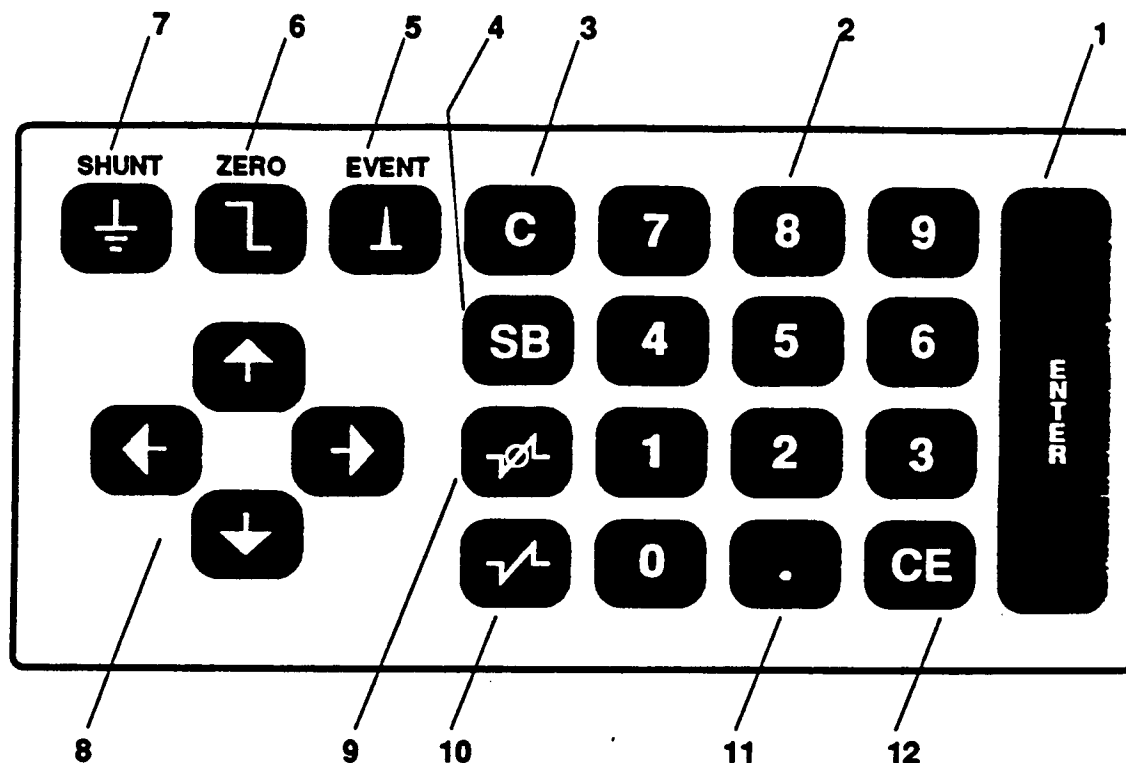


Fig. 2.1. Front panel.

The following controls are located on the front panel of the detector:

1. **ENTER KEY.** Press this key to accept a parameter value you have typed in.
2. **NUMERIC KEYS.** Use this key to enter numeric parameter values.
3. **CLEAR ALL KEY.** Use this key to erase all numeric parameters entered on a given menu. This key function for Polychrome, Schedule Spectra, and Programmed Wavelength Changes menus.
4. **STAND-BY KEY.** Press this key to place the detector in the stand-by status. Under stand-by conditions, the lamps are turned off and the galvanometer drive is locked in a safe position.
5. **EVENT KEY.** Press this key to place a mark (approximately 20% full-scale deflection) on the recorder outputs. This key is also used to advance the detector to the Chromatogram Underway status while in the Programmed Operation Mode.
6. **ZERO KEY.** Press this key to autozero the detector baseline at all wavelengths of operation.
7. **SHUNT KEY.** Press this key to output zero volts to the recorder outputs.
8. **CURSOR KEYS.** The up, down, right and left arrow keys move the cursor around in the various LCD fields. They can also be used to enter a parameter value while simultaneously advancing to the next field.

4. **VOLTAGE SETTING.** Allows the detector operating voltage to be set for 100, 110, 220, or 240 VAC, 50/60 Hz.
5. **FUSE BLOCK.** Pry out this block to allow access to the fuses and to change the operating voltage.
6. **POWER CONNECTOR.** Accepts standard modular power line cord.
7. **ON/OFF SWITCH.** Used to switch the instrument's power on or off.
8. **RS-232C SERIAL INTERFACE.** Twenty five-pin female connector allows RS-232 communication to external computer or terminal. Used with optional software packages.
9. **DISPLAY CONTRAST CONTROL.** Turn to adjust the LCD display contrast for different viewing angles or light conditions.
10. **ACCESSORY RELAY CONNECTOR.** Use these outputs to trigger external events from the detector.

Left Bulkhead

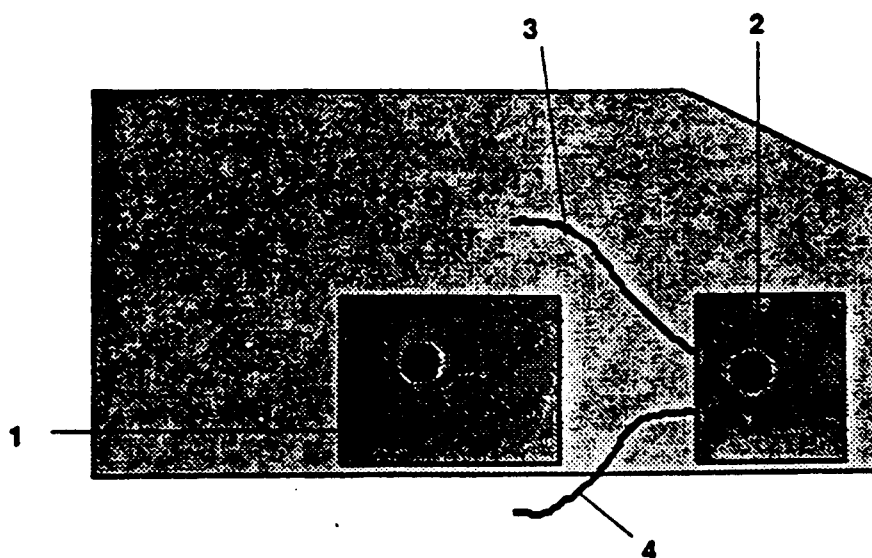


Fig. 2.3. Detector left bulkhead.

The following components are located on the left bulkhead of the detector:

1. **LAMP HOUSING.** Loosen the screw and remove the cover to allow access to the lamps.
2. **FLOW CELL HOUSING.** Loosen the thumbscrew and remove the cover to allow access to the flow cell and sample photodiode.
3. **OUTLET LINE.** A 1/16-inch OD tube for connection to a drain line, backpressure device, or waste reservoir.
4. **INLET LINE.** A 1/16-inch OD tube for connection to the column outlet.

Setting Recorder Range

The detector can accommodate up to four strip chart recorder channels connected simultaneously. The full scale voltage for each channel may be independently set at 10, 100, or 1000 mV using a bank of switches located on the rear panel immediately below the I/O connection block.

This bank contains twelve switches; three (labelled A, B, and C) for each of the recorder channels. The switches correspond to output ranges of 1 V, 100 mV, and 10 mV, respectively. For each channel the switch corresponding to the desired range should be on (facing left). The other two switches for that channel should be off (facing right). Normally, the full scale voltage is set to match the input span of the recorder being used.

To change a range, move the appropriate switches using a small screwdriver or the tip of a pen. Remember, for each recorder output, set one switch to the left and the remaining two, to the right.

Recorder Connections

The recorder cables are connected at the I/O block on the rear panel of the detector (see figure 2.2). The cables should end in 1/4 inch or so of bare wire. To make a connection, press in on the orange tab with the blade of a small screwdriver. Insert the wire end in the hole to the left of the orange tab, and release the tab. To disconnect a wire, simply press the orange tab and pull the wire end out of the hole.

The recorder used should have a "floating" (ungrounded) input. In most cases, the minus (black) input of the recorder should be connected to one of the ground (GND) terminals on the detector. The plus (red) input of the recorder should be connected to the appropriate terminal on the detector.

For the initial, check-out run, you should connect a strip chart recorder to channels one and two. The use of a multiple pen recorder is recommended.

Integrator Connection

The detector provides a fixed-span output (1V/AU) for use with integrators or data acquisition systems. Signals are provided with a resolution ranging from 7 micro AU to 7.9995 AU. Connections are made in the same way as recorder connections are made.

Other Connections

The detector provides for a number of other connections for accessories. You do not need to connect these in order to finish the initial installation and check-out. If you do not wish to connect the Autozero, Event Switch, Relays, or Serial Port at this time, you may proceed directly to INITIAL TEST.

Autozero: The AUTOZERO connection on the I/O block duplicates the function of the ZERO key on the front panel whenever it is connected to ground or a TTL low by a momentary contact closure provided by a remote instrument.

The cable should end in 1/4 inch or so of bare wire. To make the connection, press on the orange tab with the blade of a small screwdriver. Insert the wire end in the hole to the left of the orange tab and release the tab. To disconnect the wire, simply press the orange tab and pull the wire end out of the hole.

If the remote instrument will be using a TTL to trigger the autozero, then the TTL trigger should be connected to the AUTOZERO input of the detector and the grounds of the controlling device should be connected to a ground terminal on the detector. If the remote instrument will be using a contact closure to trigger the autozero, then the common terminal of the relay should be connected to one of the ground terminals of the detector. The normally open terminal on the relay should be connected to the AUTOZERO terminal on the detector.

2.7 Fluid Connections

The detector inlet is the lower tube which protrudes from the rear wall of the cell compartment. As a general rule, the less tubing between the column outlet and the detector cell, the better. Ideally, the column outlet should be connected directly to the detector inlet line. If this is not feasible (if the column is contained in an oven or heater, for example), you should use a minimum length of narrow-bore (0.005-0.010 inch ID) connecting tubing.

Because different columns use different fittings, the detector is supplied with a bare tube end to allow connection to any column accepting 1/16-inch tubing. You should use nuts and ferrules suitable to your column.

Connect the cell outlet (the upper of the two tubes which protrude from the rear wall of the cell compartment) to a line leading to an appropriate waste reservoir. If bubble formation in the detector cell causes problems, you may wish to connect the cell outlet to a restrictor or backpressure device providing 20 to 60 psi backpressure.

2.8 Operation

Turn on the HPLC system and allow the column to equilibrate with the flowing eluant (the time required will depend on you particular column and eluant). If you have not already done so, turn on the power to the detector using the switch located on the lower right panel. Adjust the contrast of the display using the DISPLAY CONTRAST adjustment as necessary. The MAIN menu should appear on the LCD.

When the UV lamp lights, press the SHUNT key and hold it down while you adjust the pen on the recorder connected to CHANNEL 1. Adjust the pen to your desired position on the recorder chart.

Release the SHUNT key and press the ZERO key. The absorbance display should be very close to the zero position which you just set.

2.9 Setting Parameters

Wavelength

Insure that the cursor is in the ABSORBANCE @ 254 NM position. Enter the wavelength appropriate for your test mix by typing in the numbers and pressing enter. Press the ZERO key to rezero the detector absorbance reading at the new wavelength.

```
Absorbance @ 254 nm = 0.00000 AU : Menu 1
range = 2.0000 AUFS: rise time 1.0 sec
Spectrum: 200 to 360 @ 10 nm ; reps 4
Method: store recall ;
Program ☐multi ☐wavelength ☐spectra
Playback: ☐chromatogram ☐spectra
Menus: ☐start chromatogram ☐auxiliary
Mode: ☐spectra only UV
```

3.2 The Detector Key Pad

There are four different types of keys on the detector key pad:

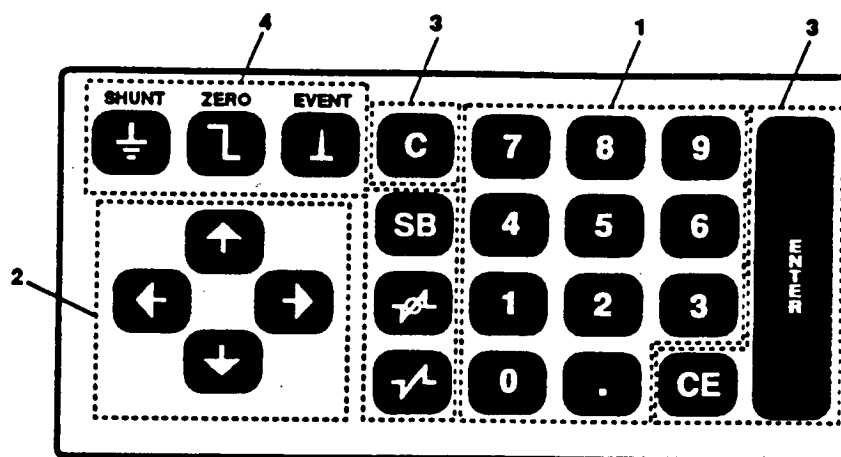


Fig. 3.1.

1. **Numeric Keys:** These keys are used to enter numerical parameters for a given field.
2. **Arrow Keys:** These keys are used to move the cursor between lines and fields on a given menu and to erase a numerical entry.
3. **Programming Keys:** There are three programming keys:
 - **Enter Key:** This key is used to enter a given value for a particular field, toggle a function on or off, and advance to another menu.
 - **Clear Entry Key:** This key is used to erase an erroneous entered value for a particular field.
 - **Clear All Key:** This key is used to erase all numeric parameters listed on a given menu; it functions only on the MULTI, SCHEDULE SPECTRA, and SCHEDULE WAVELENGTH CHANGES menus.
4. **Activity Keys:** The activity keys are used to perform particular tasks. These keys remain active on all detector menus. There are six ACTIVITY KEYS:
 - **Event Key:** The **EVENT** key is used to signal the beginning of a chromatogram. It sends out a 20% full scale (approximate) spike to the recorder outputs and advances the detector to the UNDERWAY status in the Programmed Operation Mode. If the detector is used with the optional computer software, the **EVENT** key signals the computer for the start of the run. All event marks are recorded in the the detector's memory for playback purposes. Thus, they may be used as an annotative device.
 - **Shunt Key:** The **SHUNT** key is used to set all recorder outputs to zero volts for the purpose of adjusting recorder pen position. After releasing this key, the zero volt output is maintained for approximately 3.0 seconds.
 - **Zero Key:** This key is used to zero the absorbance values at all monitored wavelengths. It does not zero the wavelengths of a programmed spectra.
 - **Take Spectrum Key:** This key is used to take a discrete spectra of the flow cell contents.
 - **Spectra Zero Key:** This key is used to store background zero values for a programmed, discrete spectra.
 - **Stand-By Key:** This key places the monitor in the stand-by status, where all lamps are shut off and the galvanometer drive is locked in a safe position.

- Zeroing the detector
- Performing a full spectral autozero
- Setting a zero offset
- Switching lamps on or off
- Setting high speed spectra parameters:
 - Upper and lower wavelength
 - Step size
 - Repetition
- Setting baseline correction parameters
 - Start time
 - End time
 - End absorbance
- Setting memory extender parameters
- Operating the accessory relay
- Playing back stored spectra or chromatograms

Note: Neither chromatograms nor spectra can be stored in the detector's memory during IMMEDIATE MODE OPERATION. Additionally, baseline correction will not function and spectral scans should not be performed while in IMMEDIATE MODE.

If you wish to store a chromatogram in the detector's memory for later replay or perform and store a spectra, you must use the PROGRAMMED OPERATION MODE. This mode also allows for scheduled events, such as Programmed Wavelength Changes and Scheduled Spectra, to be implemented. Additionally, the Baseline Correction and Memory Extender functions of the monitor are activated under this mode of operation.

While in PROGRAMMED OPERATION, you may:

- Schedule detector parameters to change:
 - Wavelength (single wavelength only)
 - Range
 - Recorder output
 - Rise time
 - Schedule end time for a run
- Zero the detector at all wavelengths
- Perform a full spectral autozero
- Perform a spectra and store its data
- Change spectra repetition parameters
- Schedule a predefined spectra to occur and store its data
- Perform a chromatogram and store its data

Rise Time

Move the cursor to the RISE TIME line and type in the desired value, then press ENTER. The new value will be displayed on the RISE TIME field. Allowable values range from 0 to 9.9 seconds. Values outside of this range will not be accepted; the message "INVALID RISE TIME" will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value.

Spectrum

Move the cursor to the SPECTRUM line. There are four sets of values on this line. The first two define the spectral start and stop wavelengths, respectively. Spectra may be taken using the deuterium lamp (190–365 nm) or the tungsten lamp (366–800 nm). Values outside of the ranges from 190 to 365 nm or 366 to 800 nm will not be accepted; the message "INVALID SCAN PARAMETERS" will appear. To clear the error message, press any key and enter the correct value.

The third number field defines the spectra step wavelength. This is the interval between scanned points. Acceptable values range from 1 to 10 nm. Values outside of this range will not be accepted; the message "INVALID SCAN PARAMETERS" will appear. To clear the error message, press any key and enter the correct value.

The scanning algorithm of the monitor mandates the selection of upper and lower scan wavelength values which are inside the ranges of 190 to 365 and 366 to 800 nm by a factor of at least the step wavelength divided by two. For example, using a step size of 6 nm, the largest scans would be 193 to 362 and 369 to 797 nm. If these parameters are incorrect, the detector will display the message "INVALID SCAN PARAMETERS" on the lower line. Depending upon the entered lower, upper, and step wavelength values, the monitor will correct one of these three when it displays the error message. To clear the error message, press any key and enter the correct value.

The fourth number field lists the repetition number for the defined scan. Valid entries are 1, 2, and 4. Values outside this range trigger an error message which informs the user of the allowed choices. This parameter should be optimized to balance signal-to-noise (the greater the repetition number, the higher the S/N) with peak width/speed.

The Program Line

This line lists fields for entering the MULTI (WAVELENGTH), WAVELENGTH (CHANGES) and (SCHEDULE) SPECTRA menus. To access a menu, for example, the MULTI-WAVELENGTH menu, move the cursor to that field and push enter. To return to the MAIN menu, move the cursor to the RETURN TO PREVIOUS MENU field and push ENTER.

MULTI-WAVELENGTH							MENU 4
☐ Return to previous menu						☐ sort	
Ratio: 254 over 350 : CH 3							
NO	nm	CH	range	RT	INT		
1	254	1	0.0020	1.0	1	☐ INS	
2	350	2	0.0020	1.0		☐ INS	

3.6 Performing a Single Wavelength Chromatogram (Monochrome) (Immediate Mode Operation)

Immediate Mode Monochrome chromatograms are performed using the MAIN menu. Turn on the HPLC system and allow the column to equilibrate with the flowing eluant (the time required will depend on your particular column and eluant). If you have not already done so, turn on the power to the detector using the switch located on the lower rear panel. Adjust the contrast of the display using the DISPLAY CONTRAST control as necessary. The MAIN menu should appear on your screen.

The lamp indicator located in the lower right corner of the screen indicates the warm up status of the deuterium lamp by displaying "UVW". When the UV lamp lights, the display changes to "UV".

Press the SHUNT key and hold it down while you adjust the recorder pen to the desired position.

Release the SHUNT key and press the ZERO key. The absorbance display should be very close to zero, and the recorder pen should be very close to the zero position which you just set.

You may now set the appropriate detector parameters.

Setting Wavelength

Insure that the cursor is on the WAVELENGTH field of the MAIN menu. Type in the desired wavelength, then press the ENTER key. You may also enter the wavelength parameter by using the down arrow key to advance to the RANGE field.

Setting Range

If you have not already done so, use the down arrow key to move to the RANGE field. Enter the RANGE appropriate for your sample concentration and flow cell path length. Use the cursor down key to advance to the RISE TIME field.

Setting Rise Time

The detector uses a digital second order Bessel filter. The response of this type of filter is described in terms of a 10% to 90% response to a step input. This response time is called RISE TIME (RT). For a given RT, the response of a Bessel filter is roughly equivalent to a RC time constant equal to 1/2 the RT. A RT of 1 second will produce approximately the same result as a 0.5 second time constant although the second order Bessel filter produces much less tailing. The default value of RT is set for 1.0 second. This value is adequate for most routine HPLC separations. If you are running a very fast analysis, a shorter RT may be necessary. For very high sensitivity detection, a longer RT may be more appropriate. In general, if the RT is set to be equal to or less than 1/3 the half height peak width, the filter will cause no peak broadening.

Type in the desired RISE TIME in increments of 0.1 second and press the ENTER key.

Recorder and Integrator Channels

While in the single wavelength mode, the monitor transmits ranged and filtered data out of RECORDER CHANNEL 1. Nonranged (unattenuated), filtered data is available from the INTEGRATOR OUTPUT. If you are using a chart recorder, insure that it is connected to RECORDER CHANNEL 1. If an integrator or data acquisition system is to be used, it should be connected to the INTEGRATOR OUTPUT.

Enter the remaining desired WAVELENGTHS, RECORDER CHANNELS, RANGES, and RISE TIMES in the same manner as used above. If you are unsure of the wavelengths to use for your sample, use 254 nm, 280 nm, and 329 nm as the remaining three wavelengths.

Move the cursor to the RATIO line. Select two of the monitored wavelengths to be ratioed. Enter these wavelengths in the NUMERATOR and DENOMINATOR fields. If you are unsure of which two wavelengths to ratio, ratio 214 nm over 280 nm.

MULTI-WAVELENGTH							MENU 4	
☐ Return to previous menu				☐ sort				
Ratio: 214 over 280 : CH 3								
NO	nm	CH	range	RT	INT			
1	214	1	0.0020	1.0	I	☐ INS		
2	254	2	0.0020	1.0		☐ INS		
3	280	3	0.0020	1.0		☐ INS		
4	329	4	0.0020	1.0		☐ INS		

Enter the RECORDER CHANNEL (CH) OUTPUT for the RATIO PLOT. It is recommended that three of the outputs be used for WAVELENGTH ABSORBANCE DATA and the fourth used for the RATIO PLOT.

The RATIO PLOT is the log of the NUMERICAL RATIO.

$$\text{NUMERICAL RATIO} = \frac{\text{ABSORBANCE WAVELENGTH 1} / \text{RANGE WAVELENGTH 1}}{\text{ABSORBANCE WAVELENGTH 2} / \text{RANGE WAVELENGTH 2}}$$

If an integrator or data acquisition system is to be used, you must identify one of the monitored wavelengths for integration. Move the cursor to the INTEGRATOR (INT) field. Using the up or down arrows, move the cursor into the line which contains the desired wavelength for integration and press ENTER. An "I" will appear in that line.

Move the cursor to the RETURN TO PREVIOUS menu field and press ENTER. The MAIN menu should now be displayed.

Move the cursor down to the SPECTRA ONLY field of the MODE line and press ENTER twice to toggle the mode to MULTIPLE WAVELENGTH.

Move the cursor to the MULTI field of the PROGRAM line and push enter. The MULTI-WAVELENGTH menu should appear.

Running the Chromatogram

Switch your recording devices on and press the ZERO key. The absorbance values should be very close to zero. Press the SHUNT key and adjust the pen position on your chart recorder(s). Position the RATIO PLOT pen at 50% full scale. If an integrating device is being used, set the device for the baseline noise of the integrated wavelength.

Inject your sample and press the EVENT key. You may change any of the PARAMETERS during the run by advancing the cursor to the appropriate field, typing in the desired change, and pressing the ENTER key or moving the cursor to another field.

As your peaks elute, notice the RATIO PLOT trace. If the peak is pure (no coelution occurs) the RATIO PLOT will trace a "square wave" pattern. Deviations from this pattern are indicative of coelution and an impure peak. Ratioing is further discussed in the "Theory of Operation" section.

Move the cursor to the START CHROMATOGRAM field of the MENUS line and push the ENTER key. The PUSH EVENT TO START CHROMATOGRAM menu will be displayed.

```
PUSH EVENT TO START CHROMATOGRAM  MENU 6
ABSORBANCE @ 254 nm = 0.00000 AU
MODE: spectra only
☐ return to main menu
non-scheduled spectra reps: 4
range 2.0000 AUFS
rise time = 1.0 sec                                UV
```

Push the ZERO key to zero the absorbance at the monitored wavelength. The absorbance value should be very close to zero.

Push the FULL SPECTRAL ZERO key to zero the background absorbance of your effluent at all of the scanned wavelengths.

Press the SHUNT key and adjust the position of the pen(s) on your recording device. If an integrator/data acquisition system is being used, set this device for the background noise at the monitored wavelength.

Running the Chromatogram

Inject your sample and press the EVENT key. The CHROMATOGRAM UNDERWAY menu will appear.

```
CHROMATOGRAM UNDERWAY  MENU 6
ABSORBANCE @ 254 nm = 0.00000 AU
MODE: spectra only      0.0
☐ STOP chromatogram
spectra reps: 4
range 0.002 AUFS
rise time = 1.0 sec
```

You may change the parameters of RISE TIME, RANGE, and SPECTRA REPS at any time. You may NOT change the monitored wavelength. The elapsed time is indicated by the clock in the upper, right corner of the menu.

When one of the peaks elute, press the TAKE SPECTRA key when the pen is close to the apex of the peak. This will store the spectra for that peak in the detector's memory.

After the last peak elutes, move the cursor to the STOP CHROMATOGRAM field and press ENTER. This halts the data storage process and rearms the detector to start recording a chromatogram. If you wish to replay the chromatogram, advance to the "Chromatogram Playback" section of this manual. If you wish to view the recorder spectra, advance to the "Spectra Playback" section of this manual.

If you wish to record another chromatogram or additional spectra of the peaks in your test mix, it is not necessary to reprogram the monitor. All entered detector parameters are retained, and you need only to inject your sample and press the EVENT key to advance the detector to the UNDERWAY mode. It should be noted that only one chromatogram, with its associated spectra, is stored in the monitor at any time. Advancing the detector to the UNDERWAY status erases all stored chromatographic and spectral data from the previous run.

Move the cursor to the NM field of the MULTI-WAVELENGTH menu NO 1 line. Enter the first of the desired wavelengths. If you are unsure of the appropriate wavelengths to use for your sample, refer to the spectra taken in the previous section, or simply use 214 nm. Enter 214 as your first wavelength. Type in the WAVELENGTH value and then press the ENTER key.

MULTI-WAVELENGTH						MENU 4	
☐ Return to previous menu				☐ sort			
Ratio: 214 over 280 : CH 3							
NO	nm	CH	range	RT	INT		
1	214	1	0.0020	1.0	1	☐ INS	
2	254	2	0.0020	1.0		☐ INS	
3	280	3	0.0020	1.0		☐ INS	
4	329	4	0.0020	1.0		☐ INS	

Move the cursor across to the CH field and type in the desired RECORDER CHANNEL NUMBER. For organizational purposes, it is recommended to use CHANNEL 1. Enter the value by advancing the cursor across to the RANGE field.

Type in an appropriate RANGE for your sample mix and flow cell path length at this wavelength. Enter the value by advancing the cursor across to the RISE TIME (RT) field.

Type in an appropriate RISE TIME value for your test mix and chromatographic conditions. For most instances, a 1.0 second RISE TIME is sufficient for peaks as short as 10 seconds at their base. A shorter RISE TIME should be used for faster peaks.

Enter the remaining desired WAVELENGTHS, RECORDER CHANNELS, RANGES, and RISE TIMES in the same manner as above. If you are unsure of the wavelengths to use for your sample, refer to the spectra gathered in the previous section or use 254, 280, and 329 nm as the remaining wavelengths.

Move the cursor to the RATIO line. Select two of the monitored wavelengths to be ratioed. Enter these wavelengths in the NUMERATOR and DENOMINATOR fields. If you are unsure of which two wavelengths to ratio, ratio 254 nm over 350 nm.

Enter the RECORDER CHANNEL (CH) OUTPUT for the RATIO PLOT. It is recommended that three of the outputs be used for WAVELENGTH ABSORBANCE DATA and the fourth be used for the RATIO PLOT OUTPUT.

The RATIO PLOT is the log of the NUMERICAL RATIO.

$$\text{NUMERICAL RATIO} = \frac{\text{ABSORBANCE WAVELENGTH 1} / \text{RANGE WAVELENGTH 1}}{\text{ABSORBANCE WAVELENGTH 2} / \text{RANGE WAVELENGTH 2}}$$

If an integrator or data acquisition system is to be used, you must identify one of the monitored wavelengths for integration. Move the cursor to the INTEGRATOR (INT) field. Using the up or down arrows, move the cursor into the line which contains the desired wavelength for integration and press ENTER. An "I" will appear in that line.

Running a Chromatogram

Inject your sample and press the EVENT key. The CHROMATOGRAM UNDERWAY menu will appear.

PUSH EVENT TO START CHROMATOGRAM						MENU 7
MODE: multi-wavelength						
☐ return to main menu ; spectra reps: 4						
Ratio: 214 over 280 : CH 4						
NO	nm	CH	range	RT	INT	
1	214	1	0.0020	1.0	I	0.00000
2	254	2	0.0020	1.0		0.00000
3	280	3	0.0020	1.0		0.00000
4	329		0.0020	1.0		0.00000

You may change the parameters of RISE TIME and RANGE at any time. You may NOT change any of the monitored wavelengths. The elapsed time is indicated by the clock in the upper, right corner of the menu.

After the last peak elutes, move the cursor to the STOP CHROMATOGRAM field and press ENTER. This halts the data storage process and rearms the detector to start recording a chromatogram. If you wish to replay the chromatogram, advance to the "Chromatogram Playback" section of this manual. If you wish to record another chromatogram, it is not necessary to reprogram the detector. All entered parameters are retained, and you need only to inject your sample and press the EVENT key to advance to the UNDERWAY status. It should be noted that only one chromatogram can be stored in the monitor's memory at any time. Advancing the detector to the UNDERWAY status erases all stored chromatographic data from the previous run.

In this example, we did not define or perform any spectra. However, it is possible to take spectra while in the multiple wavelength mode. For details on programming spectral parameters, see the "Recording and Storing a Programmed Single Wavelength Chromatogram With Spectra" section of this manual.

3.10 Playing Back a Chromatogram

A chromatogram stored in memory may be played back at any time except during the recording process. It should be noted that a chromatogram is retained in memory only until another one is recorded or the power is turned off. Just one chromatogram and its associated spectra can be stored within the detector's memory at any time. The details of playback procedure depend to some extent on the conditions under which the chromatogram was originally recorded: Single Wavelength, Multiple Wavelength, and with or without Extended Memory. Stored data may be played back any number of times. Each time the RANGE, RISE TIME or any other parameter may be changed.

MULTI-WAVELENGTH PLAYBACK						MENU 9
☐ Return		☐ Playback chromatogram				
... @		1 times real time		☐ stop PB		
Ratio: 214 over 280 : CH 4						
Baseline 0.83 to 3.33 min @ 0.00000 AU						
NO	nm	CH	range	RT	INT	
1	214	1	0.0020	1.0	I 0.00000	
2	254	2	0.0020	1.0	0.00000	
3	280	3	0.0020	1.0	0.00000	
4	329	4	0.0020	1.0	0.00000	

Move the cursor down to the 1 TIMES REAL TIME field. This field controls the rate at which the monitor will replay the recorded chromatogram. Chromatograms may only be played back as fast or faster than real-time. Acceptable values range from 1 to 20. Enter the desired playback rate.

Move the cursor down to the first wavelength line and enter the desired values for RANGE, RECORDER CHANNEL, and RISE TIME.

Move the cursor down to all subsequent lines and repeat the above procedure. If more than three wavelengths have been recorded, it is necessary to use the down arrow key to scroll the screen and expose all of the other recorded wavelengths.

If an integrator or data acquisition system is connected to the integrator output, you must specify a wavelength for integration. To do so:

Place the cursor on the wavelength line to be integrated, move it across to the INT field, and push ENTER. An "I" will appear in the INT field for the desired wavelength to be integrated.

If you desire to record an absorbance RATIO PLOT of any of the two recorded wavelengths, the following procedure is used:

Move the cursor to the RATIO line and enter the numerator and denominator wavelengths. Move the cursor across to the CH field and enter the recorder channel output for the ratio trace. If you enter a recorder channel output already being used by one of the recorded wavelengths, the monitor will prioritize this channel for the RATIO PLOT.

Ready your recording device(s) and move the cursor to the PLAYBACK CHROMATOGRAM field. Press ENTER to begin the playback process. The PLAYBACK CHROMATOGRAM UNDERWAY menu will appear.

The elapsed time and absorbencies for each recorded wavelength will be displayed. At the completion of the chromatogram replay, time and absorbance values will no longer be displayed.

To stop the playback at any time, move the cursor to the STOP PB field and press ENTER.

Replaying Memory Extended Chromatograms

The procedure used to replay a MEMORY EXTENDED chromatogram, be it single or multiple wavelength, is very similar to replaying an unextended version. Because MEMORY EXTENDER

If you have performed the spectra outlined in this tutorial and you have not shut the detector off since the acquisition of those spectra, the spectra playback menu will be displayed after the PLEASE WAIT message disappears. If you have not performed spectra or shut the monitor off since recording the spectra, no spectra will be stored in the instrument's memory, and the error message INVALID SCAN PARAMETERS will be listed.

Note the spectral parameters of lower, upper, and STEP WAVELENGTHS that were programmed for the recorded spectra. These properties are listed on the third line. The fourth line should read "SPECTRUM NO 1 TAKEN AT 1 SEC W/4 REPS." This indicates that the first spectra, number 1, taken at 1 second after the start of the programmed mode run, is ready to be played back. Also note that the REPETITION value of four is displayed, meaning that a total of four repetitive scans were performed and averaged together for the stored spectral data.

Move the cursor down to the PLAYBACK RATE field and enter an appropriate playback rate in nm/second. Choose a rate which will not exceed the response speed of your recording device while simultaneously allowing you to calibrate the wavelength axis. The latter is achieved by coordinating the playback rate with the strip chart paper speed. For example, if a playback rate of 5 nm/second is used with a chart speed of 12 cm/minute, the resultant wavelength (x) axis is scaled to 5 nm for every 2 mm of paper.

Move the cursor to the RANGE field and enter an appropriate RANGE for your test mix and flow cell path length. Allowable values range from 0.0005 to 7.9995 AUFS in multiples of 0.0005 AUFS.

Move the cursor to the CHANNEL NUMBER (CH NO) field and specify the recorder output to which your recording device is interfaced. Allowable outputs are channels 1 through 4. Spectra MAY NOT be replayed through the INTEGRATOR output.

Set the proper pen position for the recording device and start the chart drive. Move the cursor to the PLAYBACK SPECTRA field and press ENTER to begin the playback process. Wavelengths and their absorbance values should be displayed on the second line. Upon completion of the playback, these values will disappear from the screen.

To replay the next spectra, move the cursor to the NEXT SPECTRA field and press ENTER. The fourth line should note "SPECTRA NO. 2... ETC." Proceed with the rest of the playback procedure as previously outlined.

3.12 Programmed Wavelength Changes (Single Wavelength Operation)

This mode of programmed operation allows changes in WAVELENGTH, RANGE, RISE TIME, and RECORDER CHANNEL to be made at predetermined times during the chromatogram. These changes must be scheduled before the beginning of the run. Events may be scheduled in minutes or seconds. The default status is in minutes. For this tutorial, change the status to seconds by advancing to the AUXILIARY menu, scrolling to the bottom line and toggling the DISPLAY TIME BASE field to seconds.

From the MAIN menu, move the cursor to the TIMED MONO field of the PROGRAM line and push ENTER. The SCHEDULED WAVELENGTH CHANGES menu should be displayed.

SCHEDULED WAVELENGTH CHANGES						MENU 3	
☐ Return to main menu							
NO	NM	FROM	TO	CH	RANGE	FILTER	
1	254	0	1200	1	2.0000	1.0	☐ INS

Press the ZERO key to zero the absorbance at the first monitored wavelength. The absorbance value should be very close to zero.

Press the SHUNT key and adjust the position of the pen(s) on your recording device. If an integrator/data acquisition system is being used, set this device for the background noise at the monitored wavelength.

Inject your sample and press the EVENT key. The CHROMATOGRAM UNDERWAY menu will appear.

CHROMATOGRAM UNDERWAY	MENU 6
ABSORBANCE @ 254 nm = 0.00000 AU	
MODE: spectra only	0.0
☐ STOP chromatogram	
spectra reps: 4	
range 0.002 AUFS	
rise time = 1.0 sec	

Watch the menu and note that changes of wavelength, rise time, range, and channel occur. You may manually change the parameters of rise time and range at any time. You may NOT alter the wavelength value.

At the end time of the last scheduled change, the detector will stop gathering data and reset to the ARMED TO START STATUS. To replay the recorded data, simply move the cursor to the RETURN TO MAIN menu field and press ENTER. Then move the cursor to the CHROMATOGRAM field of the PLAYBACK line and press ENTER. The remaining instructions are found in the "Playback Chromatogram" section of this manual.

Editing the Wavelength Changes Menu

Editing the SCHEDULED WAVELENGTH CHANGES menu is easy. If you decide to delete a scheduled wavelength, move the cursor to any field on that wavelength's line and press the CLEAR key twice.

If you decide to insert a scheduled change between two other scheduled changes, move the cursor to the INSERT field of the scheduled change line which is just before the desired insertion time, and press the ENTER key. This automatically copies the previous line and moves the cursor to the INSERT field of the newly made line. Now, type in the desired wavelength, end time, channel, range, and filter parameters for the new line. You have just inserted a new change.

Zero Upon Wavelength Change

As a default parameter, the monitor performs an AUTOZERO after every change of wavelength, whether it is manually or automatically actuated. This function is controlled on the AUXILIARY menu. While performing SCHEDULED WAVELENGTH CHANGES, it is important to determine if any of the scheduled changes occur during times when a given peak still contributes to the absorbance. For example, if all of your peaks are well resolved, as in the chromatogram depicted in figure 3.2, wavelength changes can be scheduled during times between the peaks when the baseline is flat.

In this case the ZERO UPON WAVELENGTH CHANGE is desirable. If your chromatogram contains poorly resolved peaks, as in figure 3.3, and you schedule a wavelength change to occur between two poorly resolved peaks, a ZERO UPON WAVELENGTH CHANGE would subtract a value equivalent to the sum of the effluent and peak absorbencies. This would result in a negative displacement of the baseline.

Move the cursor to the TIME field and enter 10 seconds. Move the cursor across to the REPS field and enter 4. Press the arrow down key to advance to the third line.

Move the cursor to the TIME field and enter 50 seconds. Move the cursor across to the REPS field and enter 2. Press the arrow down key to advance to the fourth line.

SCHEDULED SPECTRA			MENU 5	
☐ return to main menu				
NO	TIME	REPS	☐ sort	
1	1.00	2	☐ zero	
2	10.00	4	☐ zero	
3	50.00	2	☐ zero	
4	200.00	4	☐ zero	UV

Move the cursor to the TIME field and enter 200 seconds. Move the cursor across to the REPS field and enter 2. The SCHEDULED SPECTRA menu should look like the one above.

To begin the SCHEDULED SPECTRA program, move the cursor to the RETURN TO MAIN menu field and press ENTER.

From the MAIN menu, move the cursor to the START CHROMATOGRAM field of the menu line and press ENTER. The ARMED TO START menu should appear.

To begin the program, press the ENTER key. All scheduled spectra will be performed as programmed. To stop the program, move the cursor to the STOP CHROMATOGRAM field and push ENTER.

Editing the Schedule

The schedule of spectra may be edited at any time except when a chromatogram is UNDERWAY. To delete any scheduled spectra, move the cursor to the spectra of interest and press the CLEAR ENTRY key (CE) twice. This can be accomplished either from the TIME or REPS fields. To erase the entire screen of SCHEDULED SPECTRA, press the CLEAR ALL (C) key.

To insert a scheduled spectra between two other spectra, simply move the cursor down to next available line and enter the desired values of TIME and REPS for the inserted spectra. Move the cursor up to the SORT field and press ENTER. The monitor will automatically sort the schedule in the proper order of increasing time.

3.14 Baseline Correction

The detector incorporates a baseline correction program in its software. The baseline correction capability is enabled whenever parameters other than zero are entered in the baseline correction fields located on the AUXILIARY menu and the monitor is activated under the PROGRAMMED OPERATION MODE.

The BASELINE CORRECTION algorithm subtracts a linear function from the measured absorbance prior to sending the chromatogram to the output channel. The BASELINE CORRECTION is defined by START and END times, and an END ABSORBANCE that you specify.

With these three parameters, the detector performs a linear interpolation and subtracts the corresponding absorbance from the measured absorbance as a function of time. This correction is performed only on REAL-TIME data acquired under the PROGRAMMED OPERATION MODE. Raw data stored by the monitor for later replay is NOT baseline corrected. However, replayed data may be baseline corrected by entering the appropriate BASELINE CORRECTION parameters on the AUXILIARY menu just prior to playback.

3.15 Memory Extender

The detector is equipped with a MEMORY EXTENDER feature which increases the monitor's capacity for absorbance vs. time data in order to allow long chromatograms to be recorded. Under the default conditions, this feature is not active and the monitor will store approximately 60 minutes of chromatographic data. After 60 minutes of operation, the detector will display "CMF" (chromatogram memory full) in place of elapsed time on the CHROMATOGRAM UNDERWAY menu. Although subsequent data are still output to the recorder in real-time, they are NOT stored. ONLY THE FIRST 60 MINUTES IS STORED FOR SUBSEQUENT PLAYBACK.

The MEMORY EXTENDER feature works by varying the data storage rate while leaving the data acquisition rate unchanged. MEMORY EXTENDER is activated by entering a MEMORY EXTENDER MULTIPLIER VALUE on the AUXILIARY menu. The default value for the multiplier is one: the detector stores every data point. When the multiplier is changed to an integer, n , the detector stores every n th point. A multiplier value of 5, for example, stores every fifth data point and effectively extends the storage time by a factor of five. For more information regarding this feature, consult the THEORY OF OPERATION section of this manual.

To enter a MEMORY EXTENDER MULTIPLIER:

From the MAIN menu, move the cursor to the AUXILIARY field of the menu line and push ENTER. The AUXILIARY menu should be displayed.

```
Baseline 0.00 to 3.33 min @ 0.00000 AU
Auto Spectra ☐OFF 1 per peak; sides EARLY
Auto scan relay ☐disabled
Zero upon wavelength change ☐ON
SCROLL to access additional parameters
Zero value @ 254 nm 0.00000 AU
Memory extend 1 times
Display time base ☐min
```

Use the down arrow key to scroll to the MEMORY EXTENDER line and move the cursor to the 1 TIMES field. Type in the desired multiplier value (for demonstrative purposes, enter 2). Using the cursor up arrow key, move the cursor to the RETURN TO MAIN menu line and push ENTER. Initiate a PROGRAMMED MODE CHROMATOGRAM in the usual fashion.

3.16 Performing Auto-Spectra

The monitor has a peak detection program that is used to trigger the acquisition of discrete spectra. One, two, or three spectra may be taken on a given peak. Spectra may be taken early or late on a peak's up-shoulder, at its apex, and early or late on the down-shoulder.

Move the cursor to the AUTO-CALIBRATE field and press ENTER.

```

Return to main menu :      MENU 2
Lamps: tungsten OFF  deuterium ON
Auto calibrate
Baseline 0.00 to 3.33 min @ 0.00000 AU
Auto Spectra OFF 1 per peak; sides EARLY
Auto scan relay           disabled
Zero upon wavelength change  ON
SCROLL to access additional parameters

```

A "PLEASE WAIT" message will appear. When it disappears, the auto-calibration procedure is complete.

4.2 The Detector Key Pad

There are four different types of keys on the detector key pad:

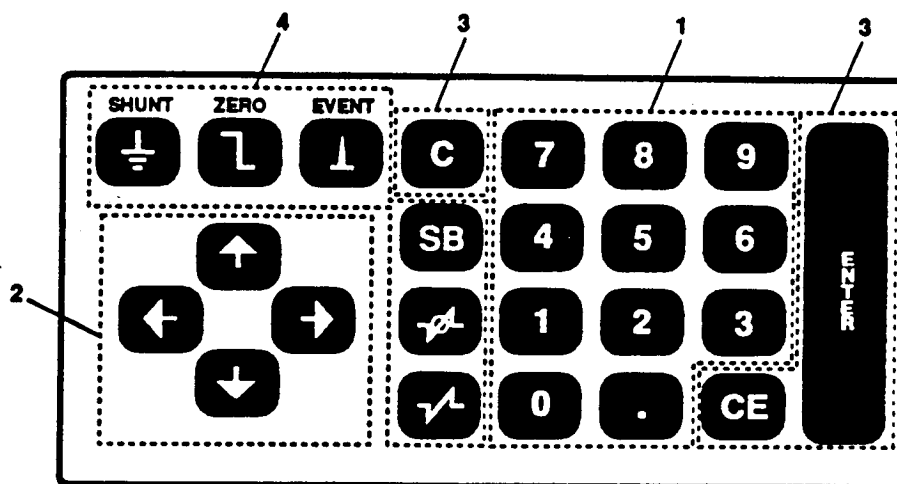


Fig. 4.1. The detector key pad.

1. Numeric Keys: These keys are used to enter numerical parameters for a given field.

2. Arrow Keys: These keys are used to move the cursor between lines and fields on a given menu and to erase a numerical entry.

3. Programming Keys: There are three programming keys:

Enter Key: This key is used to enter a given value for a particular field, to toggle a function on or off, and to advance to another menu.

Clear Entry Key: This key is used to erase an erroneous entered value for a particular field.

Clear All Key: This key is used to erase all numeric parameters listed on a given menu. This key functions only on the POLYCHROME, SCHEDULED SPECTRA, and SCHEDULED WAVELENGTH CHANGES menus.

4. Activity Keys: The activity keys are used to perform particular tasks. These keys remain active on all detector menus. There are six ACTIVITY keys:

Event Key: The EVENT key is used to signal the beginning of a chromatogram. It sends out a 20% full scale (approximate) spike to the recorder outputs and advances the detector to the UNDERWAY status in the Programmed Operation Mode. If the detector is used with the optional computer software, the EVENT key signals the computer for the start of the run. All event marks are recorded in the the detector's memory for playback purposes. Thus, they may be used as an annotative device.

Shunt Key: The SHUNT key is used to set all recorder outputs to 0 V for the purpose of adjusting recorder pen position. After releasing this key, the 0 V output is maintained for approximately 3.0 seconds.

Zero Key: This key is used to zero the absorbance values at all monitored wavelengths. It does not zero the wavelength of a programmed spectra.

4.4 Detector Operation — Immediate and Programmed Operation Modes

The detector has two modes of operation, IMMEDIATE MODE and PROGRAMMED OPERATION MODE. While in IMMEDIATE MODE, a number of operations may be carried out directly from the keyboard in real-time. These operations include:

- Setting detector parameters:
 - Single and multi-wavelength values
 - Full scale range for each monitored wavelength
 - Designation of recorder outputs for up to four of the monitored wavelengths
 - Designation of the integrator output for any one of the monitored wavelengths
 - Wavelength values for ratioing
 - Recorder output for ratio signal
 - Rise time
- Zeroing the detector
- Performing a full spectral autozero
- Setting a zero offset
- Switching lamps on or off
- Setting high speed spectra parameters:
 - Upper and lower wavelength
 - Step size
 - Repetition
- Setting baseline correction parameters:
 - Start time
 - End time
 - End absorbance
- Setting memory extender parameters
- Operating the accessory relay
- Playing back stored spectra or chromatograms

Neither chromatograms nor spectra can be stored in the detector's memory during IMMEDIATE MODE OPERATION. Additionally, baseline correction will not function and spectral scans should not be performed while in IMMEDIATE MODE.

If you wish to store a chromatogram in the detector's memory for later replay or perform and store a spectra, you must use the Programmed Operation Mode. This mode also allows for scheduled events, such as Programmed Wavelength Changes and Scheduled Spectra, to be implemented. Additionally, the Baseline Correction and Memory Extender functions of the monitor are activated under this mode of operation.

The MAIN menu fields are:

Wavelength

When you turn on the monitor, the cursor should be located in the wavelength field (ABSORBANCE @ 254 NM = 0.00000). The default wavelength which is displayed upon power-up is 254 nm. To change wavelength, type in the desired value and press ENTER. Acceptable wavelengths extend from 190 to 800 nm. Values outside of this range are not accepted; the message "INVALID WAVELENGTH ENTERED" will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value.

Range

This field is used to set the full scale range in absorbance units (AUFS) for IMMEDIATE MODE or PROGRAMMED MODE monochrome operation. It does not function for multiple wavelength (POLYCHROME) operation. In single wavelength operation, all absorbance data is transmitted out of recorder CHANNEL 1. Thus, this field controls the range for that channel. The default value for this parameter is 2.0000 AUFS. To change range, move the cursor to the RANGE field, type in the desired value, and then press ENTER. Acceptable values range from 0.0005 AUFS to 7.9995 AUFS in multiples of 0.0005 AUFS. Values may be entered to four decimal places; the detector will round the 4th decimal place to the nearest 0.0005 AUFS. Values outside of the allotted range will not be accepted; the message "INVALID RANGE ENTERED" will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value. The RANGE setting does not affect the INTEGRATOR OUTPUT.

Rise Time

This field is used to set the RISE TIME FILTER for IMMEDIATE or PROGRAMMED MONOCHROME operation. It does not function for multiple wavelength (POLYCHROME) operation. The RISE TIME parameter controls a progressive averaging, digital filter which conserves peak area at all settings. The filter may distort the shape of the peak, however area is always conserved.

As a general rule, a rise time should be selected which is 1/3 to 1/5 the half-height width of the fastest peak in a given separation. The default rise time setting of 1.0 second will suffice for peaks as fast as 3 seconds at half height. The RISE TIME filter may be totally bypassed by entering a RISE TIME value of 0.0 seconds.

Allowable RISE TIME values range from 0.0 to 9.9 seconds in intervals of 0.1 second. Values outside of this range will not be accepted; the message "INVALID RISE TIME" will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value.

The RISE TIME filter functions for both the recorder and integrator outputs.

Spectrum

The SPECTRUM line contains four fields used to program the parameters for a discrete spectra. Because the scanning speed of the monitor exceeds the response speed of analog recording devices, no real-time spectral data is transmitted out of the unit's four recorder channels. Spectra must be stored in the monitor's memory under PROGRAMMED MODE OPERATION conditions and then later replayed using the SPECTRA PLAYBACK function. Subsequently, spectra should only be taken while in the PROGRAMMED OPERATION MODE. If spectra are taken while under IMMEDIATE MODE OPERATION, the scan command will be carried out, but no spectral data will be available.

The first two fields of the SPECTRUM line define the spectral start and end wavelengths, respectively. Values outside of the allowed 190 to 365 and 366 to 800 nm range will not be accepted; the message "INVALID SCAN PARAMETERS" will appear. To clear the error message, press any key and enter the correct value. The detector scans from low to high wavelengths. Thus you must specify an end wavelength value which is greater than the start wavelength value.

The Playback Line

This line allows access to either the CHROMATOGRAM or SPECTRA PLAYBACK menus. To access the CHROMATOGRAM PLAYBACK menu, you must first store a chromatogram in the instrument's memory. Failure to do so results in a "NO CHROMATOGRAM IN MEMORY" error message. To remove the message, press any key.

To access the SPECTRA PLAYBACK menu, you must first store spectra in the instrument's memory. Failure to do so results in a "NO SPECTRA IN MEMORY" error message. To remove the message, press any key.

The menus are discussed in detail below.

The Menus Line

The MENUS line contains fields for entering the AUXILIARY menu and arming the detector for PROGRAMMED OPERATION. To call up the AUXILIARY menu, move the cursor to the AUXILIARY field and press ENTER. This menu is further discussed in later sections of this manual.

In order for the detector to enter the PROGRAMMED OPERATION MODE, in which you may run programs for SCHEDULED WAVELENGTH CHANGES and SCHEDULED SPECTRA and store chromatographic and/or spectral data for later replay, you must ARM THE INSTRUMENT TO START. This is accomplished by moving the cursor to the START CHROMATOGRAM field and pressing ENTER. You should complete all programming steps before arming the monitor.

The Mode Line

The MODE line contains the field used to toggle the detector between SPECTRA ONLY (MONOCHROME), SINGLE WAVELENGTH (PROGRAMMED WAVELENGTH CHANGES), and MULTIPLE WAVELENGTH (POLYCHROME) modes of operation. Of these three, only TIME CHANGES must be operated while in the PROGRAMMED OPERATION MODE. Toggling between these three modes is accomplished by moving the cursor to the MODE field and pressing ENTER until the desired mode is displayed.

It is important to check the status of the mode line when performing IMMEDIATE or PROGRAMMED MODE multi-wavelength. Entering the wavelength, range, recorder channel, rise time, ratio, and integrator channel outputs on the MULTI-WAVELENGTH menu does NOT automatically toggle the mode line to the MULTIPLE WAVELENGTH status. You must manually change to this mode. It is recommended that you program all desired polychrome wavelengths on the MULTI-WAVELENGTH menu before toggling the MODE line to the MULTIPLE WAVELENGTH status.

In most case, programming the SCHEDULED WAVELENGTH CHANGES menu automatically advances the mode status to TIMED CHANGES. However, it is recommended that you check the status of this field before arming the detector.

4.6 The Multi-Wavelength Menu

The MULTI-WAVELENGTH menu is used to program detector parameters for HIGH SENSITIVITY MULTI-WAVELENGTH detection in both IMMEDIATE and PROGRAMMED modes of operation. The menu may be present in two forms.

- IMMEDIATE MODE MULTI-WAVELENGTH MENU
- PROGRAMMED MODE MULTI-WAVELENGTH CHROMATOGRAM UNDERWAY MENU

NUMERICAL RATIO. By calculating the NUMERICAL RATIO as a percent of full scale and tracing it as a log function, the overall RATIO PLOT is easily displayed on a strip chart recorder, eliminating a ratio full scale setting. If the recorder's pen is positioned at 50% full scale for a ratio of 1.0, then 100% full scale indicates a ratio of 10.0 and 0% full scale corresponds to a 0.1 ratio. The NUMERICAL RATIO may be calculated from the RATIO PLOT by using the following equation.

$$\text{Numerical Ratio} = 10^{\left(\frac{X-50}{50} \right)}$$

where X is the full scale deflection of the recorder pen.

Some Examples:

During the elution of a given peak, the RATIO PLOT moved from 50% full scale to 30% full scale. The corresponding NUMERICAL RATIO (NR) is:

$$\text{Numerical Ratio} = 10^{\left(\frac{30-50}{50} \right)} = 10^{-2/5} = 0.398$$

During the elution of another peak, the RATIO PLOT moved from 50% full scale to 75% full scale. The corresponding NUMERICAL RATIO (NR) is:

$$\text{Numerical Ratio} = 10^{\left(\frac{75-50}{50} \right)} = 10^{1/2} = 3.162$$

MULTI-WAVELENGTH						MENU 4	
☐ Return to previous menu					☐ sort		
Ratio: 254 over 350					: CH 3		
NO	nm	CH	range	RT	INT		
1	254	1	0.0020	1.0	1	☐ INS	
2	350	2	0.0020	1.0		☐ INS	

The first and second fields of the RATIO line are used to enter the numerator and denominator wavelengths for the ratio, respectively. Values are entered by moving the cursor to the appropriate field, typing in the wavelength values, and pressing the ENTER key. Acceptable wavelengths are any of the monitored wavelengths in the 190 to 800 nm range.

The third field of the RATIO line controls the recorder output for the RATIO PLOT. Acceptable values are 1, 2, 3, and 4. The user should avoid entering a RATIO PLOT OUTPUT which is already specified as a RECORDER CHANNEL for one of the monitored wavelengths. If duplication occurs, the RATIO PLOT will be transmitted out of the specified RECORDER CHANNEL.

As a general rule, a rise time should be selected which is 1/10th the half-height width of the fastest peak in a given separation. The default RISE TIME setting of 1.0 second will suffice for peaks as fast as 10 seconds at half height. The RISE TIME filter may be totally bypassed by entering a RISE TIME value of 0.0 seconds.

Allowable RISE TIME values range from 0.0 to 9.9 seconds in intervals of 0.1 second. Values outside of this range will not be accepted; the message "INVALID RISE TIME" will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value.

The RISE TIME filter functions for both the recorder and integrator outputs.

MULTI-WAVELENGTH						MENU 4
☐ Return to previous menu					☐ sort	
Ratio: 254 over 350 : CH 3						
NO	nm	CH	range	RT	INT	
1	214	1	0.0020	1.0	I	☐ INS
2	254	2	0.0020	1.0		☐ INS
3	280	3	0.0020	1.0		☐ INS
4	329	4	0.0020	1.0		☐ INS

Integrator (INT)

This field selects the INTEGRATOR OUTPUT WAVELENGTH. The default status places an "I" in the line of the first wavelength. To select a different wavelength to be sent to the INTEGRATOR OUTPUT, move the cursor to the INT field of the desired wavelength and press the ENTER key. The "I" should now be displayed in the desired wavelength line. Data channeled to the INTEGRATOR OUTPUT is transmitted at a fixed 1 volt/AU scale and is filtered by the RISE TIME circuit. To obtain unfiltered data, enter a 0.0 second RISE TIME for the integrated wavelength.

The Insert Function (INS)

The INS field is used to insert a wavelength between two other monitored wavelengths. To do so:

Move the cursor to the INS field of the wavelength line that is to precede the inserted wavelength and press ENTER. An exact copy of this line will appear immediately below.

Move the cursor to the newly inserted line and enter the desired parameters for the inserted wavelength.

It is not necessary to enter wavelengths in increasing or decreasing order. The monitor automatically sorts the entered wavelengths and scans them in order of increasing wavelength (this sorting function is not indicated by the LCD). Because only five wavelengths and their absorbance values may be viewed on the MULTI-WAVELENGTH menu at any time, the INSERT function was included to allow the user to enter a wavelength within a group of other wavelengths, so that all may be simultaneously displayed.

Immediate Mode and Programmed Mode Multi-Wavelength Menus

All previous examples of the MULTI-WAVELENGTH menu displayed in this section depicted the IMMEDIATE MODE MULTI-WAVELENGTH menu. This menu is used during IMMEDIATE

The remaining fields comprise the various components for the SCHEDULED WAVELENGTH lines. A maximum of 20 wavelength changes may be entered. Attempts at entering more than 20 changes result in the error message "PROGRAM MEMORY FULL." This is displayed on the bottom line and will obscure the last (20th) scheduled wavelength change. To clear the error message, press any key.

The Wavelength Line Fields are:

Wavelength (NM)

This field is used to enter the wavelength value for a particular wavelength change. Acceptable wavelengths extend from 190 to 365 and 366 to 800 nm. Values outside of this range will not be accepted; the message "INVALID WAVELENGTH ENTERED" will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value.

From...To...

These two fields allow entry of the elapsed time range during which a given wavelength will be monitored. To change the time, type in the new value and press the ENTER key. The new value will be displayed. Allowed values are integers between 1 and 65534 second. Values outside this range are ignored.

Recorder Channel (CH)

This field is used to enter the RECORDER CHANNEL OUTPUT for real-time absorbance data monitored for the defined wavelength change. Acceptable values range from 1 to 4.

Range

This field is used to set the full scale range in absorbance units (AUFS) for a given wavelength change. The default value is 2.0000 AUFS. Acceptable values range from 0.0005 AUFS to 7.9995 AUFS in multiples of 0.0005 AUFS. Values may be entered to four decimal places; the detector will round the fourth decimal place to the nearest 0.0005 AUFS. Values outside of the allotted range will not be accepted; the message "INVALID RANGE ENTERED" will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value.

Filter (Rise Time)

This field is used to set the RISE TIME FILTER for a given scheduled wavelength. This setting controls a progressive averaging, digital filter that conserves peak area at all settings.

As a general rule, a rise time should be selected which is 1/3 to 1/5 the half-height width of the fastest peak in a given separation. The default setting of 1.0 second will suffice for peaks as fast as 3 seconds at their half height. The filter may be totally bypassed by entering a RISE TIME value of 0.0 second.

Allowable RISE TIME values range from 0.0 to 9.9 second intervals of 0.1 second. Values outside of this range will not be accepted; the message "INVALID RISE TIME" will flash on the bottom line. When this occurs, press any key to clear the error message and then enter the correct value. THE RISE TIME FILTER FUNCTIONS FOR BOTH THE RECORDER AND INTEGRATOR OUTPUT.

error message "PROGRAM MEMORY FULL." This is displayed on the bottom line and will obscure the last (32nd) scheduled spectra. To clear the error message, press any key. IT IS NOT NECESSARY TO ENTER SCHEDULED SPECTRA IN CHRONOLOGICAL ORDER; THE MONITOR HAS A SORTING FUNCTION.

The Scheduled Spectra Fields are:

Time

This field is used to enter the time, in seconds, during which a given scheduled spectra will be performed. Allowable times range from 1 to 16383 seconds. Values outside of this range will not be accepted; the error message "INVALID SPECTRA TIME" will appear on the bottom line. To remove the message, press any key.

Reps

This field controls the REPETITION number for a given scheduled spectra. This parameter determines the number of consecutive scans that will be performed during spectral acquisition. The resultant scans are averaged together and stored in the monitor's memory for later playback (see the SPECTRUM section of the MAIN menu portion of the COMMAND REFERENCE GUIDE for further details).

Allowable values for the REPS field are 1, 2, and 4. Values other than these will not be accepted; an error message listing acceptable REPS values will appear. To clear the message, press any key.

Zero

Press ENTER to schedule a Spectra Zero.

Sort

This field is used to sort the scheduled spectra in chronological order after entering them in the menu. To do so, move the cursor to the SORT field and press ENTER. The scheduled spectra should now be listed in chronological order.

4.9 The Playback Chromatogram Menu

This menu is used to play back chromatographic data stored under the PROGRAMMED OPERATION mode. There are two types of PLAYBACK CHROMATOGRAM menus: Single Wavelength and Multiple Wavelength. Both are accessed from the MAIN menu by moving the cursor to the CHROMATOGRAM field of the PLAYBACK line and pressing ENTER.

```
CHROMATOGRAM PLAYBACK          MENU 8
ABSORBANCE @ 254 nm = 0.00000 AU  0.0
MODE: spectra only
☐Return      ☐Playback chromatogram
... @          1 times real time  ☐stop PB
PB range= 2.0000 AUFS; rise time= 1.0 sec
Baseline  0.83 to 3.33 min @ 0.00000 AU
Strip chart channel no.= 1
```

The STOP PLAYBACK (STOP PB) field is used to interrupt the replay of a stored chromatogram. This is accomplished by moving the cursor to this field and pressing the ENTER key.

The sixth line contains the PLAYBACK RANGE and PLAYBACK RISE TIME parameters. Allowable values for range extend from 0.0005 to 7.9995 AUFS in multiples of 0.0005 AUFS. Allowable RISE TIME values extend from 0.0 to 9.9 seconds in multiples of 0.1 second. Values outside of these allotted ranges will trigger an error message.

The final line contains the STRIP CHART CHANNEL OUTPUT NUMBER field. This field controls the recorder outputs for the playback process. Allowable values range from 1 to 4.

If you wish to send filtered, but nonranged data to a stand-alone integrator or data acquisition system during the playback process, you must use the integrator output. The data is automatically transferred to the integrator output upon initiation of the playback process.

4.10 Multi-Wavelength Playback

MULTI-WAVELENGTH PLAYBACK						MENU 9
☐ Return		☐ Playback chromatogram				
... @		1 times real time				☐ stop PB
Ratio: 214 over 280 : CH 4						
Baseline 0.83 to 3.33 min @ 0.00000 AU						
NO	nm	CH	range	RT	INT	
1	214	1	0.0020	1.0	I	0.00000
2	254	2	0.0020	1.0		0.00000
3	280	3	0.0020	1.0		0.00000
4	329	4	0.0020	1.0		0.00000

The second line contains the RETURN and PLAYBACK CHROMATOGRAM fields. Moving the cursor to the RETURN field and pressing ENTER returns the display to the MAIN menu. Moving the cursor to the PLAYBACK CHROMATOGRAM field and pressing the ENTER key begins the replay process. Chromatogram replay is indicated by the appearance of absorbance values (in the wavelength lines) and elapsed time (in the left side of the first line) upon the menu. All times are listed in seconds.

The third line contains the MEMORY EXTENDER MULTIPLIER VALUE, PLAYBACK RATE VALUE, and STOP PLAYBACK (PB) fields. The user is instructed to consult the previous section for an explanation of these features.

The fourth line contains the RATIO WAVELENGTHS and RATIO PLOT OUTPUT fields. The first of the two RATIO WAVELENGTH fields is the numerator wavelength and the second is the denominator wavelength. For more information regarding the monitor's ratio function, the user is referred to Ratio section of the Polychrome Portion of the Command Reference Guide. The CHANNEL (CH) field of this line selects the recorder output for the RATIO PLOT. Allowable values range from 1 to 4. The user must insure against specifying a RATIO PLOT output which is concomitantly used as one of the wavelength recorder outputs. If this occurs, the recorder output will be assigned to the last field in which the repeated output value was entered and the other field will not specify an output.

The next several lines list parameters for all of the recorded wavelengths.

The final line is used to select between seconds or minutes for timing purposes.

Baseline 0.80 to 3.33 min @ 0.00000 AU
Auto Spectra ☐OFF 1 per peak; sides EARLY
Auto scan relay ☐disabled
Zero upon wavelength change ☐ON
SCROLL to access additional parameters
Zero value @ 254 nm 0.00000 AU
Memory extend 1 times
Display time base ☐min

SPECTRA PLAYBACK	MENU 10
Absorbance @ 210 nm = 0.00000 AU	
Spectrum: 200 to 360 @ 10 nm steps	
Spectrum no 1 taken at 1 sec w/4reps	
☐ Return	☐ Playback spectra
☐ Next spectra	playback rate 20 nm/sec
Range 2.0000 AUFS	ch no

The second line of this menu displays the wavelengths and their absorbance values measured in the recorded spectra. Before playback begins, this line will remain blank. Upon initiating the playback process, these fields are displayed.

The third line displays the programmed spectral parameters of lower, upper, and step wavelengths for the stored spectra. There are no user interactive fields on this line.

The fourth line contains information about the stored spectra. Stored spectra are all numbered in sequence of their acquisition and are recalled, by using the NEXT SPECTRA field, in numerical order. This line displays the spectra number, the time (in seconds) during the run in which it was taken, and the number of repetitive scans averaged in the stored data. There are no user interactive fields on this line.

The next line contains the RETURN and PLAYBACK SPECTRA fields. Moving the cursor to the RETURN field and pressing the ENTER key, returns the display to the MAIN menu. Moving the cursor to the PLAYBACK field and pressing the ENTER key begins the spectra playback process. Playback is indicated by the appearance of wavelength and absorbance data on the second line. Upon completion of the playback, these values will disappear from the menu.

The fifth line contains the NEXT SPECTRA and PLAYBACK RATE fields. The NEXT SPECTRA field is used to advance through the stored spectra in numerical order. Moving the cursor to this field and pressing the ENTER key, advances the menu to display the parameters for the next stored spectra. Spectra are stored in order of their acquisition. This newly displayed spectra is replayed by moving the cursor to the PLAYBACK SPECTRA field and pressing ENTER.

The PLAYBACK RATE field specifies the rate of spectra playback in a nm/second basis. Allowable rates extend from 1 to 255 nm/second in 1 nm/second intervals. Spectra playback rates should be chosen which do not exceed the frequency response of the recording device. Matching the playback rate to a given strip chart paper speed, allows calibration of the wavelength axis.

The final line lists the fields for SPECTRA PLAYBACK RANGE and CHANNEL NUMBER (CH NO). Spectra may be replayed at any range within the 0.0005 AUFS to 7.9995 AUFS range in multiples of 0.0005 AUFS. Spectra may be played out of any of the four recorder outputs.

4.12 The Auxilliary Menu

The AUXILIARY menu is accessed from the MAIN menu by moving the cursor to the AUXILIARY field of the MENUS line and pressing the ENTER key.

- Unscrew the two thumbscrews on the photodiode mount and pull the photodiode mount straight back. The connecting cable is long enough to allow the photodiode mount to rest on the benchtop.
- Loosen the thumbscrew holding the tubing strain relief clamp in place and gently pull the clamp toward you far enough to disengage the tubing from the clamp.
- Unscrew the two thumbscrews holding the cell holder assembly in place and pull the cell holder assembly straight back toward you to remove it.
- Replacement cells are mounted pre-aligned in a cell holder assembly. Installation reverses the removal process.
- Slide the cell holder assembly onto the alignment dowels. The inlet line should come into the bottom of the cell in order to provide efficient bubble flushing. Securely fasten the cell holder assembly with two thumbscrews.
- Slip the inlet and outlet tubes into the slots in the strain relief clamp and tighten the thumbscrew holding the clamp in place.
- Replace the photodiode mount (the cable should come from the right side) and fasten it securely with the two thumbscrews.
- Replace the cell cover (be careful not to pinch the cable or the tubing) and fasten it securely with the thumbscrew.
- Reconnect the inlet line to the column, stop-flow accessory, or the connecting tubing and reconnect the outlet tubing to function collection, back pressure device, or waste container.
- Reconnect the power cord to the rear panel of the detector.

Note: Be careful to avoid putting fingerprints on the photodiode. The photodiode surface may be cleaned with spectroscopic grade methanol and lint-free lens paper.

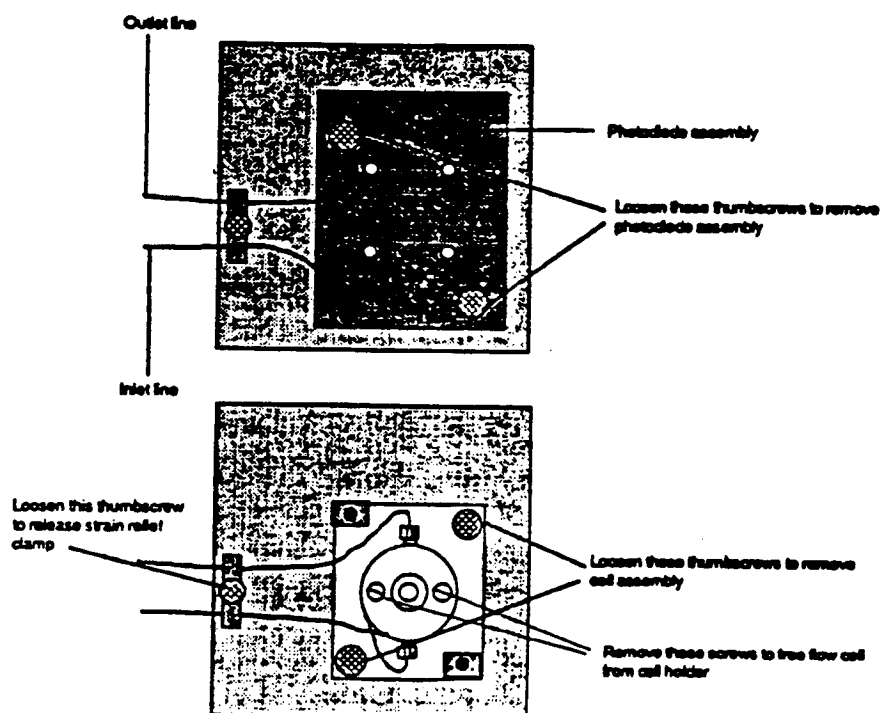


Fig. 5.2.

Disconnect the tungsten lamp lead from the detector (the lead has 2 wires and connects to the lower of the lamp terminals) by gently pulling straight back toward you. Do not twist the connector while pulling (See figure 5.3).

Unscrew the thumbscrew holding the lamp mount in place, and pull the lamp mount straight back toward you. Be careful not to lose the aluminum standoff on the thumbscrew. Be careful not to get fingerprints on the lamp.

Note: Never loosen the screws holding the lamp to the mount, and never attempt to move the lamp up or down in the mount. Doing so will degrade the system performance. The lamp is provided pre-aligned in the mount.

To install the tungsten lamp:

Slide the lamp mount onto the alignment dowels located above and below the aperture (the mount has pre-drilled holes to accommodate the dowels). The lamp leads should emerge from the right and slightly below the lamp. Use the thumbscrew and aluminum standoff to attach the lamp mount to the case. Connect the lamp lead to the lower of the two terminals in the lamp compartment.

Replace the lamp compartment cover and fasten with the screw.

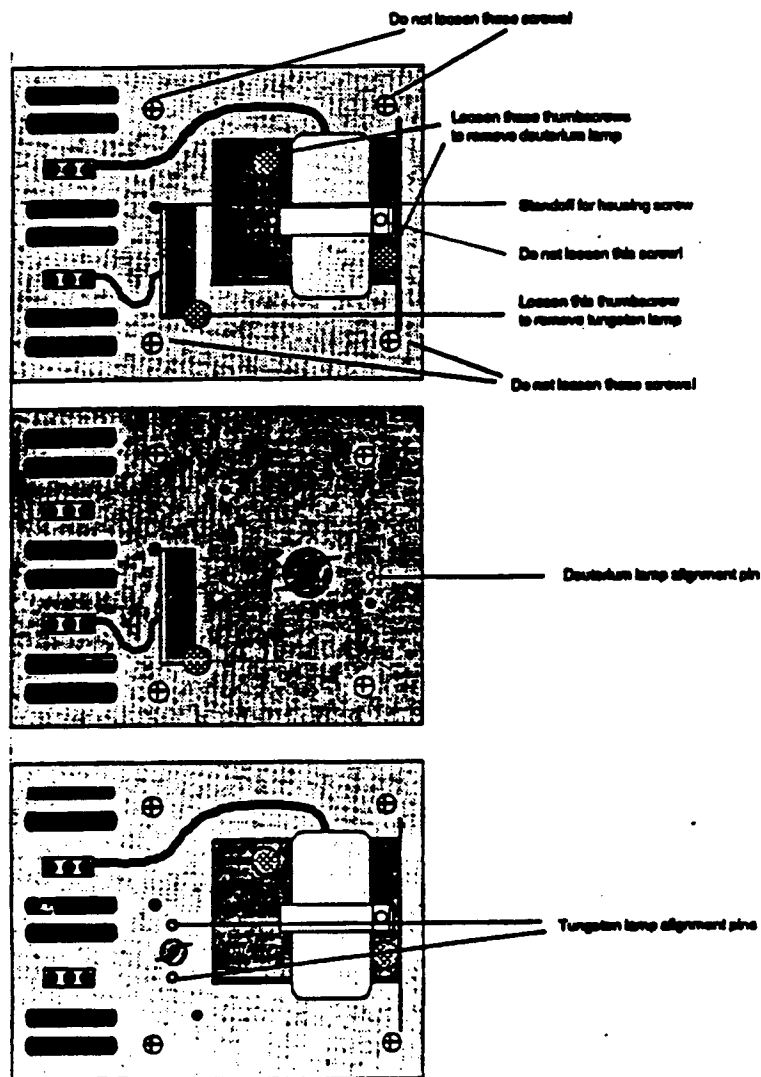


Fig. 5.3.

"CANNOT STORE/RECALL WHILE ACTIVE": The detector is unable to recall data while a method is being run. Wait until the method is over before attempting to store or recall data.

"CANNOT START DURING SPECTRA": The detector will not allow the starting of a chromatographic run while a spectral run is in progress.

"CANNOT CALIBRATE IN DYNAMIC MODES": It is not possible to calibrate the detector when running a method. You must first press stop chromatogram.

"DUAL LAMP MULTI NOT POSSIBLE": It is not possible to scan between the Deuterium lamp (190 to 365 nm) and the Tungsten lamp (366 to 800 nm) while running a method.

"D2 LINE TO FAR FROM EXPECTED VALUE": This indicates a problem with the grating being out of calibration. For a corrective procedure see LOW LIGHT FROM DEUTERIUM LAMP.

"GRATING ACTUATOR ERROR": If the grating mechanism becomes out of sync, this error appears followed by **"GRATING ERROR RECOVERY"**. This is a self correcting problem. If this message occurs frequently, contact your factory representative.

"GRATING CONTROLLER ERROR": If this message occurs, turn the detector's power off and on again. The message should not recur. If it does, contact your factory representative.

"INVALID WAVELENGTH": An attempt to enter a wavelength below 180 nm or above 800 nm.

"INVALID RANGE ENTERED": An attempt to enter a range of zero (0) or above 7.9995.

"INVALID CHANNEL NUMBER": An attempt to enter a recorder number below 1 or above 4.

"INVALID RISE TIME": An attempt to enter a rise time greater than 9.9 second.

"INVALID PLAYBACK RATE": An attempt to enter a playback rate slower than real-time or faster than an allowed time factor of twenty.

"INVALID METHOD NUMBER": An attempt to enter a method number greater than four.

"INVALID SPECTRA TIME": You have entered a time value greater than allowed.

"INVALID TIME": You have entered a time value greater than allowed.

"LOW LIGHT FROM DEUTERIUM LAMP": Either the deuterium lamp has failed to light on the second try or else the lamp has lit, but the output is below acceptable limits. The reference photodiode looks for key atomic wavelines and intensities. If these are not adequate the first time the lamp is lit, the detector will relight the lamp. If on the second attempt the lamp fails, this message is displayed.

If this occurs you should switch the power off for five seconds then power-up the detector again. If the error message recurs, disconnect the power cord, open the lamp housing (refer to "Section 5" on service) and check that the deuterium lamp lead is connected to the upper of the two receptacles. If the lead is properly connected, replace the deuterium lamp assembly.

"LOW LIGHT FROM TUNGSTEN LAMP": Follow procedure for Deuterium lamp.

"MULTI MEMORY IS FULL": No further data can be stored in the detector's memory. One solution would be to rerun the chromatogram with the memory extender feature. To clear the memory, either start a new run or turn off the power.

"NO SPECTRA IN MEMORY": An attempt to playback a spectrum when one is not in the detector's memory.

rows. The left-hand row represents the relative light intensity from the reference photodiode. The next-to-leftmost row represents the relative light intensity from the sample photodiode. You should use only the four-digit numbers.

In order to diagnose the condition of the flow cell and both lamps, this menu should be invoked at both 254 nm (deuterium lamp) and 546 nm (tungsten lamp). For the standard stainless steel, 6 mm, 4 ul HPLC flow cell, the following rules apply:

- At 254 nm:
 - Reference numbers should be greater than 1100
 - Ratio of sample to reference numbers should be equal to or greater than 0.4
- At 546 nm:
 - Reference numbers should exceed 1200
 - Ratio of sample to reference numbers should exceed or match 0.6

The reference numbers should be the same regardless of flow cell. The sample/reference ratios for the microbore, capillary, Kel-F and preparative flow cells will be lower.

In general, a dirty flow cell will yield acceptable reference levels and poor sample to reference ratios (be careful, however, excessive solvent absorbance can generate the same problems). A weak lamp will yield a poor reference level with acceptable or near-acceptable sample to reference ratio.

To exit from the diagnostic menu, move the cursor to the RETURN position and press ENTER.

Symptom	Cause	Remedy
3. Baseline drifts excessively	a. Unit is not warmed up	a. Allow 1 hour warm up for optimum performance
	b. Contamination of sample cell window has occurred	b. Clean cell by flushing with solvents (methanol, acetone, water, nitric acid (5N), water, methanol). Inspect cell and photodiode for fingerprints and smudges and clean if necessary
	c. Solvent from column is changing absorption	c. Column is filled with UV absorbers that are bleeding - replace column, impure solvent is equilibrating with the column - replace solvent with more pure grade, switch to a longer wavelength so that background absorption does not fluctuate as much
	d. Leakage in the lines from sample to flow cell	d. Check lines for leakage
	e. Tiny bubble trapped in the sample cell	e. Increase flow rate and/or back-pressure
	f. Output span of detector does not match input span of recorder or integrator	f. Press event mark to see if a spike 20% of full scale occurs
	g. Large temperature fluctuations are occurring	g. Remove detector from the source of drafts of hot and cold air
	h. Flow cell, photodiode assembly, or flow cell cover is loose	h. Tighten thumbscrews fastening flow cell holder, photodiode assembly, and flow cell cover
4. Detector's display panel is not readable	a. Display contrast potentiometer is not set correctly	a. Reach around to the rear panel and turn pot till display is readable (Figure 2.2 #9)
	b. Fuse is blown	b. Check voltage selector card on power inlet module, then check fuse if 7979 power voltage input is correct.

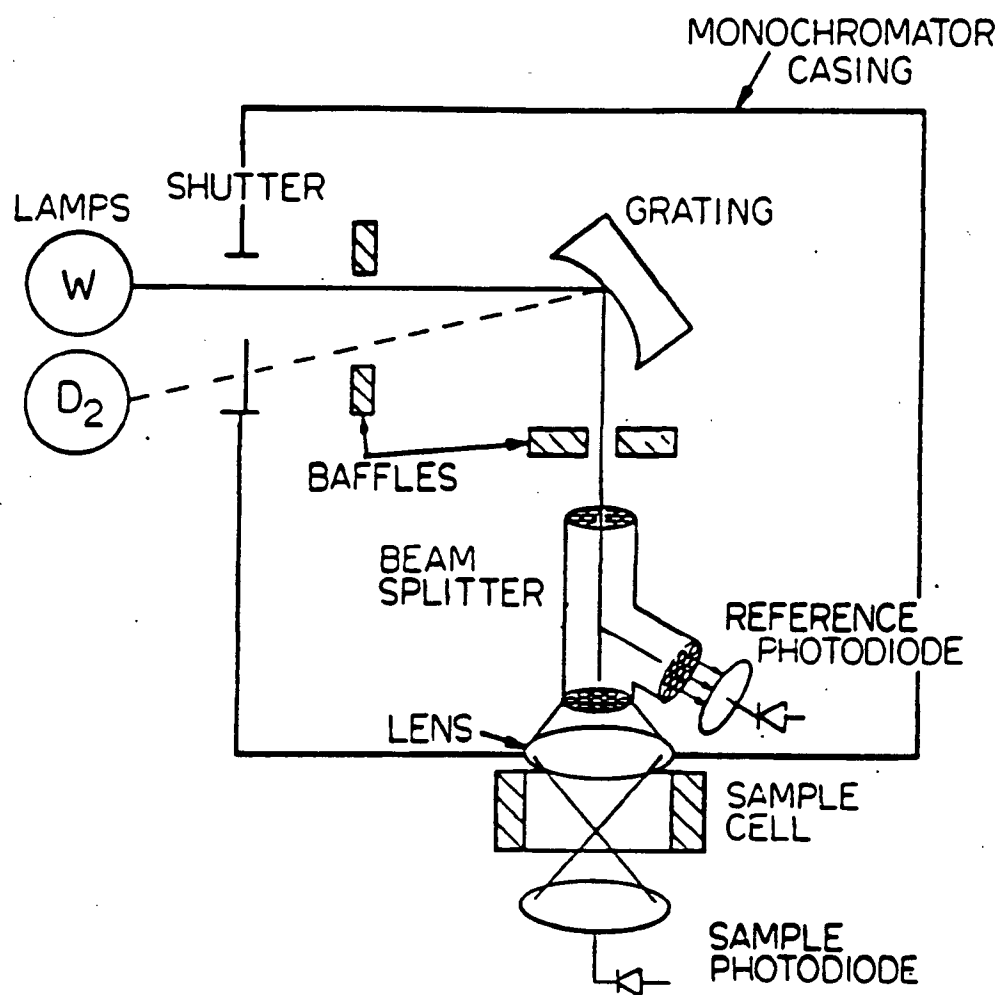


Fig. 7.1. The detector optical system.

7.4 Ratio Function

Ratioing the absorbancies at two particular wavelengths may be used as a means of determining the purity of a given peak. The ratio value at any given time is the quotient of the molar absorptivities of the peak at the two ratioed wavelengths. If the peak is pure, that is, if no compound coelution occurs, the ratio should remain constant during the peak's elution period and the resultant ratio trace would be a square wave. A changing ratio during a peak's elution period is indicative of solute coelution, and the peak is impure. Ratioing will only detect compound coelution if the impurity absorbs at both of the ratioed wavelengths. Additionally, if both the impurity and solute of interest display identical elution profiles (equal ascending and descending slopes resulting in an identical increase and decrease in concentration as a function of time), the measured ratio becomes the sum of the molar absorptivities of the solute and impurity, and no ratio change will be seen during the peak's elution period.

7.5 The Detector's Ratio Algorithm

Calculated ratios are NOT the ratios of the absolute absorbancies for the two ratioed wavelengths, but rather the ratios of their absorbancies as a percentage of full scale. Thus, the calculated numerical ratio is:

$$\text{NUMERICAL RATIO} = \frac{\text{ABSORBANCE WAVELENGTH 1} / \text{RANGE WAVELENGTH 1}}{\text{ABSORBANCE WAVELENGTH 2} / \text{RANGE WAVELENGTH 2}}$$

Since it is of no value to ratio baseline absorbance data, the ratio calculation is only performed when BOTH wavelength absorbancies exceed 3% of full scale. Ratios are no longer calculated when these values fall below a full scale level of 2%. The chart recorder ranges for the wavelengths that are being ratioed should be set so that the peaks of interest fall between 20% and 100% of full scale at both wavelengths.

The NUMERICAL RATIO which is displayed on the detector's LCD is not the ratio transmitted out of the RECORDER OUTPUT. The recorder output transmits a RATIO PLOT which is the log of the NUMERICAL RATIO. By calculating the NUMERICAL RATIO as a percent of full scale and tracing it as a log function, the overall RATIO PLOT is easily displayed on a strip chart recorder, eliminating a ratio full scale setting. If the recorder's pen is positioned at 50% full scale for a ratio of 1.0, then 100% full scale indicates a ratio of 10.0 and 0% full scale corresponds to a 0.1 ratio. The NUMERICAL RATIO may be calculated from the RATIO PLOT by using the following equation:

$$\text{Numerical Ratio} = 10^{\left(\frac{X-50}{50} \right)}$$

where X is the full scale deflection of the recorder pen.

Some Examples:

During the elution of a given peak, the RATIO PLOT moved from 50% full scale to 30% full scale. The corresponding NUMERICAL RATIO (NR) is:

$$\begin{aligned} \text{Numerical Ratio} &= 10^{\left(\frac{30-50}{50} \right)} \\ &= 10^{-2/5} = 0.398 \end{aligned}$$

During the elution of another peak, the RATIO PLOT moved from 50% full scale to 75% full scale. The corresponding NUMERICAL RATIO (NR) is:

$$\begin{aligned} \text{Numerical Ratio} &= 10^{\left(\frac{75-50}{50} \right)} \\ &= 10^{1/2} = 3.162 \end{aligned}$$

The Down-Shoulder: When the absorbance value is less than the first value following the triggering of the Up-Shoulder spectra.

The user may choose to gather one, two, three spectra for each peak.

7.8 The Auto-Calibration Function

The monitor is equipped with an AUTO-CALIBRATION program designed to insure proper wavelength accuracy. When triggered, this program searches for discrete lines of the deuterium lamp in the visible region. These lines are monitored using the reference photodiode to avoid any deleterious effects stemming from the flow cell's contents. The calibration parameters are stored in the nonvolatile memory of the detector and will remain after the detector has been disconnected from power. See Section 3.18.

It is recommended that the user allow the detector a 60 minute warm-up before activating the auto-calibration function.

7.9 Important Error Messages

If the user should enter invalid parameters on any of the detector's menus, an ERROR MESSAGE will appear on the bottom line. Each message informs the user of his error, and they may be erased by pressing any keyboard key. Most of these messages are self explanatory and do not warrant further discussion. However, two possible error messages need be addressed:

"Non-Volatile Ram Error": See Manual - This message indicates that the battery-backed up random access memory (RAM) has failed. If this occurs, turn off the monitor and reset the RAM by powering up the detector while holding down any of the front panel keys. A RAM reset is indicated by a 256 nm displayed on the MAIN MENU. After a 30 to 60 minute warm-up, activating the AUTO-CALIBRATION function is recommended.

"Autocal Value Invalid": See Manual - This message indicates that the autocalibration function did not find the proper Deuterium spectral lines. Repeat the Auto-Calibration procedure.

Continued autocalibration failure may be due to a bad deuterium lamp. See the CHECKING SAMPLE AND REFERENCE LIGHT INTENSITIES portion of the troubleshooting section for further details.

Four wavelengths 220-330 nm, 400-800 nm	$< 3 \times 10^{-5}$ AU/60 sec with 1.0 sec rise time
Data acquisition Rate	24 points/seconds/wavelength
High Speed Multiple Wavelength Operation	
Typical Noise with 2.0 sec. rise time for a scan from 195 to 800 nm	195-210 nm $< 8 \times 10^{-5}$ AU 210-300 nm $< 3 \times 10^{-5}$ AU 300-355 nm $< 4 \times 10^{-5}$ AU 355-430 nm $< 8 \times 10^{-5}$ AU 430-600 nm $< 4 \times 10^{-5}$ AU 600-800 nm $< 3 \times 10^{-5}$ AU
Data Acquisition Rate	96 points/second/wavelength
Typical Noise of Discrete Spectra	$< 5 \times 10^{-5}$ AU/nm
Typical Noise of Time Slice Spectra	Dependent upon noise of Multiple Wavelength data -4
Drift	$< 2 \times 10^{-5}$ AU/hour after 30 minute warm-up
Zero Adjust	Plus or minus any amount by auto-zero and manual entry
Chart Recorder Filter	Digital active filter with rise times selectable from 0.0 to 9.9 seconds (divide by 2 to convert to an equivalent time constant)

5. Remove the two thumbscrews securing the flow cell assembly to the detector and pull the flow cell assembly straight back along the alignment dowels until it is free from the optical bench.
6. Orient the new flow cell assembly so that the dowel rods, mounting screw holes, and tubing are in the proper position. Mount the cell so the inlet tube (wrapped around the flow cell body) is the lower of the two tubes. Avoid scratching or placing fingerprints on the flow cell windows. These windows should be cleaned with spectroscopic grade methanol and line-free lens paper. They may be dusted by blowing off debris with filtered, oil-free air.
7. Secure the flow cell assembly to the optical bench using the two thumbscrews.
8. Lock the inlet and outlet tubing into place by tightening them down with the tubing restraint.
9. Replace the photodiode assembly.
10. Replace the flow cell housing cover by replacing its retaining screw.
11. Connect the flow cell inlet and outlet tubing to the appropriate components.

Flow Cell Operation

A. Applications

The performance characteristics of the flow cell are dependent on its application. Flow cell Model 167-0910 is intended for use in analytical HPLC applications. Its 6 mm pathlength yields superior sensitivity, while its 9 ml volume maintains excellent peak shape. This cell is designed to be used with 1/16" O.D. x 0.010" ID stainless steel tubing.

B. Static Pressure Ratings and Flow-Pressure Characteristics

The flow cell is rated up to 1,000 PSI nominal static pressure. Pressures exceeding 1,000 PSI may result in cell leakage. If leakage occurs, the cell gaskets should be replaced (See the "Flow Cell Maintenance" section for cell disassembly and repair).

Pressure vs. flow rate data for flow cell is presented in figure C.2. This data reflects typical results obtained for the described flow cell with the standard length of appropriate ID stainless steel tubing used for the cell inlet and outlet lines with water as the solvent. To calculate the expected backpressure for a different solvent (P_{solvent}) in your system:

1. Determine the flow rate to be used.
2. Interpolate the pressure drop for water using the graph (PH₂O).
3. Multiply that pressure drop figure by the viscosity of the solvent in question (Cps).

$$P_{\text{solvent}} = (PH_2O) (Cps)$$

Under typical conditions, laminar flow exists in flow cell. In this case, the following relationship holds:

$$\text{Pressure} = (\text{Flow Rate}) (\text{Resistance})$$

Flow Cell Maintenance

Cell Cleaning

Do not disassemble flow cells for routine cleaning purposes. Most cells can be adequately cleaned by flushing with several milliliters of appropriate solvent. We recommend the following solvents for this purpose.

- Methanol
- Tetrahydrofuran
- Methylene chloride
- HPLC grade water
- 6N nitric acid followed by flushing with HPLC grade water

Cell Disassembly

If flushing proves inadequate for cleaning, or if the flow cell becomes leaky requiring gasket replacement, the following procedure should be followed for flow cell disassembly: (see Figure C.3)

1. Stainless Steel Heat Exchanger Inlet Tube
2. 1/16" Male Tubing Nut
3. 1/16" Stainless Steel Ferrule
4. Heat Exchanger Retaining Washer
5. Heat Exchange Retaining Screw
6. Cell Body
7. Cell Gasket
8. Cell Window
9. Cell Nut
10. Spring Washer

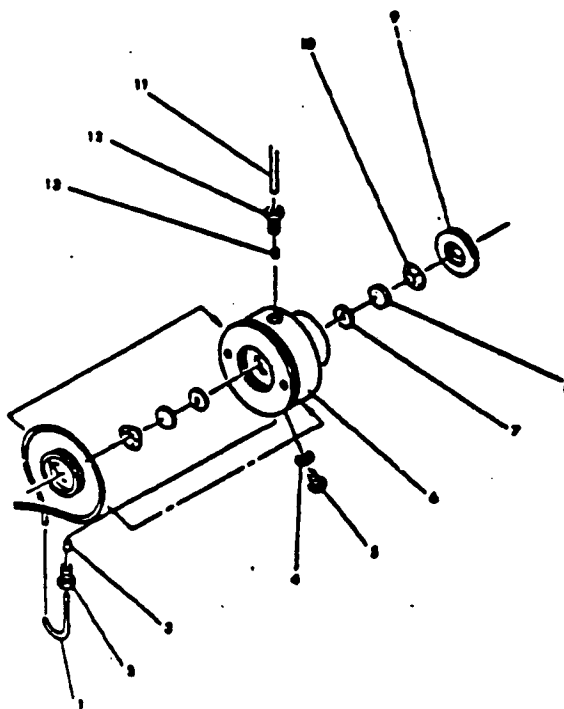


Fig. C.3.

1. Remove the flow cell assembly from the detector as described in "Installation."
2. Remove the two screws which secure the flow cell body to the flow cell mount and free the flow cell from its mount.
3. Using a wide, flat-blade screwdriver, remove the window retaining nut.

Flow Cell Reassembly (See Figure C.3)

Note: For best results, replace the flow cell gaskets each time the cell is disassembled.

1. With a pair of fine forceps, replace the cell gasket in its proper orientation so that both the optical bore (large hole) and fluid bore (small hole) are exposed by the tear drop.
 2. Carefully replace the cell window.
 3. Install the retaining washer so that its concave surface faces away from the cell body.
 4. Replace the window retaining nut. Tighten to 14 inch pounds. To avoid possible damage to the windows, do not exceed 14 inch pounds of torque.
 5. Repeat steps 1-4 for the other side of the flow cell.
 6. If the cell tubing was removed, reinstall as follows:
 - a. Replace the cell inlet tubing by replacing its fitting in the inlet hold (the one next to the hold for the heat exchanger restraint).
 - b. Bend the tubing around the cell so that it circumscribes the cell body in the heat exchanger groove before it approaches the hole for the heat exchanger restraint.
 - c. To improve heat exchange between the inlet tubing and flow cell body, a heat conductive epoxy should be applied into the groove before the tubing is installed.
- Note:** We recommend the use of Loctite Thermal Conductive Adhesive (17099) and Activator (17100), Loctite Corp., Newington, CT 06111.
- d. After the epoxy has been applied, secure the tubing in the heat exchanger restraining screw and washer.
 - e. Replace the cell outlet tubing hole.
 9. Reinstall the flow cell body on the flow cell holder using the two mounting screws. Insure that the protruding nose piece of the flow cell is inserted into the cell mount.
 10. Reinstall the flow cell onto the detector as outlined in the installation section of this manual.

Flow Cell Troubleshooting Guide

This section is organized in a symptom, problem, and remedy format.

Symptom	Cause	Solution
1. Noisy baseline	a. Bubble trapped in flow cell	a. Increase flow rate and/or back-pressure.
	b. Exterior surface of flow	b. Clean flow cell window with